

LEA-6 / NEO-6 / MAX-6

u-blox 6 GPS Modules

Hardware Integration Manual

Abstract

This document describes the features and specifications of the cost effective and high-performance LEA-6, NEO-6 and MAX-6 GPS modules featuring the u-blox 6 positioning engine.

These compact, easy to integrate stand-alone GPS receiver modules combine exceptional GPS performance with highly flexible power, design, and connectivity options. Their compact form factors and SMT pads allow fully automated assembly with standard pick & place and reflow soldering equipment for cost-efficient, high-volume production enabling short time-to-market.



Document Information	
Title	LEA-6 / NEO-6 / MAX-6
Subtitle	u-blox 6 GPS Modules
Document type	Hardware Integration Manual
Document number	GPS.G6-HW-09007-H
Document status	Preliminary

Document status information	
Objective Specification	This document contains target values. Revised and supplementary data will be published later.
Advance Information	This document contains data based on early testing. Revised and supplementary data will be published later.
Preliminary	This document contains data from product verification. Revised and supplementary data may be published later.
Released	This document contains the final product specification.

This document applies to the following products:

Name	Type number	ROM/FLASH version
LEA-6H	All	FW6.02, FW 7.01, FW 7.03
LEA-6S	All	ROM6.02, ROM7.03
LEA-6A	All	ROM6.02, ROM7.03
LEA-6T-0	All	ROM6.02, ROM7.03
LEA-6T-1	All	FW 7.03
LEA-6R	All	FW DR 1.0, FW 7.03 DR2.0
NEO-6G	All	ROM6.02, ROM7.03
NEO-6Q	All	ROM6.02, ROM7.03
NEO-6M	All	ROM6.02, ROM7.03
NEO-6P	All	ROM6.02
NEO-6T	All	ROM7.03
NEO-6V	All	ROM7.03
MAX-6G	All	ROM7.03
MAX-6Q	All	ROM7.03

This document and the use of any information contained therein, is subject to the acceptance of the u-blox terms and conditions. They can be downloaded from www.u-blox.com.

u-blox makes no warranties based on the accuracy or completeness of the contents of this document and reserves the right to make changes to specifications and product descriptions at any time without notice.

u-blox reserves all rights to this document and the information contained herein. Reproduction, use or disclosure to third parties without express permission is strictly prohibited. Copyright © 2011, u-blox AG.

u-blox® is a registered trademark of u-blox Holding AG in the EU and other countries. ARM® is the registered trademark of ARM Limited in the EU and other countries.



Preface

u-blox Technical Documentation

As part of our commitment to customer support, u-blox maintains an extensive volume of technical documentation for our products. In addition to our product-specific technical data sheets, the following manuals are available to assist u-blox customers in product design and development.

- **GPS Compendium:** This document, also known as the GPS book, provides a wealth of information regarding generic questions about GPS system functionalities and technology.
- **Receiver Description including Protocol Specification:** Messages, configuration and functionalities of the u-blox 6 software releases and receivers are explained in this document.
- **Hardware Integration Manual:** This Manual provides hardware design instructions and information on how to set up production and final product tests.
- **Application Note:** document provides general design instructions and information that applies to all u-blox GPS receivers. See Section Design-in for a list of Application Notes related to your GPS receiver.

How to use this Manual

The LEA-6 / NEO-6 / MAX-6 Hardware Integration Manual provides the necessary information to successfully design in and configure these u-blox 6-based GPS receiver modules. For navigating this document please note the following:

This manual has a modular structure. It is not necessary to read it from the beginning to the end. To help in finding needed information, a brief section overview is provided below:

1. **Hardware description:** This chapter introduces the basics of function and architecture of the u-blox 6 modules.
2. **Design-in:** This chapter provides the Design-In information necessary for a successful design.
3. **Product handling:** This chapter defines packaging, handling, shipment, storage and soldering.
4. **Product testing:** This chapter provides information about testing of OEM receivers in production.
5. **Appendix:** The Appendix includes guidelines on how to successfully migrate to u-blox 6 designs, and useful information about the different antenna types available on the market and how to reduce interference in your GPS design.

The following symbols are used to highlight important information within the manual:



An index finger points out key information pertaining to module integration and performance.



A warning symbol indicates actions that could negatively impact or damage the module.

Questions

If you have any questions about u-blox 6 Hardware Integration, please:

- Read this manual carefully.
- Contact our information service on the homepage <http://www.u-blox.com>
- Read the questions and answers on our FAQ database on the homepage <http://www.u-blox.com>

Technical Support

Worldwide Web

Our website (www.u-blox.com) is a rich pool of information. Product information, technical documents and helpful FAQ can be accessed 24h a day.

By E-mail

If you have technical problems or cannot find the required information in the provided documents, contact the nearest of the Technical Support offices by email. Use our service pool email addresses rather than any personal email address of our staff. This makes sure that your request is processed as soon as possible. You will find the contact details at the end of the document.

Helpful Information when Contacting Technical Support

When contacting Technical Support please have the following information ready:

- Receiver type (e.g. LEA-6A-0-000), Datacode (e.g. 160200.0300.000) and firmware version (e.g. FW6.02)
- Receiver configuration
- Clear description of your question or the problem together with a u-center logfile
- A short description of the application
- Your complete contact details

Contents

Preface	4
Contents.....	6
1 Hardware description	10
1.1 Overview	10
1.2 Architecture.....	10
1.3 Power management	11
1.3.1 Connecting power	11
1.3.2 Operating modes	12
1.4 Antenna supply - V_ANT (LEA-6)	12
1.5 System functions	13
1.5.1 System monitoring	13
1.6 Interfaces.....	13
1.6.1 UART.....	13
1.6.2 USB (LEA-6/NEO-6)	13
1.6.3 Display Data Channel (DDC)	14
1.6.4 SPI (NEO-6, LEA-6R).....	16
1.7 I/O pins.....	19
1.7.1 RESET_N.....	19
1.7.2 EXTINT - External interrupt pin.....	19
1.7.3 AADET_N (LEA-6)	19
1.7.4 Configuration pins (LEA-6S/6A, NEO-6)	19
1.7.5 Second time pulse for LEA-6T.....	19
1.7.6 TX ready signal (FW 7.0x)	20
1.7.7 ANTOFF (NEO-6).....	20
1.7.8 Antenna supervision signals for LEA-6T-0	20
1.7.9 LEA-6R considerations	21
2 Design-in.....	22
2.1 Checklist.....	22
2.1.1 Design-in checklist.....	22
2.1.2 Design considerations.....	24
2.1.3 Automotive Dead Reckoning (ADR) solutions	25
2.2 LEA-6 design	26
2.2.1 LEA-6 passive antenna design.....	26
2.2.2 GLONASS HW design recommendations	27
2.2.3 LEA-6R design	31
2.2.1 Pin description for LEA-6 designs.....	35
2.3 NEO-6 design	36
2.3.1 Passive antenna design (NEO-6).....	36

2.3.2	Pin description for NEO-6 designs.....	38
2.4	MAX-6 design	39
2.4.1	MAX-6 passive antenna design.....	39
2.4.2	Pin description for MAX-6 designs.....	40
2.5	Layout	40
2.5.1	Footprint and paste mask	40
2.5.2	Placement	42
2.5.3	Antenna connection and grounding plane design	43
2.5.4	Antenna micro strip.....	44
2.6	Antenna and antenna supervisor	45
2.6.1	Passive antenna.....	46
2.6.2	Active antenna (LEA-6)	46
2.6.3	Active antenna bias power (LEA-6)	47
2.6.4	Active antenna supervisor (LEA-6).....	48
2.6.5	Active antenna (NEO-6 and MAX-6)	51
2.6.6	External active antenna supervisor using ANTOFF (NEO-6)	53
2.6.7	External active antenna supervisor using ANTON (MAX-6).....	54
2.6.8	External active antenna control (NEO-6)	55
2.6.9	External active antenna control (MAX-6)	56
2.6.10	GPS antenna placement for LEA-6R.....	56
3	Product handling	57
3.1	Packaging, shipping, storage and moisture preconditioning	57
3.2	Soldering	57
3.2.1	Soldering paste.....	57
3.2.2	Reflow soldering	57
3.2.3	Optical inspection.....	58
3.2.4	Cleaning.....	59
3.2.5	Repeated reflow soldering.....	59
3.2.6	Wave soldering.....	59
3.2.7	Hand soldering.....	59
3.2.8	Rework.....	59
3.2.9	Conformal coating	60
3.2.10	Casting.....	60
3.2.11	Grounding metal covers	60
3.2.12	Use of ultrasonic processes.....	60
3.3	EOS/ESD/EMI Precautions.....	60
3.3.1	Abbreviations	60
3.3.2	Electrostatic discharge (ESD)	60
3.3.3	ESD handling precautions.....	61
3.3.4	ESD protection measures.....	62
3.3.5	Electrical Overstress (EOS).....	62
3.3.6	EOS protection measures.....	62
3.3.7	Electromagnetic interference (EMI)	63

3.3.8	Applications with wireless modules LEON / LISA	64
3.3.9	Recommended parts	66
4	Product testing	67
4.1	u-blox in-series production test.....	67
4.2	Test parameters for OEM manufacturer	67
4.3	System sensitivity test	68
4.3.1	Guidelines for sensitivity tests	68
4.3.2	'Go/No go' tests for integrated devices.....	68
4.3.3	Testing LEA-6R designs.....	68
4.3.4	Testing NEO-6V designs	69
Appendix		70
A	Abbreviations	70
B	Migration to u-blox-6 receivers.....	70
B.1	Checklist for migration	70
B.2	Software migration.....	72
B.2.1	Software migration from ANTARIS 4 or u-blox 5 to a u-blox 6 GPS receiver	72
B.2.2	Software migration from 6.02 to 7.03.....	73
B.3	Hardware Migration	73
B.3.1	Hardware Migration: ANTARIS 4 → u-blox 6	73
B.3.2	Hardware Migration: u-blox 5 → u-blox 6	73
B.4	Migration of LEA modules	74
B.4.1	Migration from LEA-4 to LEA-6	74
B.4.2	Migration of LEA-4R designs to LEA-6R	75
B.4.3	Migration from LEA-5 to LEA-6	76
B.5	Migration of NEO modules	76
B.5.1	Migration from NEO-4S to NEO-6.....	76
B.5.2	Migration from NEO-5 to NEO-6	77
C	Interface Backgrounder	78
C.1	DDC Interface.....	78
C.1.1	Addresses, roles and modes	78
C.1.2	DDC troubleshooting	79
C.2	SPI Interface.....	80
C.2.1	SPI basics.....	80
D	DR calibration	83
D.1	Constraints	83
D.2	Initial calibration drive.....	83
Related documents.....		85
Revision history.....		86

Contact.....	87
--------------	----

1 Hardware description

1.1 Overview

The u-blox 6 leadless chip carrier (LCC) modules are standalone GPS receivers featuring the high performance u-blox 6 positioning engine. These compact, easy to integrate modules combine exceptional GPS performance with highly flexible power, design, and connectivity options. Their compact form factors and SMT pads allow fully automated assembly with standard pick & place and reflow-soldering equipment for cost-efficient, high-volume production enabling short time-to-market.

u-blox GPS modules are not designed for life saving or supporting devices or for aviation and should not be used in products that could in any way negatively impact the security or health of the user or third parties or that could cause damage to goods.

1.2 Architecture

u-blox 6 LCC modules consist of two functional parts - the RF and the Baseband sections. See Figure 1 for block diagrams of the modules.

The RF Front-End includes the input matching elements, the SAW bandpass filter, the u-blox 6 RF-IC (with integrated LNA) and the frequency source.

The Baseband section contains the u-blox 6 Baseband processor, the RTC crystal and additional elements such as the optional FLASH Memory for enhanced programmability and flexibility.

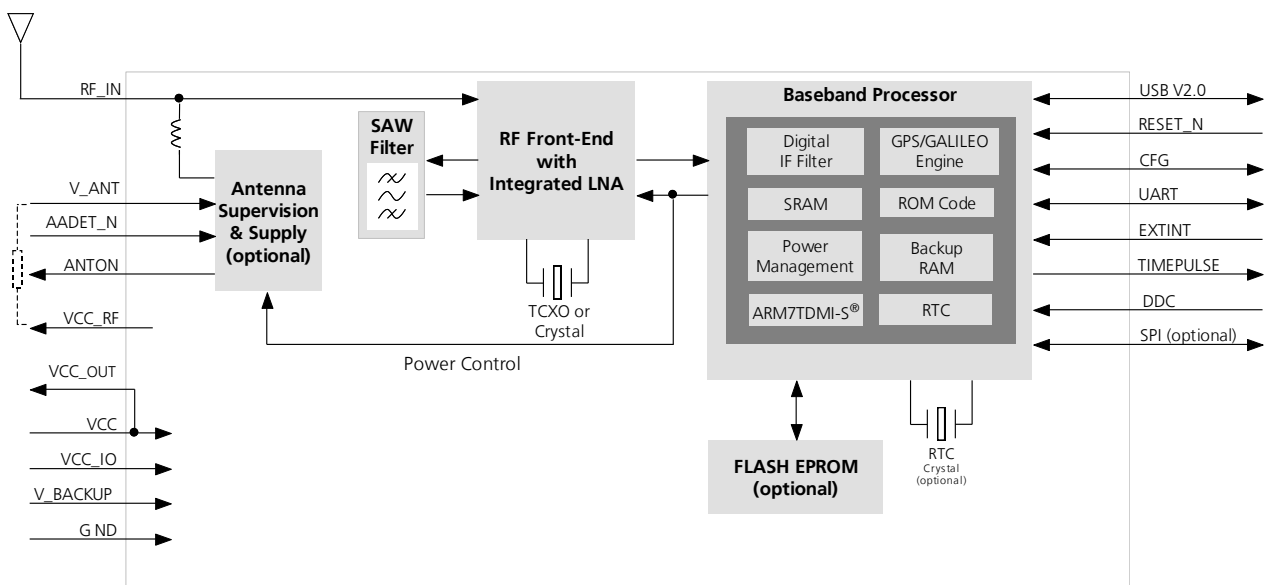


Figure 1: u-blox-6 block diagram

1.3 Power management

1.3.1 Connecting power

u-blox 6 receiver modules have three power supply pins: **VCC**, **V_BCKP** and **VDDUSB**.

(No VDDUSB for MAX-6)

1.3.1.1 VCC - main power

The main power supply is fed through the **VCC** pin. During operation, the current drawn by the u-blox 6 GPS module can vary by some orders of magnitude, especially, if low-power operation modes are enabled. It is important that the system power supply circuitry is able to support the peak power (see datasheet for specification) for a short time. In order to define a battery capacity for specific applications the sustained power figure shall be used.



When switching from backup mode to normal operation or at start-up u-blox 6 modules must charge the internal capacitors in the core domain. In certain situations this can result in a significant current draw. For low power applications using Power Save and backup modes it is important that the power supply or low ESR capacitors at the module input can deliver this current/charge.

1.3.1.2 V_BCKP - backup battery

In case of a power failure on pin **VCC**, the real-time clock and backup RAM are supplied through pin **V_BCKP**. This enables the u-blox 6 receiver to recover from a power failure with either a Hotstart or a Warmstart (depending on the duration of **VCC** outage) and to maintain the configuration settings saved in the backup RAM. If no backup battery is connected, the receiver performs a Coldstart at power up.



If no backup battery is available connect the V_BCKP pin to GND.

As long as **VCC** is supplied to the u-blox 6 receiver, the backup battery is disconnected from the RTC and the backup RAM in order to avoid unnecessary battery drain (see Figure 2). Power to RTC and BBR is supplied from **VCC** in this case.



Avoid high resistance on the on the V_BCKP line: During the switch from main supply to backup supply a short current adjustment peak can cause high voltage drop on the pin and possible malfunctions.

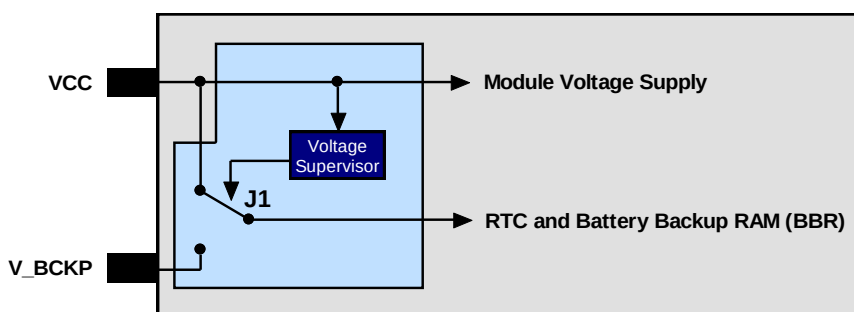


Figure 2: Backup Battery and Voltage

1.3.1.3 VDD_USB - USB interface power supply

On LEA-6 and NEO-6 **VDD_USB** supplies the USB interface. If the USB interface is not used, the **VDD_USB** pin must be connected to GND. For more information regarding the correct handling of **VDD_USB** see section 1.6.2.1

1.3.2 Operating modes

u-blox 6 modules with FW 7.0x or ROM6.02 have two continuous operating modes (Maximum Performance and Eco) and one intermittent operating mode (Power Save mode). Maximum Performance mode freely uses the acquisition engine, resulting in the best possible TTFF, while Eco mode optimizes the use of the acquisition engine to deliver lower current consumption. At medium to strong signals, there is almost no difference for acquisition and tracking performance in these modes.

1.3.2.1 Maximum Performance mode

In Maximum Performance mode, u-blox 6 receivers use the acquisition engine at full performance to search for all possible satellites until the Almanac is completely downloaded.

As a consequence, tracking current consumption level will be achieved when:

- A valid GPS position is fixed
- Almanac is entirely downloaded
- Ephemeris for all satellites in view are valid

1.3.2.2 Eco mode

In Eco mode, u-blox 6 receivers use the acquisition engine to search for new satellites **only when needed** for navigation:

- In cold starts, u-blox 6 searches for enough satellites to navigate and optimizes use of the acquisition engine to download their ephemeris.
- In non-cold starts, u-blox 6 focuses on searching for visible satellites whose orbits are known from the Almanac.

In Eco mode, the u-blox 6 acquisition engine limits use of its searching resources to minimize power consumption. As a consequence the time to find some satellites at weakest signal level might be slightly increased in comparison to the Maximum Performance mode.

u-blox 6 deactivates the acquisition engine as soon as a position is fixed and a sufficient number (at least 4) of satellites are being tracked. The tracking engine continues to search and track new satellites without orbit information.

1.3.2.3 Power Save mode

u-blox 6 receivers include a Power Save Mode. Its operation is called cyclic tracking and allows reducing the average power consumption significantly. The Power Save Mode can be configured for different update periods. u-blox recommends an update period of 1s for best GPS performance. For more information, see the u-blox 6 Receiver Description including Protocol Specification [3]



Dead Reckoning, PPP and Precision Timing features should not be used together with Power Save Mode.

1.4 Antenna supply - V_ANT (LEA-6)

LEA-6 modules support active antenna supply and supervision use the pin **V_ANT** to supply the active antenna. Use a 10 Ω resistor in front of **V_ANT**. For more information about antenna and antenna supervisor see chapter 2.6.



If not used, connect the V_ANT pin to GND.

1.5 System functions

1.5.1 System monitoring

The u-blox 6 receiver modules provide system monitoring functions that allow the operation of the embedded processor and associated peripherals to be supervised. These System Monitoring functions are output as part of the UBX protocol, class 'MON'.

Please refer to the u-blox 6 Receiver Description including Protocol Specification [3]. For more information on UBX messages, serial interfaces for design analysis and individual system monitoring functions.

1.6 Interfaces

1.6.1 UART

u-blox 6 modules include a Universal Asynchronous Receiver Transmitter (UART) serial interface. **RxD1/TxD1** supports data rates from 4.8 to 115.2 kBit/s. The signal output and input levels are 0 V to VCC. An interface based on RS232 standard levels (+/- 12 V) can be realized using level shifters such as Maxim MAX3232. Hardware handshake signals and synchronous operation are not supported.

For more information see the LEA-6 Data Sheet [1], NEO-6 Data Sheet [2] or MAX-6 Data Sheet [8]

1.6.2 USB (LEA-6/NEO-6)

The u-blox 6 Universal Serial Bus (USB) interface supports the full-speed data rate of 12 Mbit/s.

1.6.2.1 USB external components

The USB interface requires some external components in order to implement the physical characteristics required by the USB 2.0 specification. These external components are shown in Figure 3 and listed in Table 1.

In order to comply with USB specifications, VBUS must be connected through a LDO (U1) to pin **VDD_USB** of the module.

If the USB device is **self-powered** it is possible that the power supply (VCC) is shut down and the Baseband-IC core is not powered. Since VBUS is still available, it still would be signaled to the USB host that the device is present and ready to communicate. This is not desired and thus the LDO (U1) should be disabled using the enable signal (EN) of the VCC-LDO or the output of a voltage supervisor. Depending on the characteristics of the LDO (U1) it is recommended to add a pull-down resistor (R11) at its output to ensure **VDD_USB** is not floating if LDO (U1) is disabled or the USB cable is not connected i.e. VBUS is not supplied.

If the device is **bus-powered**, LDO (U1) does not need an enable control.

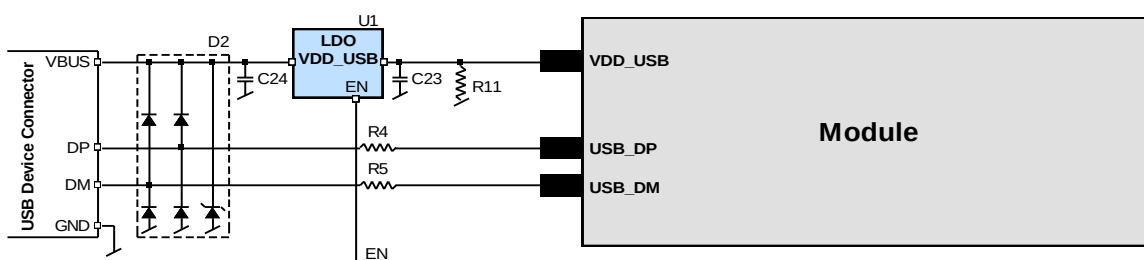


Figure 3: USB Interface

Name	Component	Function	Comments
U1	LDO	Regulates VBUS (4.4 ... 5.25 V) down to a voltage of 3.3 V.	Almost no current requirement (~1 mA) if the GPS receiver is operated as a USB self-powered device, but if bus-powered LDO (U1) must be able to deliver the maximum current of ~70 mA. A low-cost DC/DC converter such as LTC3410 from Linear Technology may be used as an alternative.
C23, C24	Capacitors		Required according to the specification of LDO U1
D2	Protection diodes	Protect circuit from overvoltage / ESD when connecting.	Use low capacitance ESD protection such as ST Microelectronics USBLC6-2.
R4, R5	Serial termination resistors	Establish a full-speed driver impedance of 28...44 Ω	A value of 22 Ω is recommended.
R11	Resistor		10 k Ω is recommended for USB self-powered setup. For bus-powered setup R11 can be ignored.

Table 1: Summary of USB external components

1.6.3 Display Data Channel (DDC)

An I²C compatible Display Data Channel (DDC) interface is available with LEA-6, NEO-6 and MAX-6 modules for serial communication. For more information about DDC implementation refer to the u-blox 6 Receiver Description including Protocol Specification [3]. Background information about the DDC interface is available in Appendix C.1.



u-blox 6 GPS receivers normally run in I²C slave mode. Master Mode is only supported when external EEPROM is used to store configuration. No other nodes may be connected to the bus. In this case, the receiver attempts to establish presence of such a non-volatile memory component by writing and reading from a specific location.



TX ready indicator (data ready) for FW 7.0x see 1.7.6



The u-blox 6 DDC interface supports serial communication with u-blox wireless modules. See the specification of the applicable wireless module to confirm compatibility.



With u-blox 6, when reading the DDC internal register at address 0xFF (messages transmit buffer), the master must not set the reading address before every byte accessed as this could cause a faulty behavior. Since after every byte being read from register 0xFF the internal address counter is incremented by one saturating at 0xFF, subsequent reads can be performed continuously.

Pins SDA2 and SCL2 have internal 13 k Ω pull-ups. If capacitive bus load is very large, additional external pull-ups may be needed in order to reduce the pull-up resistance.

Table 2 lists the maximum total pull-up resistor values for the DDC interface. For small loads, e.g. if just connecting to an external EEPROM, these built-in pull-ups are sufficient.

Load Capacitance	Pull-Up Resistor Value R20, R21
50 pF	N/A
100 pF	18 k Ω
250 pF	4.7 k Ω

Table 2: Pull-up resistor values for DDC interface

1.6.3.1 Communicating to an I²C EEPROM with the GPS receiver as I²C master

Serial I²C memory can be connected to the DDC interface. This can be used to save configuration permanently. It will automatically be recognized by firmware. The memory address must be set to 0b10100000 (0xA0) and the size fixed to 4 kB.

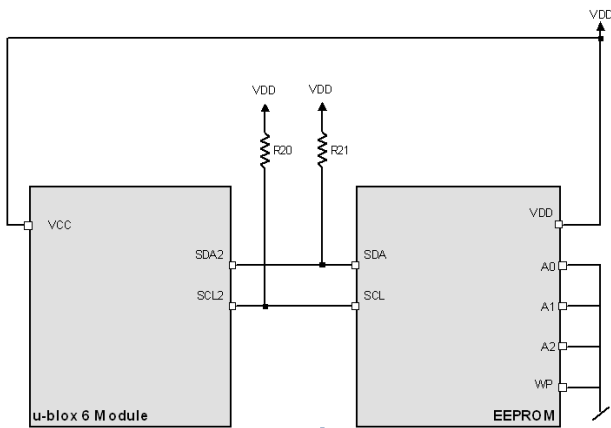


Figure 4: Connecting external serial I²C memory used by the GPS receiver (see EEPROM data sheet for exact pin orientation)

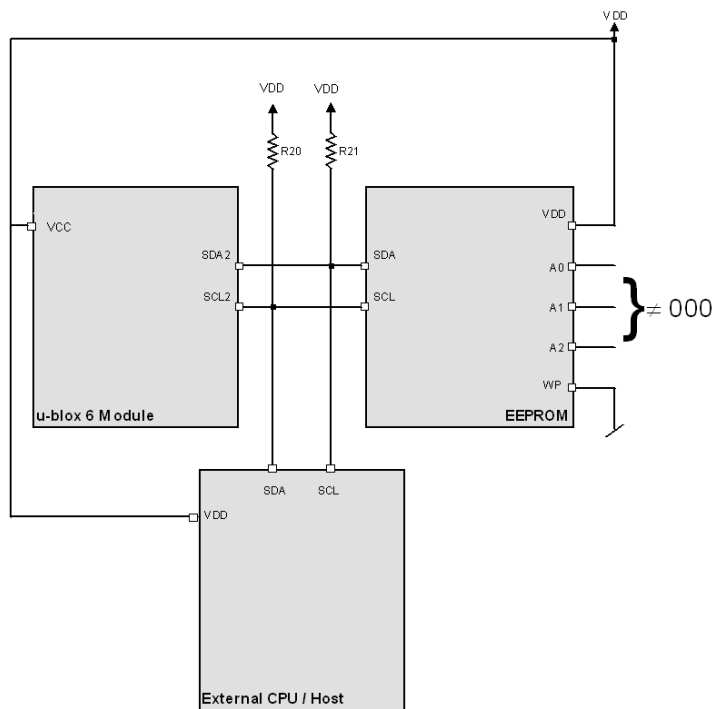


Figure 5: Connecting external serial I²C memory used by external host (see data sheet for exact pin orientation)

Note that the case shown on Figure 4 is different than the case when EEPROM is present but used by external host / CPU as indicated on Figure 5. This is allowed but precaution is required to ensure that the GPS receiver does not detect the EEPROM device, which would effectively configure the GPS receiver to be MASTER on the bus causing collision with the external host.

To ensure that the EEPROM device (connected to the bus and used by the host) is not detected by the GPS receiver it is important to set the EEPROM's address to a value different than 0xA0. This way EEPROM remains free to be used for other purposes and the GPS receiver will assume the SLAVE mode.



Ensure that at the start up the host allows enough time for the receiver to communicate over the bus to establish presence of the EEPROM. It is only when this interrogation is complete that the host can exercise full control over the bus (MASTER mode).



Also note that the FLASH based modules do not attempt to store any information in the external EEPROM and as such do not attempt to communicate to the external EEPROM. The ROM based receivers always interrogate external EEPROM at the start-up. The interrogation process is guaranteed to complete within 250ms upon start up. This is the time the external host has to give to the ROM based GPS receiver to complete the EEPROM interrogation.



Although the FLASH based modules do not attempt to detect the EEPROM at the start up, an attempt to communicate to the GPS receiver via DDC before 250 msec expires is not advised because the GPS receiver is unable to respond due to other start up activities.

Following I2C serial EEPROM are supported:

Manufacturer	Order No.
ST	M24C32-R
Microchip	24AA32A
Catalyst	CAT24C32
Samsung	S524AB0X91

Table 3: Recommend parts list for I2C Serial EEPROM memory

1.6.4 SPI (NEO-6, LEA-6R)

A Serial Peripheral Interface (SPI) is available with u-blox 6 NEO modules. The SPI allows for the connection of external devices with a serial interface, e.g. FLASH memories or A/D converters, or to interface to a host CPU.

LEA-6R includes a Serial Peripheral Interface (SPI) for connecting external sensors. The interface can be operated in SPI master mode only. Two chip select signals are available to select external slaves. See 2.2.3.1



TX ready indicator (data ready) for LEA-6H (FW 7.0x) see 1.7.6

Background information about the SPI interface is available in Appendix C.2.

1.6.4.1 Connecting SPI FLASH memory (NEO-6 modules)

SPI FLASH memory can be connected to the SPI interface to save Assist Now Offline data and/or receiver configuration. It will automatically be recognized by firmware when connected to SS_N.

Figure 6 shows how external memory can be connected. Minimum SPI FLASH memory size is 1 Mbit.

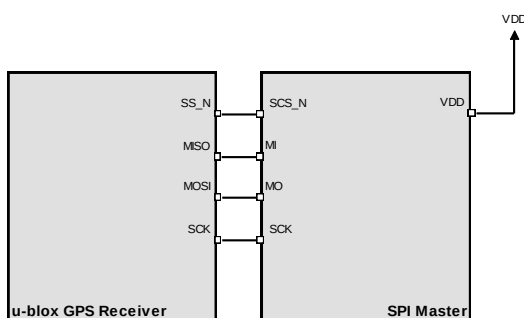


Figure 6: Connecting external SPI Memory to u-blox GPS receivers

Following SPI serial Flash are supported:

Manufacturer	Order No.
Winbond	W25X10A
Winbond	W25X20A
AMIC	A25L010
AMIC	A25L020

Table 4: Supported SPI FLASH memory devices



Only use serial FLASH types listed in Table 4. For new designs confirm if the listed type is still available. It is not possible to use other serial FLASH types than those listed in Table 4 with u-blox 6 receivers.

1.6.4.2 SPI communication (connecting to an SPI master) NEO-6

Figure 7 shows how to connect a u-blox GPS receiver to a host/master. The signal on the pins must meet the conditions specified in the Data Sheet.

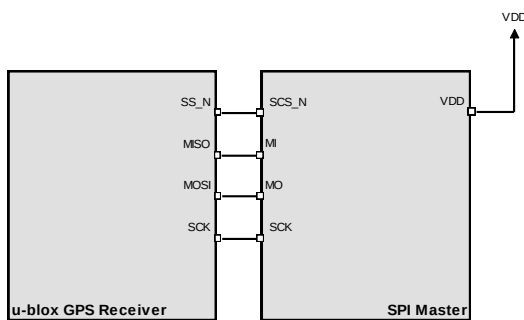


Figure 7: Connecting to SPI Master



For those u-blox 6 modules supporting SPI the SPI MOSI, MISO and SCK pins share a configuration function at start up. To secure correct receiver operation make sure that the SS_N pin is high at start up. Afterwards the SPI function will not affect the configuration pins.

1.6.4.3 Pin configuration with module as one of several slaves

The buffers enabled by the CS_N signal make sure that the GPS receiver starts up with a known defined configuration, since the SPI pins (MOSI, MISO and SCK) are at start up also configuration pins.

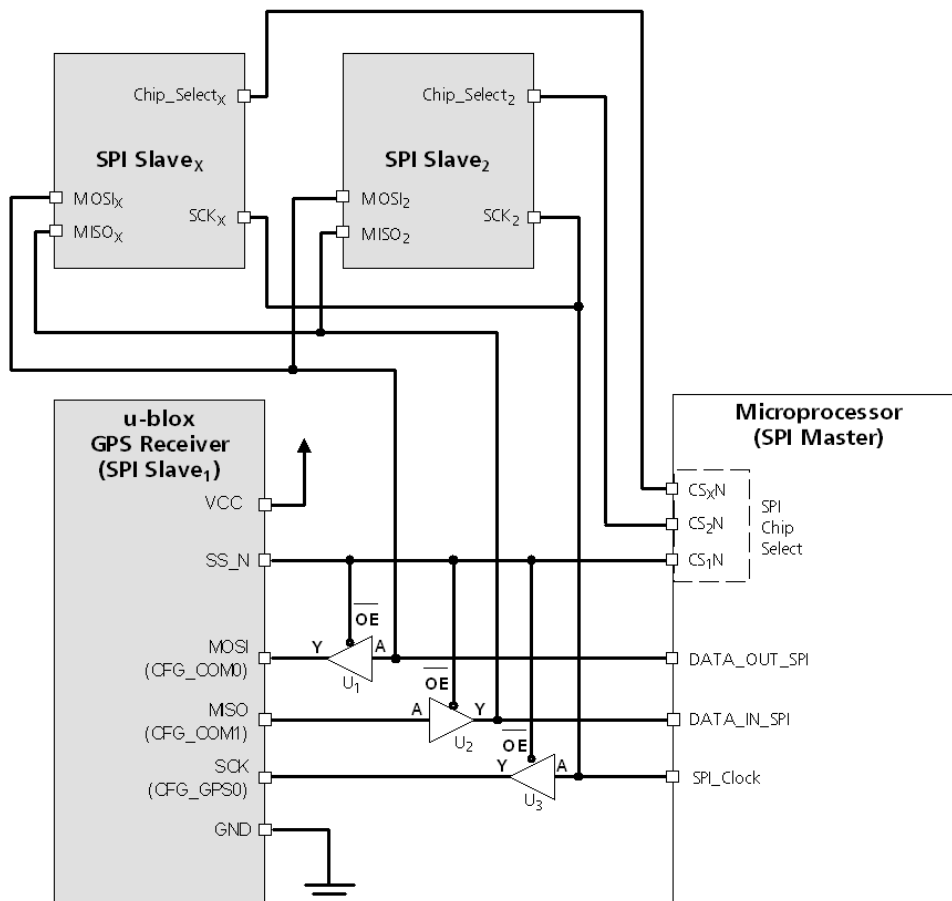


Figure 8: Diagram of SPI Pin Configuration

Component	Description	Model	Supplier
U ₁ – U ₃	Buffer	NC7SZ125	Fairchild

Table 5: Recommended components for SPI pin configuration



Use same power voltage to supply U1 – U3 and VCC.

1.7 I/O pins

1.7.1 RESET_N

LEA-6 modules include a **RESET_N** pin. Driving **RESET_N** low activates a hardware reset of the system. **RESET_N** is only an input and will not reset external circuitry.

Use components with open drain output (i.e. with buffer or voltage supervisor).

There is an internal pull up resistor of 3.3 k Ω to VCC inside the module that requires that the reset circuitry can deliver enough current (e.g. 1 mA).

Do not drive **RESET_N** high.

NEO-6 and MAX-6 modules do not include a **RESET_N** pin. However, this functionality can be implemented for these modules by connecting the NEO-6 and MAX-6 pin 8 to pin 9 with a 3.3 k Ω resistor, instead of connecting them directly. Pin 8 (NEO-6) or pin 9 (MAX-6) can then be used as a **RESET_N** input with the same characteristics as the reset pin on LEA-6 modules.



Use caution when implementing RESET_N on NEO-6 and MAX-6 modules since forward compatibility is not guaranteed.

1.7.2 EXTINT - External interrupt pin

EXTINT0 is an external interrupt pin with fixed input voltage thresholds with respect to VCC (see the data sheet for more information). It can be used for the time mark function on LEA-6T or for wake-up functions in Power Save Mode on all u-blox 6 LCC modules. Leave open if unused.

1.7.3 AADET_N (LEA-6)

AADET_N is an input pin and is used to report whether an external circuit has detected an external antenna or not. Low means the antenna has been detected. High means no external antenna has been detected.

See chapter 2.6.4 for an implementation example.

1.7.4 Configuration pins (LEA-6S/6A, NEO-6)

ROM-based modules provide up to 3 pins (**CFG_COM0**, **CFG_COM1**, and **CFG_GPS0**) for boot-time configuration. These become effective immediately after start-up. Once the module has started, the configuration settings can be modified with UBX configuration messages. The modified settings remain effective until power-down or reset. If these settings have been stored in battery-backup RAM, then the modified configuration will be retained, as long as the backup battery supply is not interrupted.

The module data sheets indicate the meaning of the configuration pins when they are high (1) or low (0). In fact no configuration pins need to be pulled high. All have internal pull ups and therefore default to the high (1) state when left open or connected to a high impedance output. They should be left open unless there is a need to pull them low to alter the initial configuration.

Some configuration pins are shared with other functions. During start-up, the module reads the state of the configuration pins. Afterwards the other functions can be used.

The configuration pins of u-blox 6 use an internal pull-up resistor, which determines the default setting.



For more information about settings and messages see the module data sheet.



MAX-6 doesn't have pins for boot-time configuration

1.7.5 Second time pulse for LEA-6T

LEA-6T includes a second time pulse pin (TIMEPULSE2). For more information and configuration see the LEA-6 Data Sheet [1] and also the u-blox 6 Receiver Description including Protocol Specification [3].

1.7.6 TX ready signal (FW 7.0x)

The TX ready signal indicates that the receiver has data to transmit. A listener can wait on the TX ready signal instead of polling the DDC or SPI interfaces. The UBX-CFG-PRT message lets you configure the polarity and the number of bytes in the buffer before the TX ready signal goes active. The TX ready signal can be mapped to GPIO 05 (TXD1). The TX ready pin is disabled by default.



u-blox wireless modules (LEON and LISA) configure and enable the TX ready functionality automatically.

For more information on configuration and remap of this pin see the LEA-6 Data Sheet [1] and see also the u-blox 6 Receiver Description including Protocol Specification [3].

1.7.7 ANT OFF (NEO-6)

The ANT OFF signal can be mapped to GPIO22 (Pin 17). The ANT OFF signal is disabled by default.



To configure the ANT OFF function refer to the u-blox 6 Receiver Description including Protocol Specification [3].



Use caution when implementing ANT OFF configuration since forward compatibility is not guaranteed

1.7.8 Antenna supervision signals for LEA-6T-0

With LEA-6T-0, the antenna supervisor GPIOs are numbered differently than the other LEA-6 modules and are wired to specific PIOs:

- ANT OFF is internally mapped to GPIO13
- ANT SHORT is internally mapped to GPIO17
- ADET_N (Active Antenna Detect) is mapped to GPIO8 (Pin 20)

If the unit is reverted to the default configuration, there is no antenna supply.

The CFG-ANT command sets the PIOs and enables Power Control, Short Circuit Detection, Power Down on Short and Short Circuit Recovery.

To store the settings permanently send the UBX-CFG-CFG command with the option 'save current parameters' to BBR AND SPI Flash (!)

See also schematic of open circuit detection Figure 46 .



To configure this function refer to the u-blox 6 Receiver Description including Protocol Specification [3].

1.7.9 LEA-6R considerations

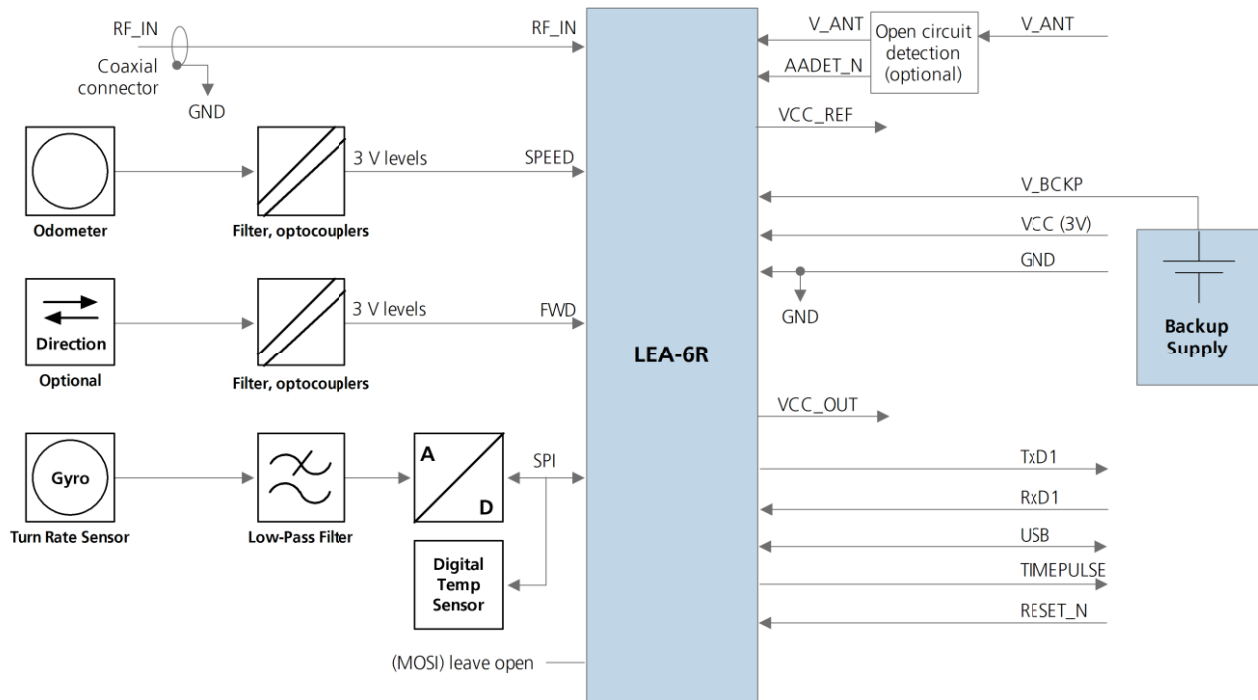


Figure 9: Block schematic of complete LEA-6R design

LEA-6R includes the following special pins: **SPI_MOSI**, **SPI_MISO**, **SPI_SCS2_N**, **FWD**, **SPI_SCS1_N**, **SPI_SCK**, and **SPEED**.

Pin	Signal name	Direction	Usage
27	SPEED	Input	Odometer Speedpulses
23	SCK	Output	SPI clock
22	SPI_SCS1_N	Output	Chip Select signal for ADC/turn rate sensor
21	FWD	Input	Direction indication (1 = forward)
9	SPI_SCS2_N	Output	Chip Select signal for temperature sensor
2	MISO	Input	Serial data (<u>Master In</u> / Slave Out)
1	MOSI	Output	Serial data (<u>Master Out</u> / Slave In), leave open

Table 6: LEA-6R special pins

2 Design-in



For migrating existing ANTARIS®4 product designs to u-blox 6 please refer to Appendix B.

In order to obtain good performance with a GPS receiver module, there are a number of points that require careful attention during the design-in. These include:

- Power Supply: Good performance requires a clean and stable power supply.
- Interfaces: Ensure correct wiring, rate and message setup on the module and your host system.
- Antenna interface: For optimal performance seek short routing, matched impedance and no stubs.

2.1 Checklist

Good performance requires a clean and stable power supply with minimal ripple. Care needs to be exercised in selecting a strategy to achieve this. Series resistance in the Vcc supply line can negatively impact performance. For better performance, use an LDO to provide a clean supply at Vcc and consider the following:

- Wide power lines or even power planes are preferred.
- Place LDO near the module.
- Avoid resistive components in the power line (e.g. narrow power lines, coils, resistors, etc.).
- Placing a filter or other source of resistance at Vcc can create significantly longer acquisition times.

2.1.1 Design-in checklist

Designing-in a u-blox 6 module is easy, especially when based on a u-blox reference design. Nonetheless, it pays to do a quick sanity check of the design. This section lists the most important items for a simple design check. The Design-In Checklist also helps to avoid an unnecessary respin of the PCB and helps to achieve the best possible performance.



Follow the design-in checklist when developing any u-blox 6 GPS applications. This can significantly reduce development time and costs.

Have you chosen the optimal module?

u-blox 6 modules have been intentionally designed to allow GPS receivers to be optimally tailored to specific applications. Changing between the different variants is easy.

- ☐ Do you need TCXO performance – Then choose an **H¹**, **S²**, **Q³** or **G⁴** variant.
- ☐ Do you want to be able to upgrade the firmware? Then you will have to use a Programmable receiver module: choose an **H¹** variant.
- ☐ Do you need USB? All LEA-6 and NEO-6 modules support USB.
- ☐ Do you need Dead Reckoning – Then choose a LEA-6**R** or NEO-6**V** (see section 2.1.3)
- ☐ Do you need Precise Point Positioning – Then choose a NEO-6**P**
- ☐ Do you need Precision Timing – Then choose a LEA-6**T** or NEO-6**T**.
- ☐ Do you need onboard Antenna Supervisor circuitry - Then choose the LEA form factor.
- ☐ Do you need onboard Antenna control - Then choose the MAX form factor.
- ☐ Do you need smallest size and forward compatibility- Then choose the MAX form factor.
- ☐ Do you need low power - Then choose 1.8V **6G** module variant.

¹ LEA-6H

² LEA-6S

³ NEO-6Q / MAX-6Q

⁴ NEO-6G / MAX-6G

Check Power Supply Requirements and Schematic:

- ☐ Is the power supply within the specified range (see data sheet)?
- ☐ Is the voltage **VDDUSB** within the specified range?
- ☐ Compare the peak current consumption of your u-blox 6 module (**~70 mA**) with the specification of the power supply.
- ☐ GPS receivers require a stable power supply, avoid ripple on **VCC** (<50 mVpp)
- ☐ For low power applications using Power Save and backup modes, ensure that the power supply or low ESR capacitors at the module input can deliver the required current/charge for switching from backup mode to normal operation. In certain situations charging the internal capacitors in the core domain can result in a significant instantaneous current draw.

Backup Battery

- ☐ For achieving a minimal Time To First Fix (TTFF) in Hotstart or a Warmstart, connect a backup battery to **V_BCKP**.
- ☐ Time information is a requirement for AssistNow Offline, AssistNow Autonomous and when in Power Save Mode with update period longer than 10 s.

Antenna

- ☐ The total noise figure should be well below 3 dB.
- ☐ If a patch antenna is the preferred antenna, choose a patch of at least 15x15x4mm. For smaller antennas an LNA with a noise figure <2 dB is recommended. To optimize TTFF make use of u-blox' free A-GPS services AssistNow Online and AssistNow Offline.
- ☐ Make sure the antenna is not placed close to noisy parts of the circuitry. (e.g. micro-controller, display, etc.)
- ☐ For active antennas add a 10 Ω resistor in front of **V_ANT**⁵ input for short circuit protection or use the antenna supervisor circuitry.
- ☐ To optimize performance in environments with out-band jamming sources, use an additional SAW filter.



For information on ESD protection for patch antennas and removable antennas, see section 3.3.4 and if you use GPS for design in combination with GSM or other radio then check sections 3.3.6 to 3.3.8.



For more information dealing with interference issues see the GPS Antenna Application Note [4].

Schematic

- ☐ If required, does your schematic allow using different module variants?
- ☐ Don't drive **RESET_N** high!
- ☐ Don't drive configuration pins high, they already have internal pull-ups.
- ☐ Plan the use of 2nd interface (Testpoints on UART, DDC or USB) for firmware updates or as a service connector.

Layout optimizations (Section 2.5)

- ☐ Is the GPS module placed according to the recommendation in Section 2.5.2?
- ☐ Has the Grounding concept been followed (see Section 2.5.3)?
- ☐ Has the micro strip been kept as short as possible?
- ☐ Add a ground plane underneath the GPS module to reduce interference.
- ☐ For improved shielding, add as many vias as possible around the micro strip, around the serial communication lines, underneath the GPS module etc.
- ☐ Have appropriate EOS/ESD/EMI protection measures been included (see Section 3.3)? This is especially important for designs including 2G, 3G modules.

⁵ Only available with LEA-6 modules

Calculation of the micro strip (Section 2.5.4)

- ❑ The micro strip must be 50 Ω and be routed in a section of the PCB where minimal interference from noise sources can be expected.
- ❑ In case of a multi-layer PCB, use the thickness of the dielectric between the signal and the 1st GND layer (typically the 2nd layer) for the micro strip calculation.
- ❑ If the distance between the micro strip and the adjacent GND area (on the same layer) does not exceed 5 times the track width of the micro strip, use the "Coplanar Waveguide" model in AppCad to calculate the micro strip and not the "micro strip" model.

2.1.2 Design considerations

For a minimal design with a u-blox 6 GPS module the following functions and pins need to be considered:

- Connect the Power supply to **VCC**.
- **VDDUSB**: Connect the USB power supply to a LDO before feeding it to **VDDUSB** and **VCC**. Or connect to GND if USB is not used.
- Assure a optimal ground connection to all ground pins of the module
- Connect the antenna to **RF_IN** over a matching 50 Ω micro strip and define the antenna supply (**V_ANT**)⁶ for active antennas (internal or external power supply)
- Choose the required serial communication interface (UART, USB, SPI or DDC) and connect the appropriate pins to your application
- If you need Hot- or Warmstart in your application, connect a backup battery to **V_BCKP**
- Decide whether **TIMEPULSE** or **RESET_N**⁶ options are required in your application and connect the appropriate pins on your module

⁶ Only available with LEA-6 modules, but see section 1.7.1 for NEO-6 modules.

2.1.3 Automotive Dead Reckoning (ADR) solutions

u-blox' ADR supports different sensor inputs. The classical setup, called "Gyroscope plus Wheel Tick" (**GWT**), consists of a gyroscope providing the heading information and wheel tick providing the speed information.

Alternatively, sensor information from left and right wheels (front or rear) or all wheels are used differentially to deduce heading, called "Differential Wheel Tick" (**DWT**). This results in slightly lower performance compared to GWT, but has the big advantage of saving the cost of a gyroscope.

2.1.3.1 Software sensor interface

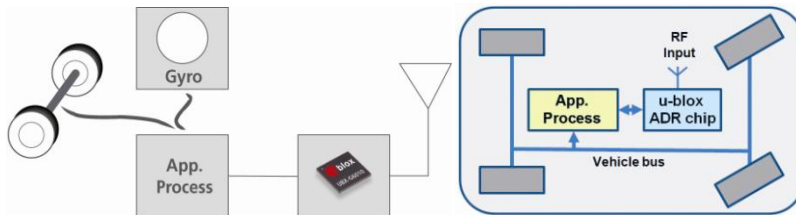


Figure 10: Software sensor interface

The industry proven u-blox ADR solution is highly flexible. The application processor can support a vast array of sensors, and must only convert the sensor data into UBX messages and pass them to the GPS receiver via a standard serial interface (USB, SPI, UART, DDC). This makes the u-blox ADR solution very portable between various vehicle platforms and reduces development effort and time-to-market. u-blox ADR is completely self-calibrating, and requires only pre-configuration to the specific vehicle platform.

u-blox' ADR with software sensor interface is available as NEO-6V module. These components are ideal for factory installed navigation since they use sensor data (wheel tick and gyroscope data) taken directly from the CAN bus.

2.1.3.2 Hardware sensor interface

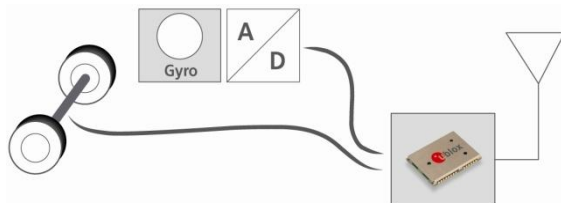


Figure 11: Hardware sensor interface

The standard quality grade LEA-6R module is a dedicated ADR solution (GWT only) for aftermarket installations with no access to the vehicle bus and no application processor for sensor data processing. Sensors are connected directly to the module: gyroscopes via SPI and ADC and the speed pulse information from the tachometer.

2.2 LEA-6 design

2.2.1 LEA-6 passive antenna design

This is a minimal setup for a PVT GPS receiver with a LEA-6 module.

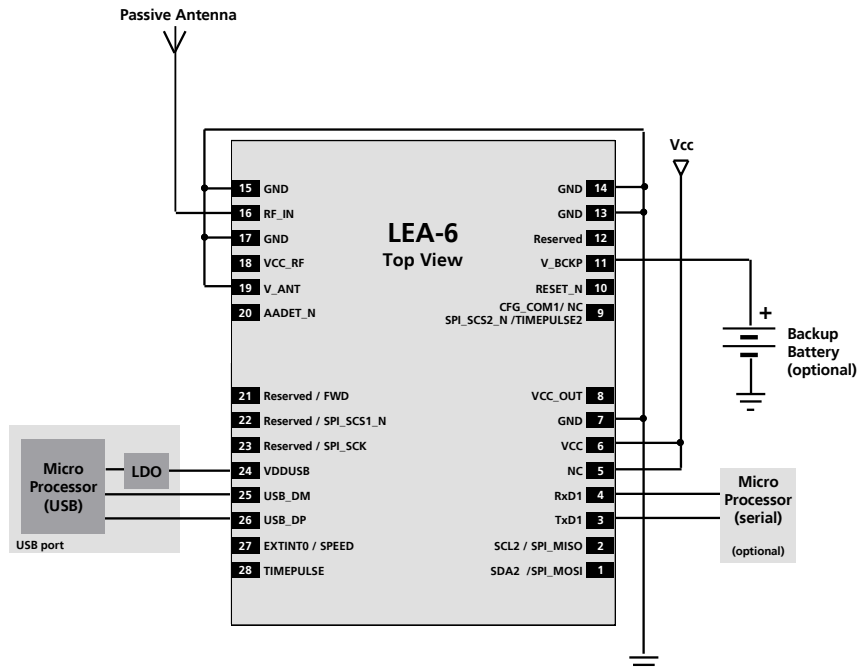


Figure 12: LEA-6 passive antenna design with USB port

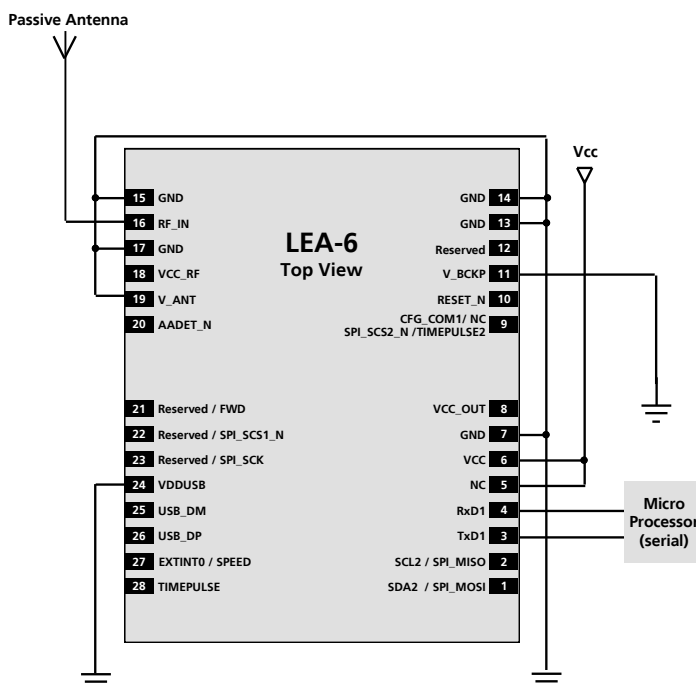


Figure 13: LEA-6 passive antenna design with no USB port or backup battery



For best performance with passive antenna designs use an external LNA to increase the sensitivity up to 2 dB. See Figure 12 and Figure 15.

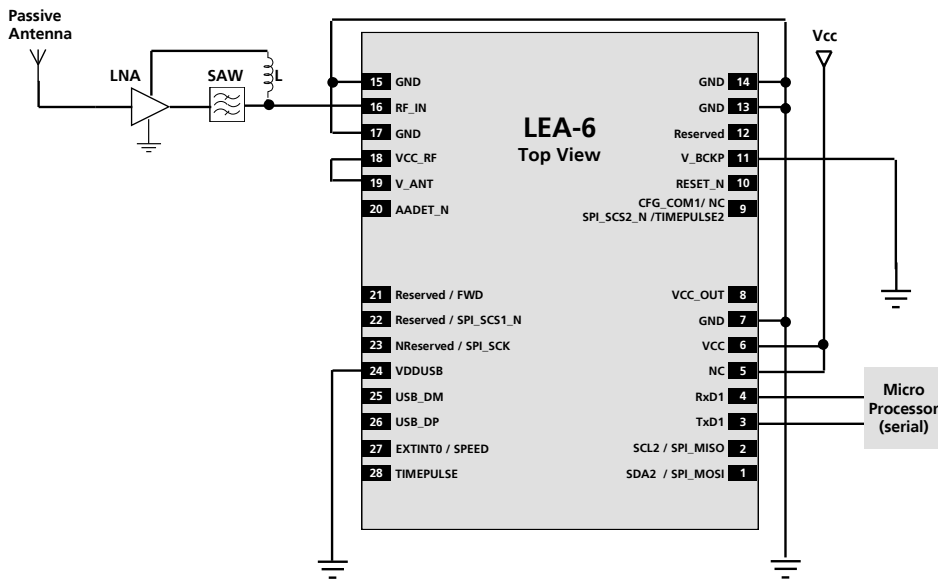


Figure 14: LEA-6 passive antenna design for best performance (with external LNA and SAW)

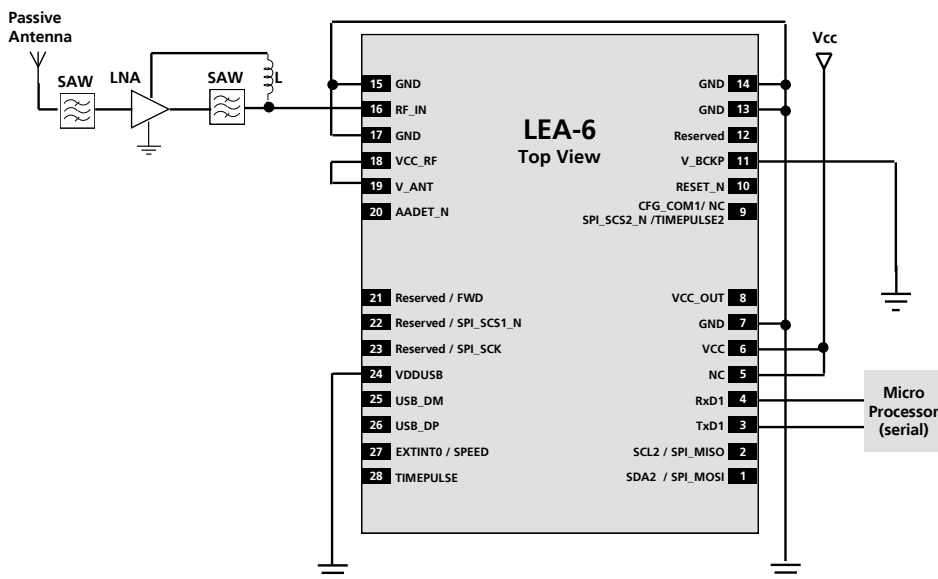


Figure 15: LEA-6 passive antenna design for best performance and increased immunity to jammers such as GSM



For information on increasing immunity to jammers such as GSM see section 3.3.8.

2.2.2 GLONASS HW design recommendations

u-blox 6 designs for GLONASS require:

- u-blox 6 Module with **Flash** (LEA-6H GLONASS ready module)
- **Wide RF path** (antenna, external SAW) to allow GLONASS & GPS signals to pass (see Figure 16)

2.2.2.1 u-blox 6 GLONASS module

u-blox will provide a GLONASS / GPS Flash firmware image. For this reason it is absolutely necessary to choose the LEA-6H GLONASS ready module which can be updated when the GLONASS / GPS firmware is available. A

GLONASS module which will be pin-compatible to the GLONASS ready LEA-6H module when the GLONASS / GPS firmware is available.

2.2.2.2 Wide RF path

As seen in Figure 16, the GLONASS / GPS satellite signals are not at the same frequency. For this reason the RF path, LNA, filter, and antenna must be modified accordingly to let both signals pass.

2.2.2.3 Filter

- Use a GPS & GLONASS SAW filter (see Figure 16) that lets both GPS and GLONASS signals pass (see the recommended parts list in section 3.3.9)
- In case an active antenna is used, make sure the filter (if any) inside is wide enough.

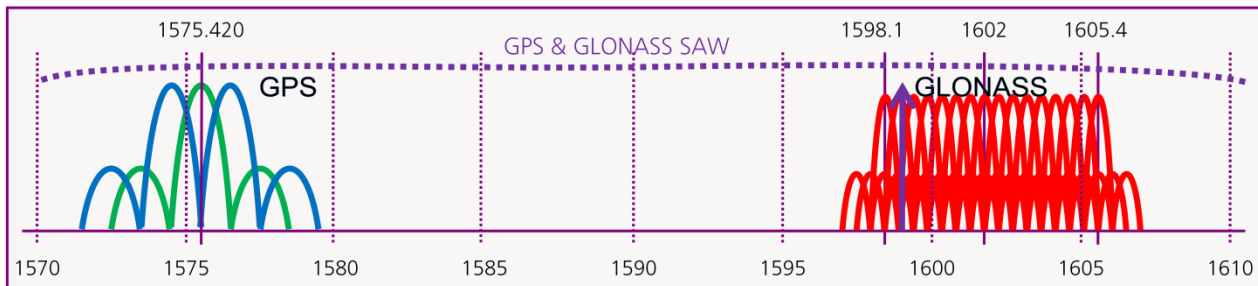


Figure 16: GPS & GLONASS SAW filter

2.2.2.4 Antenna

2.2.2.5 Active antenna

Usually an active GPS antenna includes a GPS band pass filter that filters out the GLONASS signal (see Figure 16). For this reason make sure that the filter in the active antenna is wide enough.

In combined GPS & GLONASS antennas, the antenna is tuned for receiving both signals and the filter has a larger bandwidth to provide optimal GPS & GLONASS signal reception (see Figure 16).



Use a good performance GPS & GLONASS active antenna (for recommended components see section 3.3.9.1).

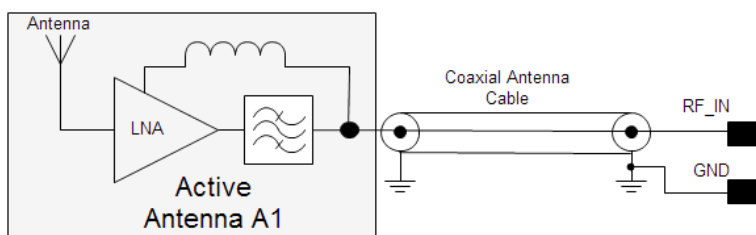


Figure 17: GPS & GLONASS active antenna

2.2.2.6 Passive Antenna

The bandwidth of a ceramic patch antenna narrows with size (see Table 7).

size	Typical bandwidth
36*36*4 mm	40 MHz
25*25*4 mm	20 MHz
18*18*4 mm	10 MHz
15*15*4 mm	8 MHz
12*12*4 mm	7 MHz
10*10*4 mm	5 MHz

Table 7: Typical bandwidths for GPS patch antennas

Figure 18 shows a 12*12*4 mm patch antenna with 20*20 mm ground plane, tuned to GPS. This patch bandwidth is so narrow that it cannot be simultaneously matched to GPS and GLONASS.

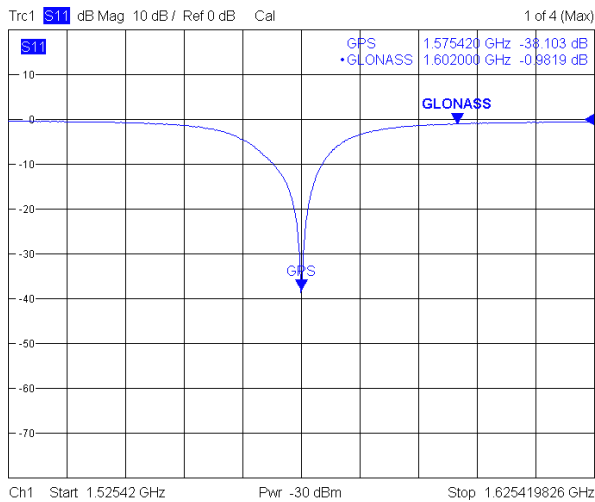


Figure 18: 12*12*4 patch antenna on 20*20 mm GND plane

Figure 19 show a 25*25*4 mm patch antenna with 60*60 mm ground plane. Due to the larger bandwidth, it can be matched to GPS and GLONASS.

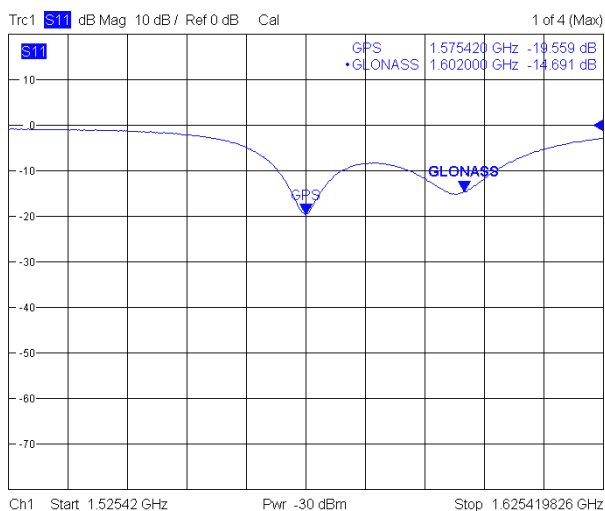


Figure 19: 25*25*4 mm patch antenna on 60*60 mm GND plane

Figure 20 show a 36*36*4 mm patch antenna. Due to the large bandwidth, the antenna is also tolerant with changes in the ground plane.

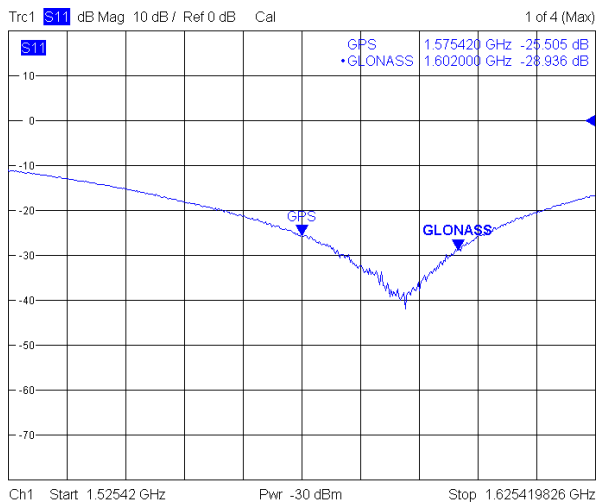


Figure 20 36*36*4 mm patch antenna



Use at least a 25*25*4 mm patch antenna, (a 36*36*4 mm patch antenna is better) and tune it so that GPS & GLONASS signals are received.

2.2.2.7 Module designs

For GPS & GLONASS designs chose the LEA-6H GLONASS ready module, which has a wide RF path and includes an internal Flash.

2.2.2.8 Module design with active antenna

Figure 21 shows a GPS & GLONASS active antenna design with the LEA-6H GLONASS ready module.

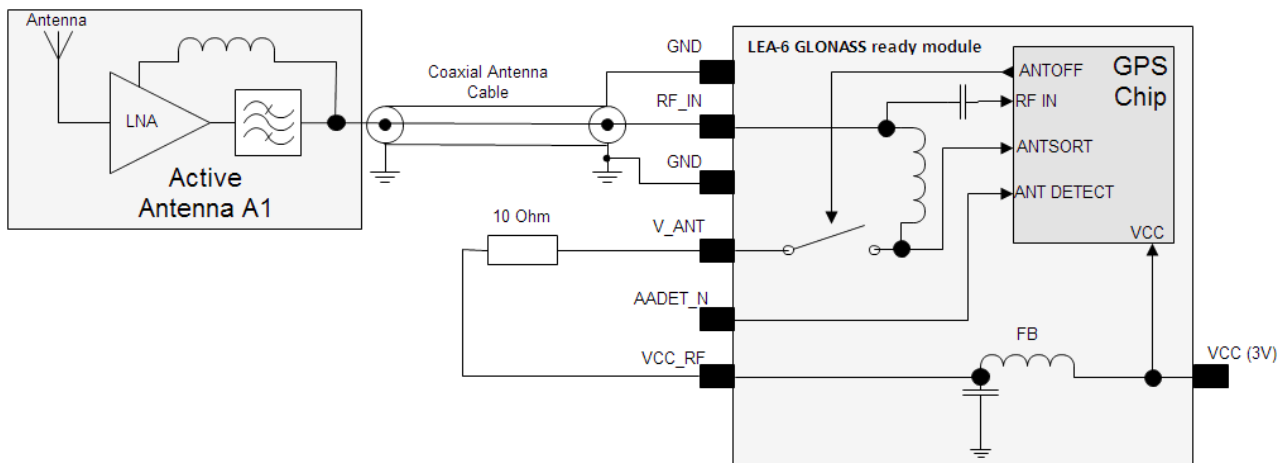


Figure 21: Module design with active antenna



Use a good performance GPS & GLONASS active antenna (for recommended components see section 3.3.9.1).

2.2.2.9 Module design with passive antenna

Figure 22 shows a GPS & GLONASS passive antenna design with the LEA-6H GLONASS ready module. For best performance with passive antenna designs use an external LNA.

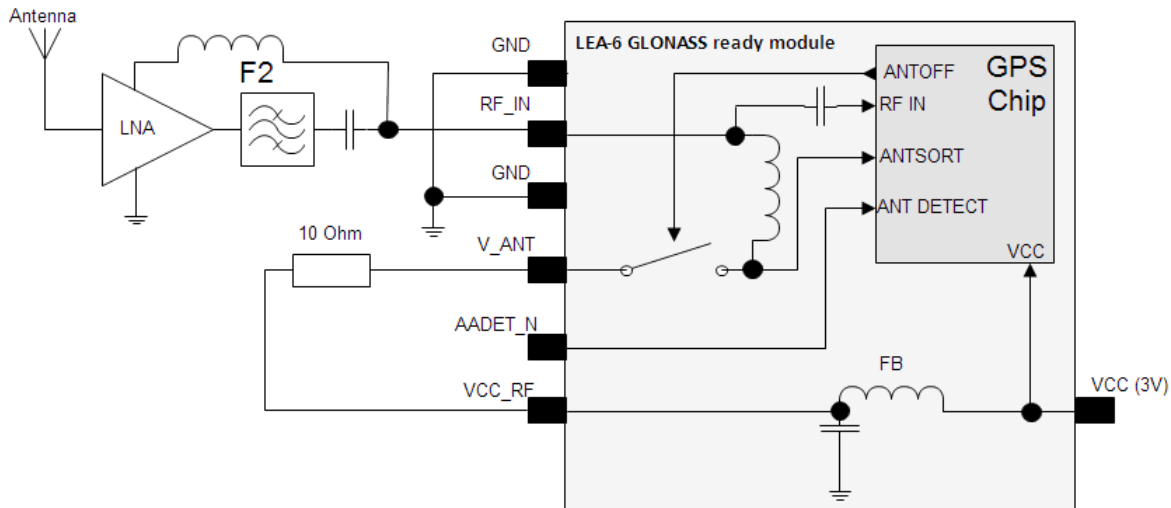


Figure 22: Module design with passive antenna

A standard GPS LNA has enough bandwidth to amplify GPS and GLONASS.



For recommended SAW Filters for GPS & GLONASS (Part F2 in Figure 20) see section 3.3.9.

2.2.2.10 GLONASS SW integration

To activate GLONASS system modes on u-blox GLONASS ready module, the customer application will have to:

- Upgrade the (Flash) firmware once available.
- Send UBX proprietary commands for activating and switching to GLONASS reception (SW command will be documented in the u-blox 6 Receiver Description including Protocol Specification [3]).

2.2.3 LEA-6R design

2.2.3.1 Connecting gyroscope and temperature sensor to the LEA-6R

The LEA-6R acts as SPI master. Following signals are used by the SPI:

Pin	Signal name	Direction	Usage
23	SPI_SCK	Output	SPI clock
22	SPI_SC51_N	Output	Chip Select signal for ADC/turn rate sensor
9	SPI_SC52_N	Output	Chip Select signal for temperature sensor
2	SPI_MISO	Input	Serial data (<u>Master In</u> / Slave Out)
1	SPI_MOSI	Output	Serial data (<u>Master Out</u> / Slave In), leave open

Table 8: SPI pins for LEA-6R

The following block schematic specifies the A/D converter and temperature sensor for the LEA-6R.



The LTC1860 and LM70-5 function at 5 V. A level translation with open-drain buffers and pull-up resistors on the outputs is required.

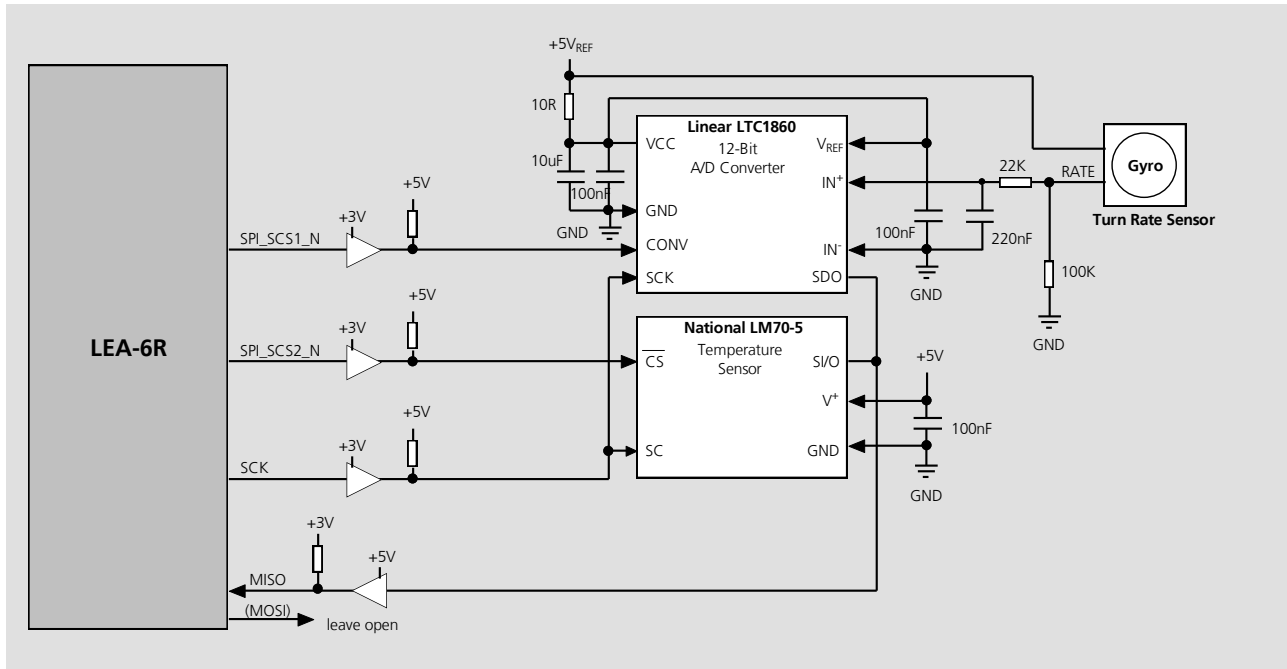


Figure 23: Attaching A/D converter and temperature sensor using the SPI

Add appropriate coupling capacitances according to the recommendations in the data sheets of the illustrated semiconductor products. All shown resistors shall have 5% accuracy or better. All shown capacitors (X7R types) shall have 10% accuracy or better.

For correct operation with the LEA-6R firmware, this circuit must be adopted without making any modifications such as, but not limited to, using different types of semiconductor devices and changing signal assignment.



LEA-6R default SPI clock is 870 kHz. As LEA-4R default value is 460 kHz, migrating from LEA-4R to LEA-6R will require a bandwidth verification of the SPI circuits and shall be designed for a bandwidth of 4 MHz.

2.2.3.2 Gyroscope requirements

Gyroscopes should meet the requirements listed below:

Parameter	Specification
Supply Voltage	5.0 V $\pm$ 0.25 V
Zero Point	2.5 V $\pm$ 0.4 V
Scale Factor	25 mV/(°/s) $\pm$ 5 mV/(°/s)
Dynamic Range	$\pm$ 60 °/s to $\pm$ 125 °/s
Linearity	$\pm$ 0.5 % (full scale)
Recommended operating temperature range	-40 to +85°C

Table 9: Requirements for gyroscopes



Follow the gyroscope manufacturer design recommendations for proper analog signal conditioning.

2.2.3.3 Supported A/D converters

The following table lists the supported A/D converters:

Manufacturer	Device
Linear Technology	LTC1860

Table 10: Supported A/D converters

2.2.3.4 Supported temperature sensors

The following table lists the supported temperature sensors:

Manufacturer	Device
National Semiconductor	LM70

Table 11: Supported temperature sensors



Note, that the temperature sensor inside the EPSON XV-8000 gyroscope sensor is not supported.

2.2.3.5 Forward / Backward indication

Use of the forward / backward indication signal **FWD** is optional but strongly recommended for good dead reckoning performance. It has an internal pull-up and therefore can be left open or connected to **VCC_OUT** or **VCC** if not used.

You need to check the voltage levels and the quality of the vehicle signals. They may be of different voltage levels, for example 12V nominal with a certain degree of variation. Use of optocouplers or other approved EMI protection and filtering is strongly recommended.



If no direction signal is available, the direction must be set to forward by configuring the meaning of the direction pin appropriately, otherwise DR positioning will be incorrect due to the wrong direction. GPS only navigation is not affected by this configuration.



As the forward/backward direction signal is not available in all cars, try to make use of the reverse gear light.

Pin	Signal name	Direction	Usage
21	FWD	Input	Direction indication (1 = forward)

Table 12: LEA-6R Forward / Backward indication

2.2.3.6 Odometer / Speedpulses

DR receivers use signals from sensors in the car to establish the velocity and distance traveled. These sensors are referred to as the odometer and the signals can be designated odometer pulses, speedpulses, speed ticks, wheel pulses or wheel ticks. These terms are often used interchangeably which can sometimes lead to confusion. For the sake of consistency, in this document we will be referring to these signals as speedpulses.

Pin	Signal name	Direction	Usage
27	SPEED	Input	Odometer Speedpulses

Table 13: LEA-6R Odometer / Speedpulses

The speedpulse signal required for DR modules must have a frequency range from 1 Hz to 2 kHz (0 Hz is equal to a speed of 0 km/hour) and must be linear to the driven speed.



For DR calibration see Section D.

2.2.1 Pin description for LEA-6 designs

Function	PIN	No	I/O	Description	Remarks
Power	VCC	6	I	Supply Voltage	Provide clean and stable supply.
	GND	7, 13-15, 17	I	Ground	Assure a good GND connection to all GND pins of the module, preferably with a large ground.
	VCC_OUT	8	O		Leave open if not used.
	V_BCKP	11	I	Backup voltage supply	It's recommended to connect a backup battery to V_BCKP in order to enable Warm and Hot Start features on the receivers. Otherwise connect to GND .
	VDDUSB	24	I	USB Power Supply	To use the USB interface connect this pin to 3.0 – 3.6V derived from VBUS. If no USB serial port used connect to GND.
Antenna	RF_IN	16	I	GPS/GALILEO signal input from antenna	Use a controlled impedance transmission line of 50 Ohm to connect to RF_IN. Don't supply DC through this pin. Use V_ANT pin to supply power.
	VCC_RF	18	O	Output Voltage RF section	Can be used to power an external active antenna (VCC_RF connected to V_ANT with 10 Ω). The max power consumption of the Antenna must not exceed the datasheet specification of the module. Leave open if not used.
	V_ANT	19	I	Antenna Bias voltage	Connect to GND (or leave open) if Passive Antenna is used. If an active Antenna is used, add a 10 Ω resistor in front of V_ANT input to the Antenna Bias Voltage or VCC_RF
	AADET_N	20	I	Active Antenna Detect	Input pin for optional antenna supervisor circuitry. Leave open if not used.
UART	TxD1	3	O	Serial Port 1	Communication interface can be programmed as TX ready for I2C interface. Leave open if not used.
	RxD1	4	I	Serial Port 1	Serial port input with internal pull-up resistor to VCC. Leave open if not used. Don't use external pull up resistor.
USB	USB_DM	25	I/O	USB I/O line	USB2.0 bidirectional communication pin. Leave open if unused.
	USB_DP	26	I/O	USB I/O line	Implementations see Section 1.6.2.
System	RESET_N	10	I	Hardware Reset (Active Low)	Leave open if not used. Do not drive high.
	TIMEPULSE	28	O	Timepulse Signal	Configurable Timepulse signal (one pulse per second by default). Leave open if not used.
	EXTINT0 /SPEED	27	I	Ext. Interrupt /Odometer	Ext. Interrupt Pin. Int. pull-up resistor to VCC. Leave open if unused. LEA-6R: Odometer speed
	CFG_COM1 /NC /SPI_SCS2_N /TIMEPULSE2	9	I	Config. Pin /NC /SPI /TIMEPULSE2	LEA-6S, LEA-6A: Leave open for default configuration. LEA-6H: Do not connect LEA-6R: SPI select 2 LEA-6T: TIMEPULSE2
	SDA2 /SPI_MOSI	1	I/O	DDC Pins /SPI	DDC Data. Leave open if not used. LEA-6R: SPI MOSI
	SCL2 /SPI_MISO	2	I/O	DDC Pins /SPI	DDC Clock. Leave open if not used. LEA-6R: SPI MISO
	Reserved	12	I		Leave open, do not drive low.
	NC	5			Leave open for only LEA-6x design. Connect to VCC for backward compatibility to LEA-5x.
	NC /FWD	21		Not Connect /Direction	Leave open LEA-6R: Forward / Backward indication
	NC /SPI_SCS1_N	22		Not Connect /SPI	Leave open LEA-6R: SPI select 1
	NC /SPI_SCK	23		Not Connect /SPI	Leave open LEA-6R: SPI clock

Table 14: Pin description LEA-6

2.3 NEO-6 design

2.3.1 Passive antenna design (NEO-6)

This is a minimal setup for a PVT GPS receiver with a NEO-6 module.

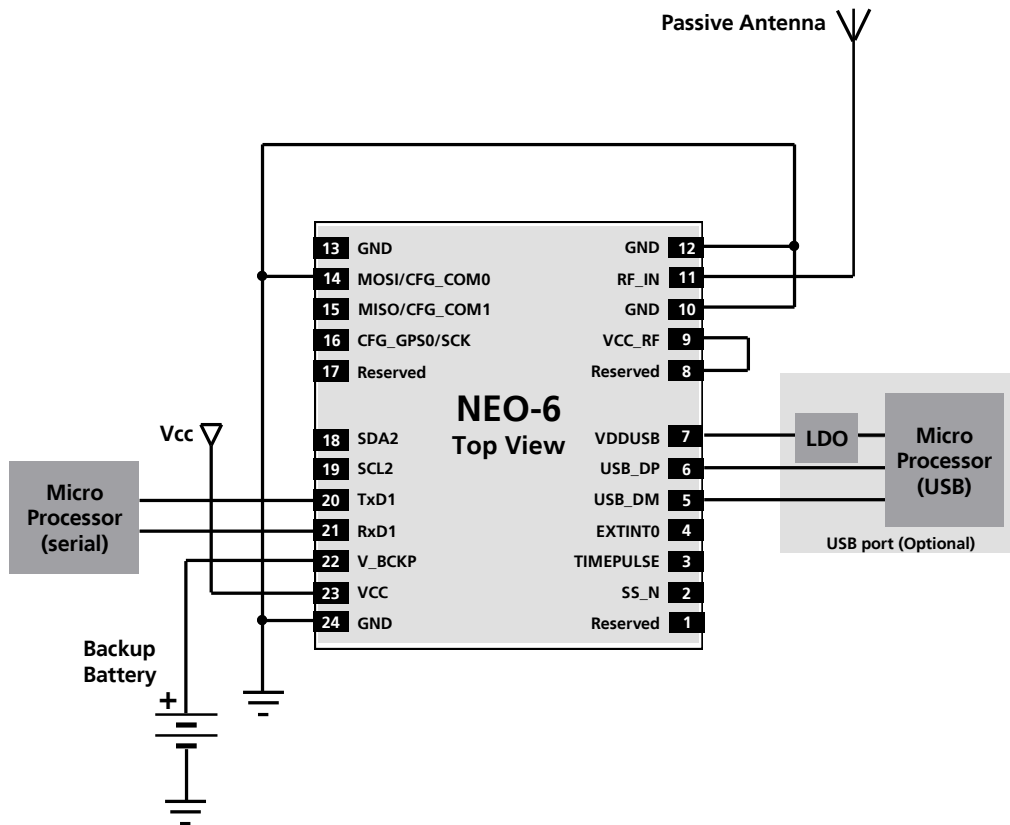


Figure 24: NEO-6 passive antenna design (with USB)



The above design is for the USB in self-powered mode. For bus-powered mode pin 14 (CFG_COM0) must be left open and VCC must be connected to VDDUSB. NMEA baud rate is 38400 when in self-powered mode.



For best performance with passive antenna designs use an external LNA to increase the sensitivity up to 2 dB. See Figure 25 and Figure 26.

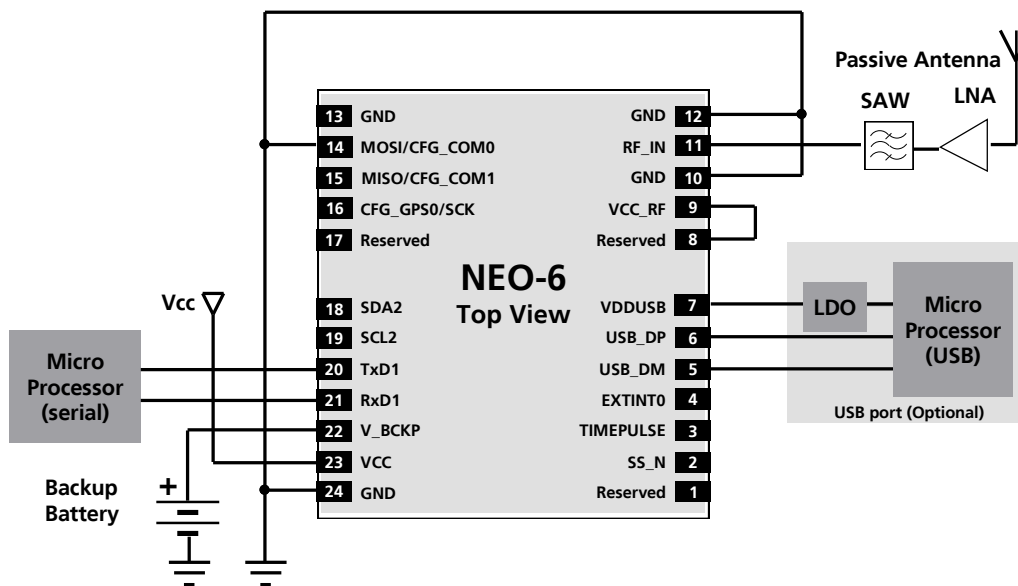


Figure 25: NEO-6 passive antenna design for best performance (with external LNA and SAW)

Figure 26 below shows a passive antenna design for NEO-6 GPS modules with an external SAW-LNA-SAW for best performance and increased immunity to jammers such as GSM. For lowest power in backup mode use ANTOFF

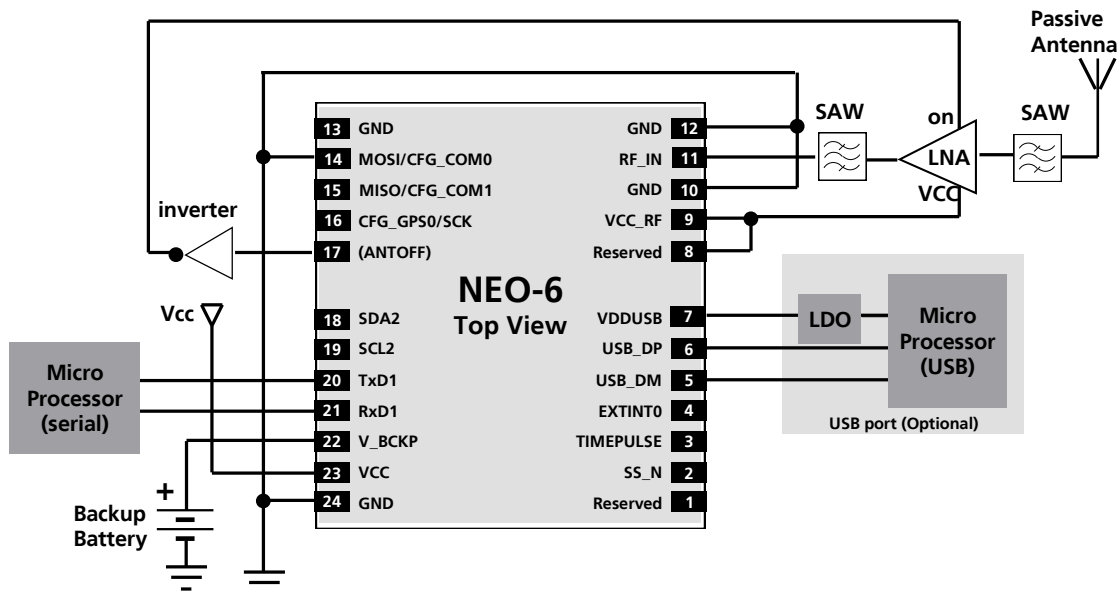


Figure 26: NEO-6 passive antenna design for best performance and increased immunity to jammers such as GSM



For information on increasing immunity to jammers such as GSM see section 3.3.8.



When using an external LNA in PSM on / off mode, pin 17 can be programmed as ANTOFF.



To configure the ANTOFF function refer to the u-blox 6 Receiver Description including Protocol Specification [3].



Use caution when implementing ANTOFF configuration since forward compatibility is not guaranteed

2.3.2 Pin description for NEO-6 designs

Function	PIN	No	I/O	Description	Remarks
Power	VCC	23	I	Supply Voltage	Max allowed ripple on VCC=50 mVpp
	GND	10,12,13,24	I	Ground	Assure a good GND connection to all GND pins of the module, preferably with a large ground plane.
	V_BCKP	22	I	Backup voltage supply	It's recommended to connect a backup battery to V_BCKP in order to enable Warm and Hot Start features on the receivers. Otherwise connect to GND .
	VDDUSB	7	I	USB Power Supply	To use the USB interface connect this pin to 3.0 – 3.6 V. If no USB serial port used connect to GND.
Antenna	RF_IN	11	I	GPS signal input from antenna	The connection to the antenna has to be routed on the PCB. Use a controlled impedance of 50 Ω to connect RF_IN to the antenna or the antenna connector.
	VCC_RF	9	O	Output Voltage RF section	Pins 8 and 9 must be connected together. VCC_RF can also be used to power an external active antenna.
UART	TxD1	20	O	Serial Port 1	Communication interface, can be programmed as TX ready for I2C interface.
	RxD1	21	I	Serial Port 1	Serial input. Internal pull-up resistor to VCC. Leave open if not used.
USB	USB_DM	5	I/O	USB I/O line	USB2.0 bidirectional communication pin. Leave open if unused.
	USB_DP	6	I/O	USB I/O line	Implementation see Section 0
System	TIMEPULSE	3	O	Timepulse Signal	Configurable Timepulse signal (one pulse per second by default). Leave open if not used.
	EXTINT0	4	I	External Interrupt	External Interrupt Pin. Internal pull-up resistor to VCC . Leave open if not used.
	SDA2	18	I/O	DDC Pins	DDC Data. Leave open, if not used.
	SCL2	19	I/O	DDC Pins	DDC Clock. Leave open, if not used.
	Reserved	17	I/O	Reserved	Can be configured as ANTOFF
	CFG_COM1 /MISO	15	I/O	Config. Pin /SPI MISO	Leave open if not used.
	CFG_COM0 /MOSI	14	I/O	Config. Pin /SPI MOSI	Leave open if not used. Note Connect to GND to use USB in Self Powered mode. See Section 1.7.4 and the NEO-6 Data Sheet [2]
	Reserved	8	I	Reserved	Pins 8 and 9 must be connected together. Can be used as a RESET_N input. See Section 1.7.1
	Reserved	1	-	Reserved	Leave open.
	SS_N	2	I/O	SPI Select	Leave open if not used.
	CFG_GPS0 /SCK	16	I/O	Config. Pin /SPI SCK	Leave open if not used.

Table 15: Pinout NEO-6

2.4 MAX-6 design

MAX-6 modules provide the following signals:

- ANTON Signal (to turn on and off external LNA). To save power consumption in Power Save mode. See Section 2.6.9
- TX ready Signal (to trigger a host, e.g. a u-blox LEON wireless module, when data at DDC interface is ready to be picked up). To save power consumption on Host side.

2.4.1 MAX-6 passive antenna design

This is a minimal setup for a PVT GPS receiver with a MAX-6 module.

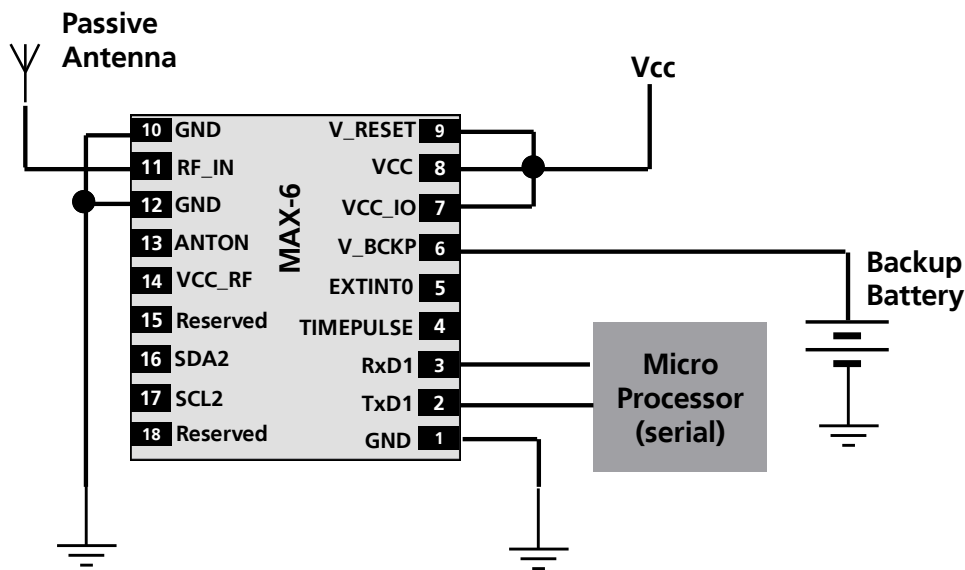


Figure 27: MAX-6 passive antenna design

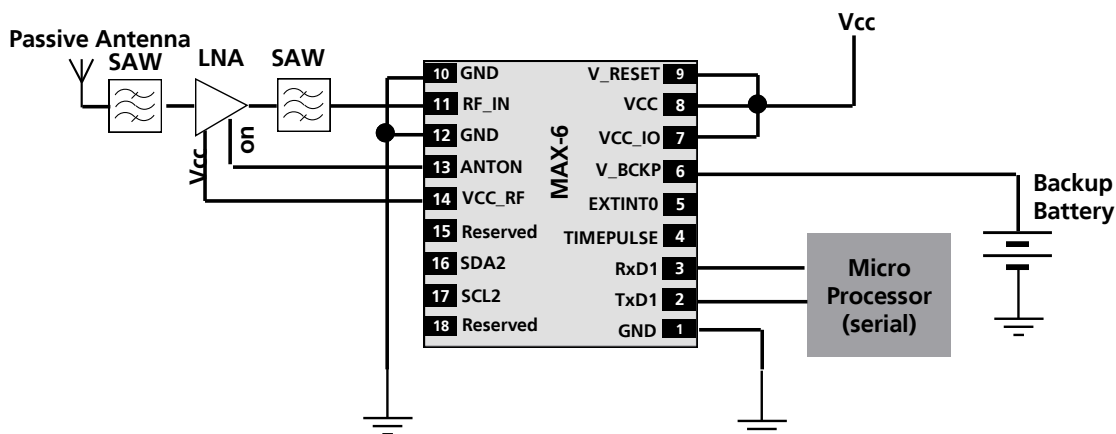


Figure 28: MAX-6 passive antenna design for best performance and increased immunity to jammers such as GSM



For information on increasing immunity to jammers such as GSM see section 3.3.8.

2.4.2 Pin description for MAX-6 designs

Function	PIN	No	I/O	Description	Remarks
Power	VCC	8	I	Supply Voltage	Max allowed ripple on VCC=50 mVpp
	GND	1,10,12	I	Ground	Assure a good GND connection to all GND pins of the module, preferably with a large ground plane.
	V_BCKP	6	I	Backup voltage supply	Backup voltage input pin. Connect to GND if not used.
Antenna	RF_IN	11	I	GPS signal input from antenna	The connection to the antenna has to be routed on the PCB. Use a controlled impedance of 50 Ω to connect RF_IN to the antenna or the antenna connector. DC block inside.
	VCC_RF	14	O	Output Voltage RF section	Can be used for active antenna or external LNA supply.
	ANTON	13	O		Active antenna or ext. LNA control pin in power save mode. Int. pull-up resistor to VCC
UART	TxD1	2	O	Serial Port 1	UART, leave open if not used, Voltage level referred VCC_IO. Can be configured as TX ready indication for the DDC interface.
	RxD1	3	I	Serial Port 1	UART, leave open if not used, Voltage level referred VCC_IO
System	TIMEPULSE	4	O	Timepulse Signal	Leave open if not used, Voltage level referred VCC_IO
	EXTINT0	5	I	External Interrupt	Leave open if not used, Voltage level referred VCC_IO
	SDA2	16	I/O	DDC Pins	DDC Data. Leave open, if not used.
	SCL2	17	I/O	DDC Pins	DDC Clock. Leave open, if not used.
	Reserved	18		Reserved	Leave open
	VCC_IO	7	I		IO supply voltage Input must be always supplied. Usually connect to VCC Pin 8. If I/O level should be different from VCC, supply VCC_IO with the I/O level required.
	V_RESET	9	I	VRESET	Must be connected to VCC always. Can be used as reset input pin with additional circuit (connected to VCC by 3.3 k Ω resistor). See Section 1.7.1
	Reserved	15		Reserved	Leave open

Table 16: Pinout MAX-6

2.5 Layout

This section provides important information for designing a reliable and sensitive GPS system.

GPS signals at the surface of the Earth are about 15 dB below the thermal noise floor. Signal loss at the antenna and the RF connection must be minimized as much as possible. When defining a GPS receiver layout, the placement of the antenna with respect to the receiver, as well as grounding, shielding and jamming from other digital devices are crucial issues and need to be considered very carefully.

2.5.1 Footprint and paste mask

Figure 29 - Figure 34 describe the footprint and provide recommendations for the paste mask for u-blox 6 LCC modules. These are recommendations only and not specifications. Note that the Copper and Solder masks have the same size and position.

To improve the wetting of the half vias, reduce the amount of solder paste under the module and increase the volume outside of the module by defining the dimensions of the paste mask to form a T-shape (or equivalent) extending beyond the Copper mask. For the stencil thickness see 3.2.1.

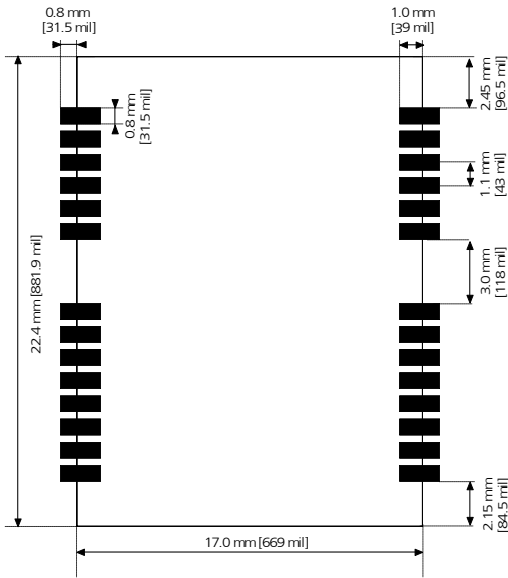


Figure 29: LEA-6 footprint

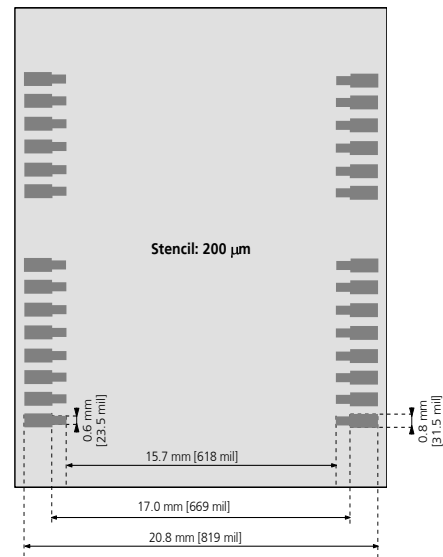


Figure 30: LEA-6 paste mask

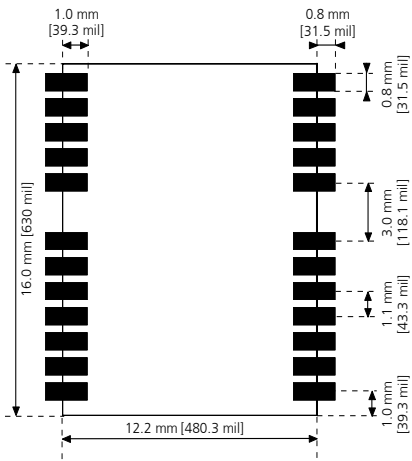


Figure 31: NEO-6 footprint

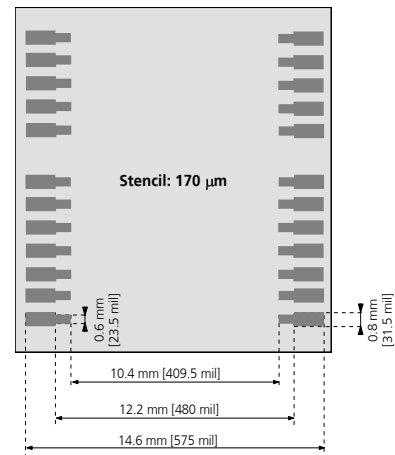


Figure 33: NEO-6 paste mask

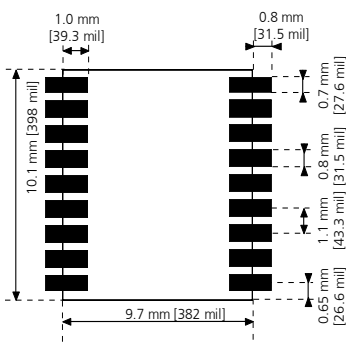


Figure 32: MAX-6 footprint

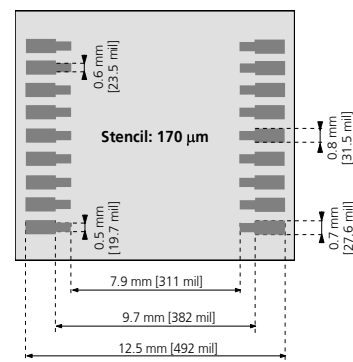


Figure 34: MAX-6 paste mask



MAX Form Factor (10.1 x 9.7 x 2.5): Same Pitch as NEO: 1.1 mm, but 4 pads in each corner only 0.7 mm wide instead 0.8 mm



The paste mask outline needs to be considered when defining the minimal distance to the next component. The exact geometry, distances, stencil thicknesses and solder paste volumes must be adapted to the specific production processes (e.g. soldering etc.) of the customer.

2.5.2 Placement

A very important factor in achieving maximum performance is the placement of the receiver on the PCB. The connection to the antenna must be as short as possible to avoid jamming into the very sensitive RF section.

Make sure that RF critical circuits are clearly separated from any other digital circuits on the system board. To achieve this, position the receiver digital part towards your digital section of the system PCB. Care must also be exercised with placing the receiver in proximity to circuitry that can emit heat. The RF part of the receiver is very sensitive to temperature and sudden changes can have an adverse impact on performance.



The RF part of the receiver is a temperature sensitive component. Avoid high temperature drift and air vents near the receiver.

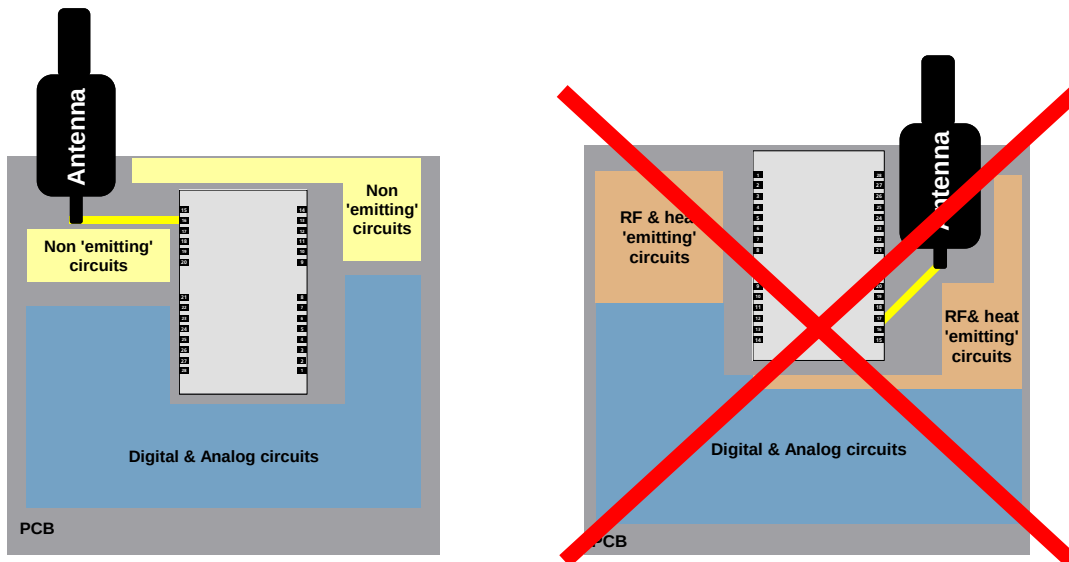


Figure 35: Placement (for exact pin orientation see data sheet)

2.5.3 Antenna connection and grounding plane design

u-blox 6 modules can be connected to passive patch or active antennas. The RF connection is on the PCB and connects the **RF_IN** pin with the antenna feed point or the signal pin of the connector, respectively. Figure 36 illustrates connection to a typical five-pin RF connector. One can see the improved shielding for digital lines as discussed in the GPS Antenna Application Note [4]. Depending on the actual size of the ground area, additional vias should be placed in the outer region. In particular, the edges of the ground area should be terminated with a dense line of vias.

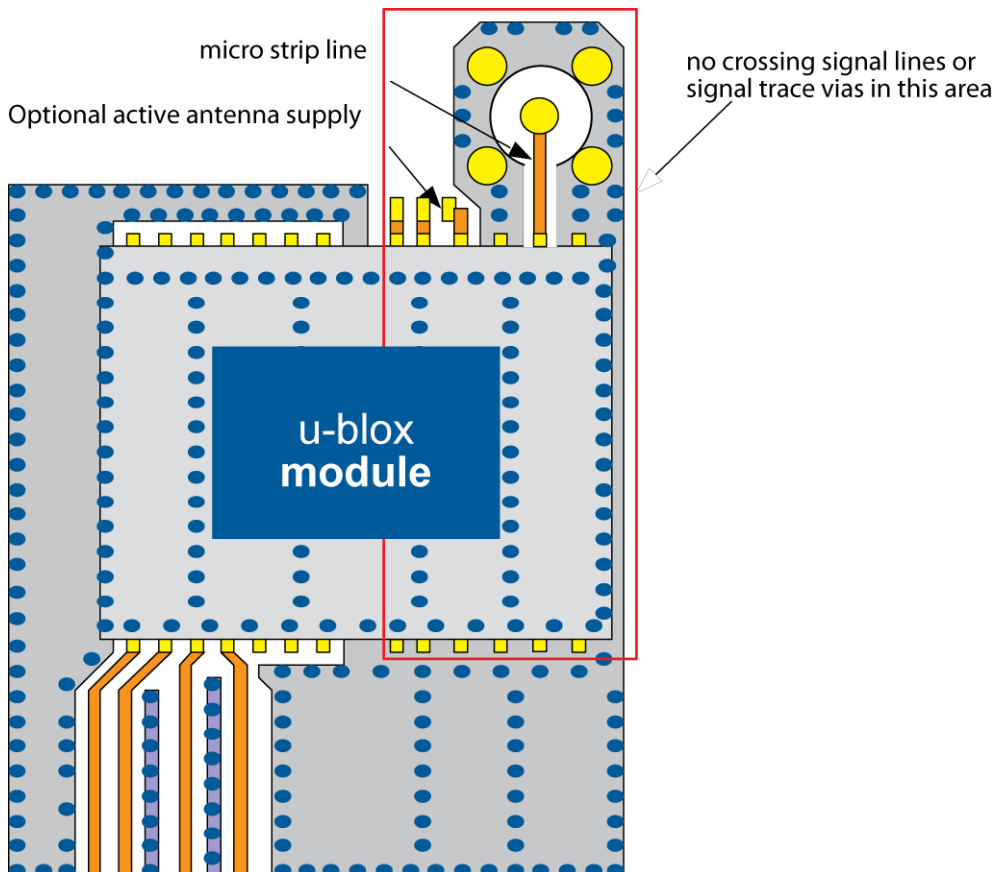


Figure 36: Recommended layout (for exact pin orientation see data sheet)

As seen in Figure 36, an isolated ground area is created around and below the RF connection. This part of the circuit **MUST** be kept as far from potential noise sources as possible. Make certain that no signal lines cross, and that no signal trace vias appear at the PCB surface within the area of the red rectangle. The ground plane should also be free of digital supply return currents in this area. On a multi layer board, the whole layer stack below the RF connection should be kept free of digital lines. This is because even solid ground planes provide only limited isolation.

The impedance of the antenna connection has to match the $50\ \Omega$ impedance of the receiver. To achieve an impedance of 50 Ohms, the width W of the micro strip has to be chosen depending on the dielectric thickness H , the dielectric constant ϵ_r of the dielectric material of the PCB and on the build-up of the PCB (see Section 2.5.4). Figure 37 shows two different builds: A 2 Layer PCB and a 4 Layer PCB. The reference ground plane is in both designs on layer 2 (red). Therefore the effective thickness of the dielectric is different.

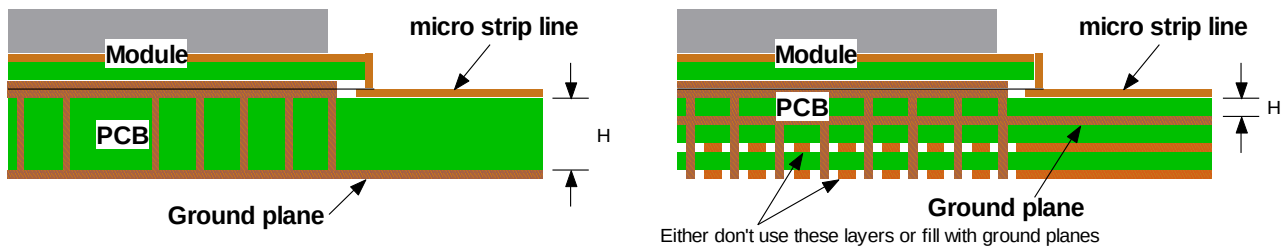


Figure 37: PCB build-up for micro strip line. Left: 2-layer PCB, right: 4-layer PCB

General design recommendations:

- The length of the micro strip line should be kept as short as possible. Lengths over 2.5 cm (1 inch) should be avoided on standard PCB material and without additional shielding.
- For multi layer boards the distance between micro strip line and ground area on the top layer should at least be as large as the dielectric thickness.
- Routing the RF connection close to digital sections of the design should be avoided.
- To reduce signal reflections, sharp angles in the routing of the micro strip line should be avoided. Chamfers or fillets are preferred for rectangular routing; 45-degree routing is preferred over Manhattan style 90-degree routing.

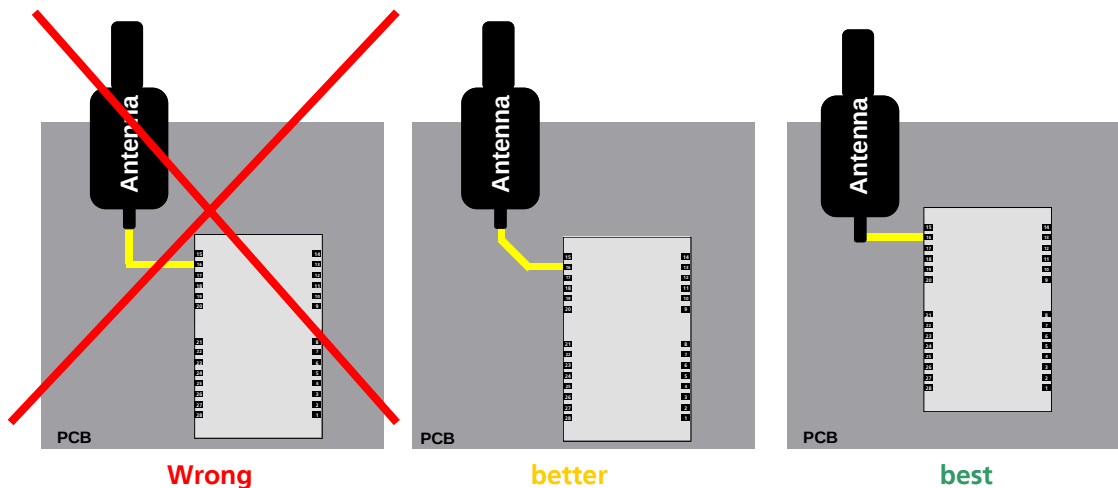


Figure 38: Recommended micro strip routing to RF pin (for exact pin orientation see data sheet)

- Do not route the RF-connection underneath the receiver. The distance of the micro strip line to the ground plane on the bottom side of the receiver is very small (some 100 μm) and has huge tolerances (up to 100%). Therefore, the impedance of this part of the trace cannot be controlled.
- Use as many vias as possible to connect the ground planes.
- In order to avoid reliability hazards, the area on the PCB under the receiver should be entirely covered with solder mask. Vias should not be open. Do not route under the receiver.

2.5.4 Antenna micro strip

There are many ways to design wave-guides on printed circuit boards. Common to all is that calculation of the electrical parameters is not straightforward. Freeware tools like AppCAD from Agilent or TXLine from Applied Wave Research, Inc. are of great help. They can be downloaded from www.agilent.com or <http://www.hp.woodshot.com/> and www.mwoffice.com.

The micro strip is the most common configuration for printed circuit boards. The basic configuration is shown in Figure 39 and Figure 40. As a rule of thumb, for a FR-4 material the width of the conductor is roughly double the thickness of the dielectric to achieve 50 Ω line impedance.

For the correct calculation of the micro strip impedance, one does not only need to consider the distance between the top and the first inner layer but also the distance between the micro strip and the adjacent GND plane on the same layer



Use the Coplanar Waveguide model for the calculation of the micro strip.

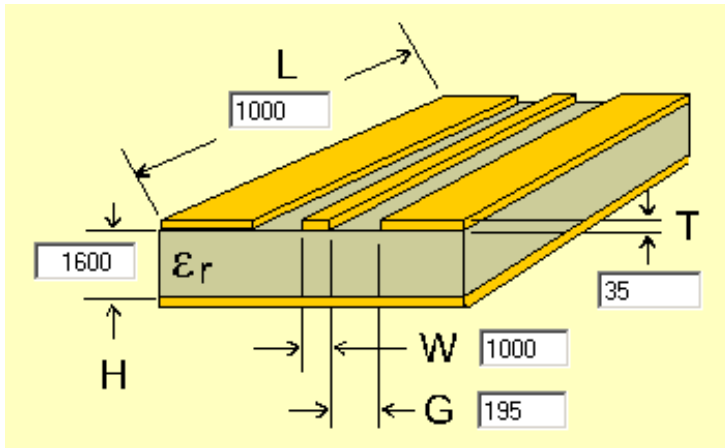


Figure 39: Micro strip on a 2-layer board (Agilent AppCAD Coplanar Waveguide)

Figure 39 shows an example of a 2-layer FR4 board with 1.6 mm thickness and a 35 μm (1 ounce) copper cladding. The thickness of the micro strip is comprised of the cladding (35 μm) plus the plated copper (typically 25 μm). Figure 40 is an example of a multi layer FR4 board with 18 μm ($\frac{1}{2}$ ounce) cladding and 180 μ dielectric between layer 1 and 2.

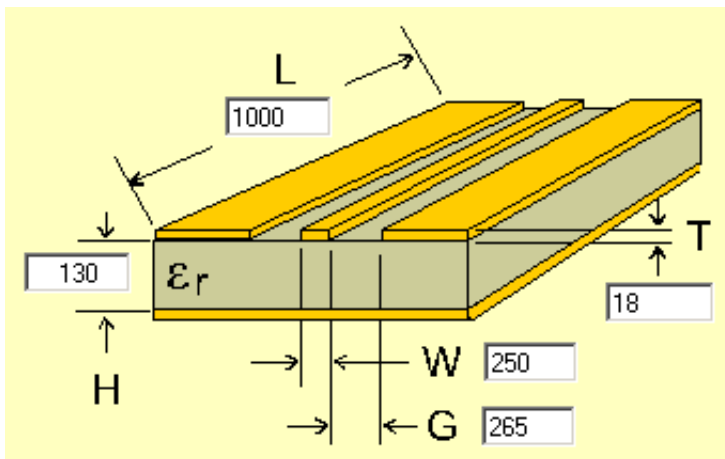


Figure 40: Micro strip on a multi layer board (Agilent AppCAD Coplanar Waveguide)

2.6 Antenna and antenna supervisor

u-blox 6 modules receive L1 band signals from GPS and GALILEO satellites at a nominal frequency of 1575.42 MHz. The RF signal is connected to the **RF_IN** pin.

u-blox 6 modules can be connected to passive or active antennas.



For u-blox 6 receivers, the total preamplifier gain (minus cable and interconnect losses) must not exceed 50 dB. Total noise figure should be below 3 dB.

u-blox 6 Technology supports short circuit protection of the active antenna and an active antenna supervisor circuit (open and short circuit detection). For further information refer to Section 2.6.2).

2.6.1 Passive antenna

A design using a passive antenna requires more attention regarding the layout of the RF section. Typically a passive antenna is located near electronic components; therefore care should be taken to reduce electrical 'noise' that may interfere with the antenna performance. Passive antennas do not require a DC bias voltage and can be directly connected to the RF input pin **RF_IN**. Sometimes, they may also need a passive matching network to match the impedance to 50 Ω .



Some passive antenna designs present a DC short to the RF input, when connected. If a system is designed with antenna bias supply AND there is a chance of a passive antenna being connected to the design, consider a short circuit protection.



All u-blox 6 receivers have a built-in LNA required for passive antennas.



Consider optional ESD protection (see Section 3.3).

2.6.2 Active antenna (LEA-6)

Active antennas have an integrated low-noise amplifier. They can be directly connected to **RF_IN**. If an active antenna is connected to **RF_IN**, the integrated low-noise amplifier of the antenna needs to be supplied with the correct voltage through pin **V_ANT**. Usually, the supply voltage is fed to the antenna through the coaxial RF cable. Active antennas require a power supply that will contribute to the total GPS system power consumption budget with additional 5 to 20 mA typically. Inside the antenna, the DC component on the inner conductor will be separated from the RF signal and routed to the supply pin of the LNA (see Figure 41).

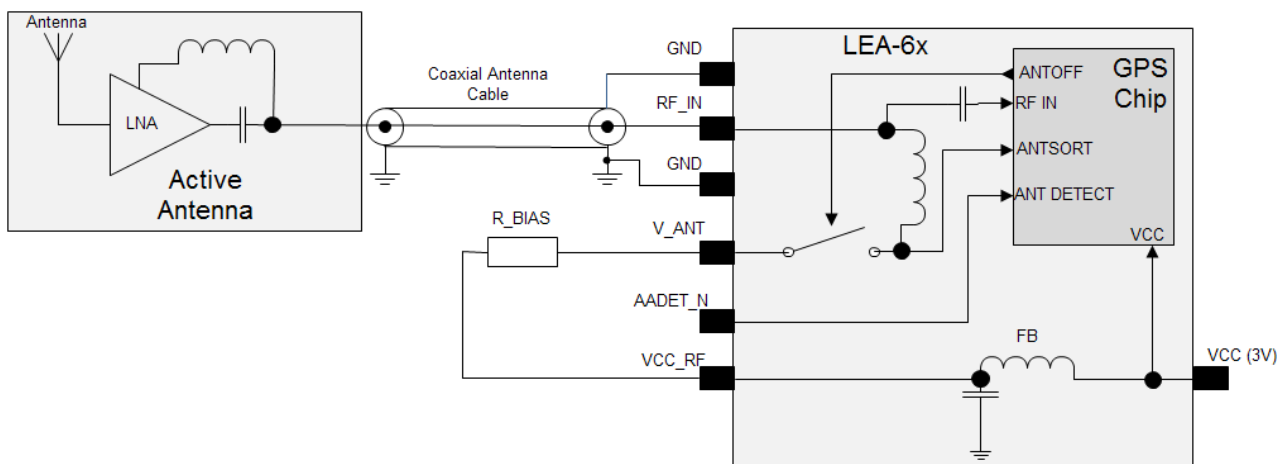


Figure 41: Active antenna biasing (for exact pin orientation see data sheet)

Generally an active antenna is easier to integrate into a system design, but an active antenna must also be placed far from any noise sources to have good performance.



Antennas should only be connected to the receiver when the receiver is not powered. Do not connect or disconnect the Antenna when the u-blox 6 receiver is running as the receiver calibrates the noise floor on power-up. Connecting the antenna after power-up can result in prolonged acquisition time.



Never feed supply voltage into RF_IN on u-blox LEA-6 modules. Always feed via V_ANT.



To test GPS signal reacquisition, it is recommended to physically block the signal to the antenna, rather than disconnecting and reconnecting the receiver.



Consider optional ESD protection; see section 3.3 for more information.

2.6.3 Active antenna bias power (LEA-6)

There are two ways to supply the bias voltage to pin **V_ANT**. For Internal supply, the **VCC_RF** output must be connected to **V_ANT** to supply the antenna with a filtered supply voltage. However, the voltage specification of the antenna has to match the actual supply voltage of the u-blox 6 Receiver (e.g. 3.0 V).

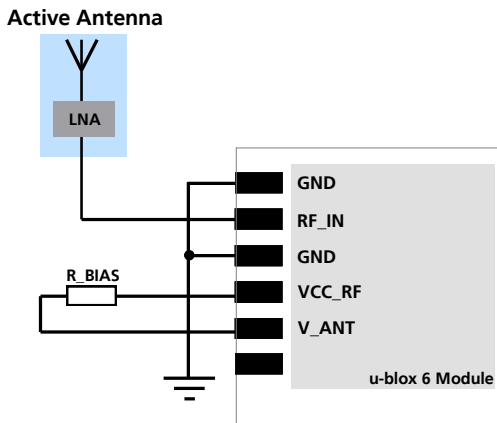


Figure 42: Internal supply Antenna bias voltage (for exact pin orientation see data sheet)

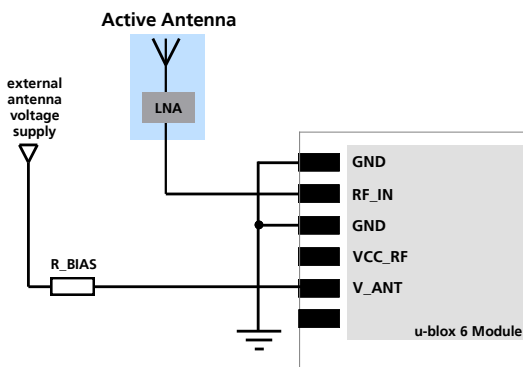


Figure 43: External supplying Antenna bias voltage (for exact pin orientation see data sheet)

Since the bias voltage is fed into the most sensitive part of the receiver, i.e. the RF input, this supply should be virtually free of noise. Usually, low frequency noise is less critical than digital noise with spurious frequencies with harmonics up to the GPS/GALILEO band of 1.575 GHz. Therefore, it is not recommended to use digital supply nets to feed pin **V_ANT**.

An internal switch (under control of the u-blox 6 software) can shut down the supply to the external antenna whenever it is not needed. This feature helps to reduce power consumption.

2.6.3.1 Short circuit protection

If a reasonably dimensioned series resistor **R_BIAS** is placed in front of pin **V_ANT**, a short circuit situation can be detected by the baseband processor. If such a situation is detected, the baseband processor will shut down supply to the antenna. The receiver is by default configured to attempt to reestablish antenna power supply periodically.



To configure the antenna supervisor use the UBX-CFG-ANT message. For further information refer to the u-blox 6 Receiver Description including Protocol Specification [3].

References	Value	Tolerance	Description	Manufacturer
R_BIAS	10 Ω	$\pm 10\%$	Resistor, min 0.250 W	

Table 17: Short circuit protection, bill of material



Short circuits on the antenna input without limitation (R_BIAS) of the current can result in permanent damage to the receiver! Therefore, it's recommended to implement an R_BIAS in all risk applications, such as situations where the antenna can be disconnected by the end-user or that have long antenna cables.



An additional R_BIAS is not required when using a short and open active antenna supervisor circuitry as defined in Section 2.6.4.1, as the R_BIAS is equal to R2.

2.6.4 Active antenna supervisor (LEA-6)

u-blox 6 Technology provides the means to implement an active antenna supervisor with a minimal number of parts. The antenna supervisor is highly configurable to suit various different applications.

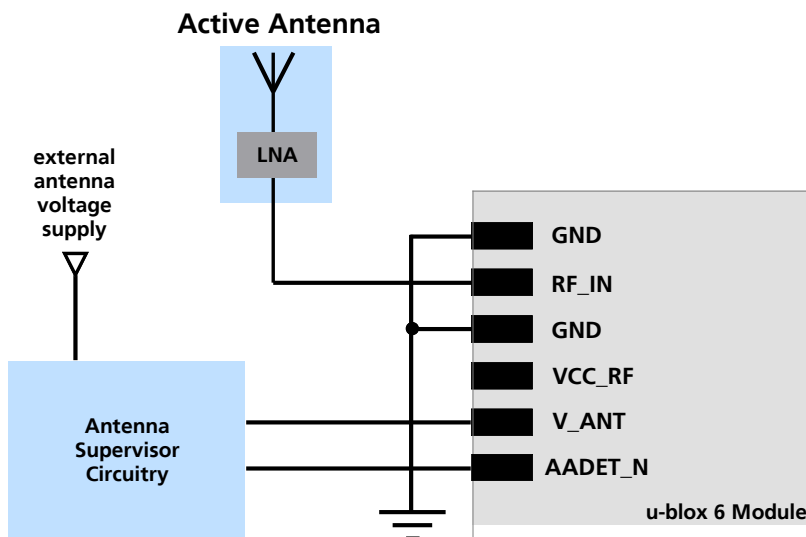


Figure 44: External antenna power supply with full antenna supervisor (for exact pin orientation see data sheet)

2.6.4.1 Short and open circuit active antenna supervisor

The Antenna Supervisor can be configured by a serial port message (using only UBX binary message). When enabled the active antenna supervisor produces serial port messages (status reporting in NMEA and/or UBX binary protocol) which indicates all changes of the antenna circuitry (**disabled** antenna supervisor, antenna circuitry **ok**, **short** circuit, **open** circuit) and shuts the antenna supply down if required. Active antenna status can be determined also polling UBX-MON-HW.

The active antenna supervisor provides the means to check the active antenna for open and short circuits and to shut the antenna supply off, if a short circuit is detected. The state diagram in Figure 45 applies. If an antenna is connected, the initial state after power-up is "Active Antenna OK".

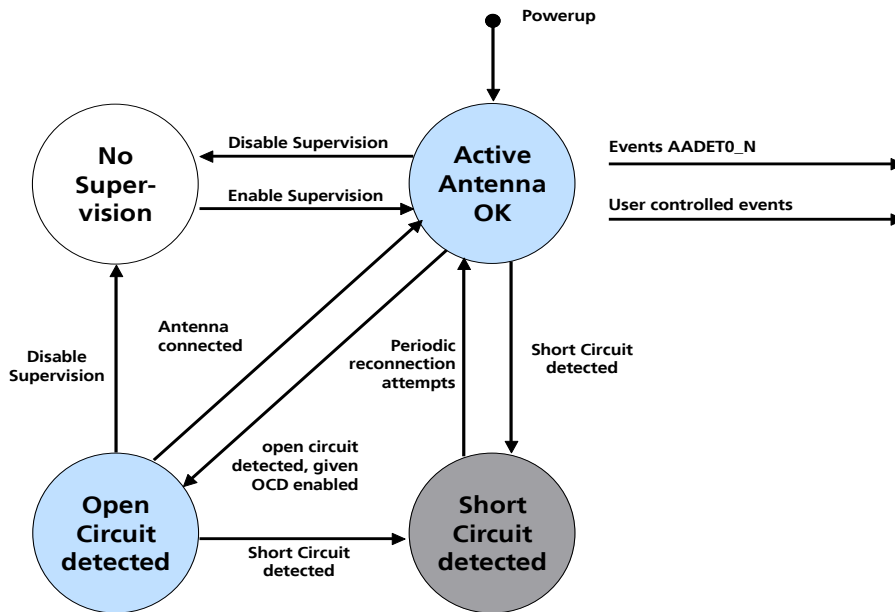


Figure 45: State diagram of active antenna supervisor

Firmware supports an active antenna supervisor circuit, which is connected to the pin **AADET_N**. An example of an open circuit detection circuit is shown in Figure 46 and Figure 47. High on **AADET_N** means that an external antenna is not connected.

Short Circuit Detection (SCD)

A short circuit in the active antenna pulls **V_ANT** to ground. This is detected inside the u-blox 6 module and the antenna supply voltage will be immediately shut down.



Antenna short detection (SCD) and control is enabled by default.

Open Circuit Detection (OCD)

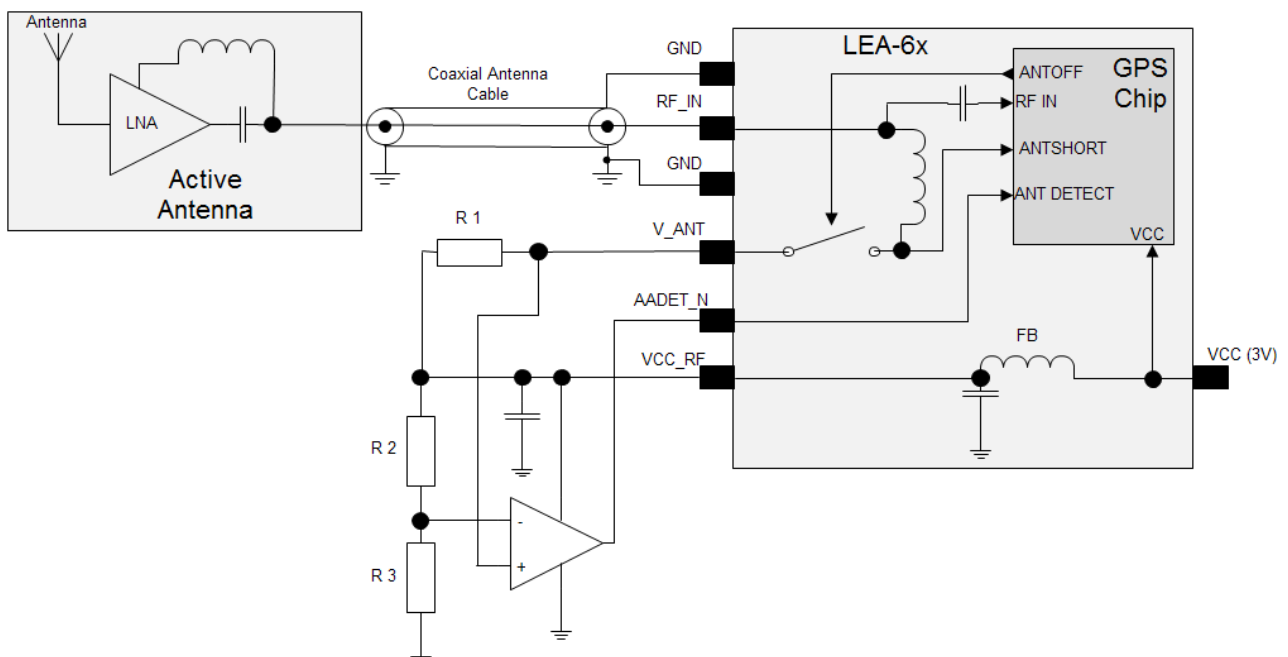


Figure 46: Schematic of open circuit detection variant A (for exact pin orientation see data sheet)

References	Value	Tolerance	Description	Remarks
R1	10 Ω	$\pm 5\%$	Resistor, min 0.250 W	
R2	560 Ω	$\pm 5\%$	Resistor	
R3	100 k Ω	$\pm 5\%$	Resistor	
U1	LT6000		Rail to Rail Op Amp	Linear Technology

Table 18: Active antenna supervisor, bill of material

$$I = \frac{\left(\frac{R2}{R2 + R3} \right)}{R1} \bullet V_{CC_RF}$$

Equation 1: Calculation of threshold current for open circuit detection



If the antenna supply voltage is not derived from V_{CC_RF}, do not exceed the maximum voltage rating of AADET_N.

The open circuit detection circuit uses the current flow to detect an open circuit in the antenna. The threshold current can be calculated using Equation 1.

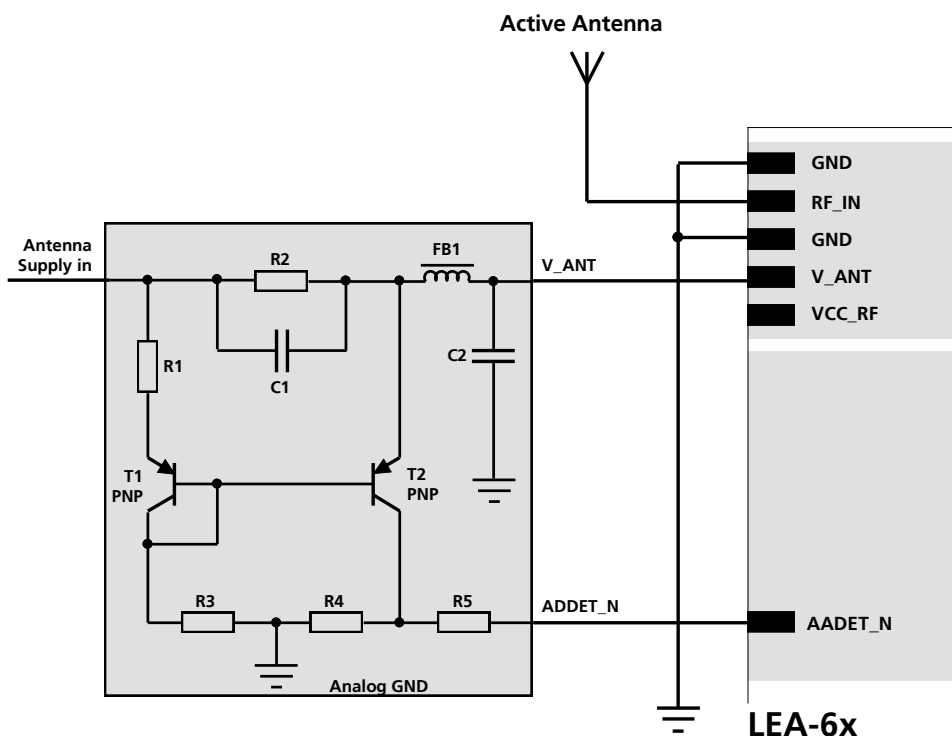


Figure 47: Schematic of open circuit detection variant B (for exact pin orientation see data sheet)



The open circuit supervisor circuitry shown in Figure 47 has a quiescent current of approximately 2mA. This current can be reduced with an advanced circuitry such as shown in Figure 47.

References	Value	Tolerance	Description	Remarks / Manufacturer
C1	2.2 μ F	$\pm 10\%$	Capacitor, X7R, min 10 V	
C2	100 nF	$\pm 10\%$	Capacitor, X7R, min 10 V	
FB1	600 Ω		Ferrite Bead	e.g. Murata BLM18HD601SN1
R1	15 Ω	$\pm 10\%$	Resistor, min 0.063 W	
R2	10 Ω	$\pm 10\%$	Resistor, min 0.250 W	
R3, R4	10 k Ω	$\pm 10\%$	Resistor, min 0.063 W	
R5	33 k Ω	$\pm 10\%$	Resistor, min 0.063 W	
T1, T2	BC856B		PNP Transistor	e.g. Philips Semiconductors ⁷

Table 19: Active antenna supervisor, bill of material

Status reporting

At startup and on every change of the antenna supervisor configuration the u-blox 6 GPS/GALILEO module will output a NMEA (**\$GPTXT**) or UBX (**INF-NOTICE**) message with the internal status of the antenna supervisor (disabled, short detection only, enabled).

None, one or several of the strings below are part of this message to inform about the status of the active antenna supervisor circuitry (e.g. "**ANTSUPERV= AC SD OD PdoS**").

Abbreviation	Description
AC	Antenna Control (e.g. the antenna will be switched on/ off controlled by the GPS receiver)
SD	Short Circuit Detection Enabled
SR	Short Circuit Recovery Enabled
OD	Open Circuit Detection Enabled
PdoS	Power Down on short

Table 20: Active Antenna Supervisor Message on startup (UBX binary protocol)



To activate the antenna supervisor use the **UBX-CFG-ANT** message. For further information refer to the u-blox 6 Receiver Description including Protocol Specification [3].

Similar to the antenna supervisor configuration, the status of the antenna supervisor will be reported in a NMEA (**\$GPTXT**) or UBX (**INF-NOTICE**) message at start-up and on every change.

Message	Description
ANTSTATUS=DONTKNOW	Active antenna supervisor is not configured and deactivated.
ANTSTATUS=OK	Active antenna connected and powered
ANTSTATUS=SHORT	Antenna short
ANTSTATUS=OPEN	Antenna not connected or antenna defective

Table 21: Active antenna supervisor message on startup (NMEA protocol)

2.6.5 Active antenna (NEO-6 and MAX-6)

NEO-6 and MAX-6 modules do not provide the antenna bias voltage for active antennas on the **RF_IN** pin. It is therefore necessary to provide this voltage outside the module via an inductor L as indicated in Figure 48. u-blox recommends using an inductor from Murata (LQG15HS27NJ02). Alternative parts can be used if the inductor's resonant frequency matches the GPS frequency of 1575.42 MHz.

⁷ Transistors from other suppliers with comparable electrical characteristics may be used.

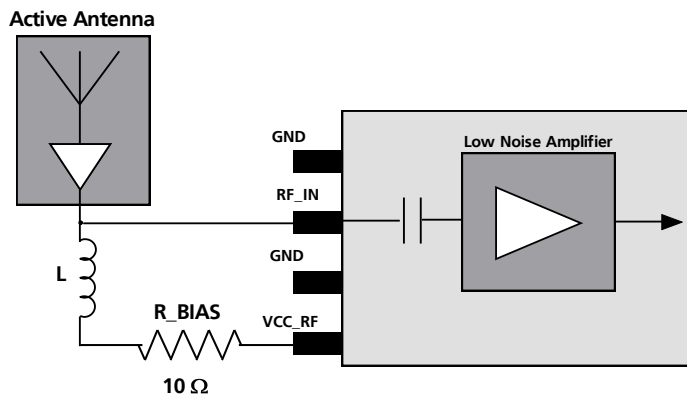


Figure 48: Internal antenna bias voltage for active antennas

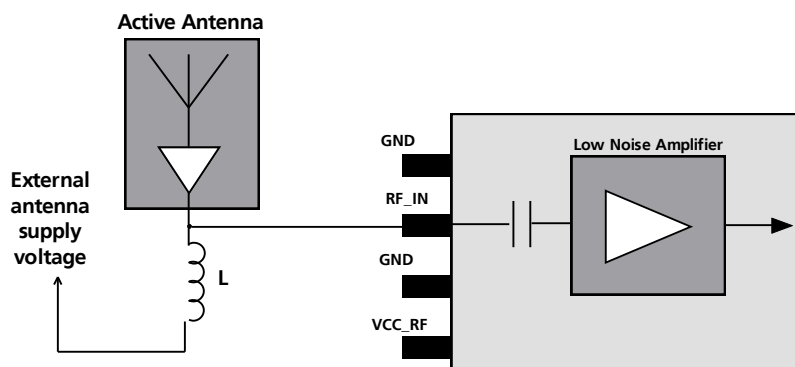


Figure 49: External antenna bias voltage for active antennas

For optimal performance, it is important to place the inductor as close to the microstrip as possible. Figure 50 illustrates the recommended layout and how it should not be done.

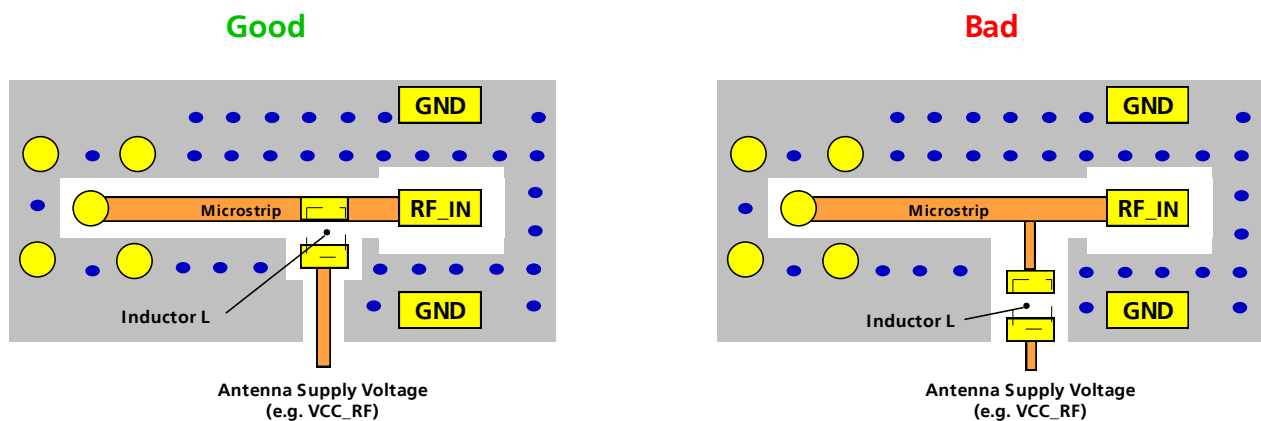


Figure 50: Recommended layout for connecting the antenna bias voltage for LEA-6M and NEO-6

2.6.6 External active antenna supervisor using ANTOFF (NEO-6)

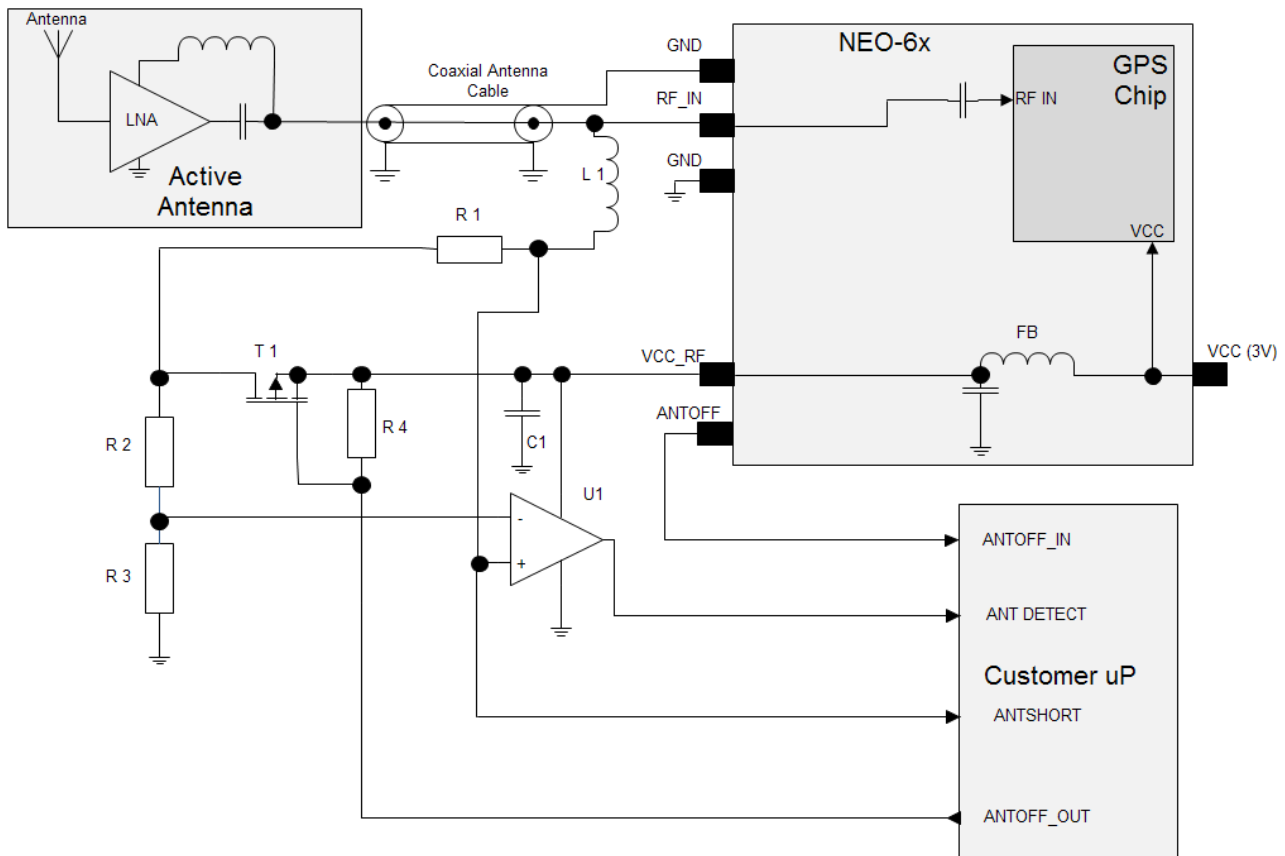


Figure 51: External active antenna supervisor using ANTOFF (NEO-6)

References	Value	Tolerance	Description	Remarks / Manufacturer
R1	10 Ω	$\pm 5\%$	Resistor, min 0.25 W	
R2	560 Ω	$\pm 5\%$	Resistor	
R3	100 k Ω	$\pm 5\%$	Resistor	
R4	100 k Ω	$\pm 5\%$	Resistor	
U1	LT6000		Rail to Rail Op Amp	Linear Technology
T1	Si1016X-T1-E3		Transistor	Vishay
L1	LQG15HS27NJ02		Inductor	muRata
C1	X5R 100N 10V	10%	Decoupling Capacitor	muRata

Table 22: Active antenna supervisor, bill of material

$$I = \frac{\left(\frac{R2}{R2 + R3} \right)}{R1} \bullet V_{CC_RF}$$

Equation 2: Calculation of threshold current for open circuit detection



State diagram of active antenna supervisor see Figure 45. When using an external LNA in PSM on / off mode, pin 17 can be programmed as ANTOFF (see 1.7.7). Use ANTOFF_IN and ANTOFF_OUT signals to command antenna power supply when going into Power Save Mode (Backup mode).



Use caution when implementing ANTOFF configuration since forward compatibility is not guaranteed

2.6.7 External active antenna supervisor using ANTON (MAX-6)

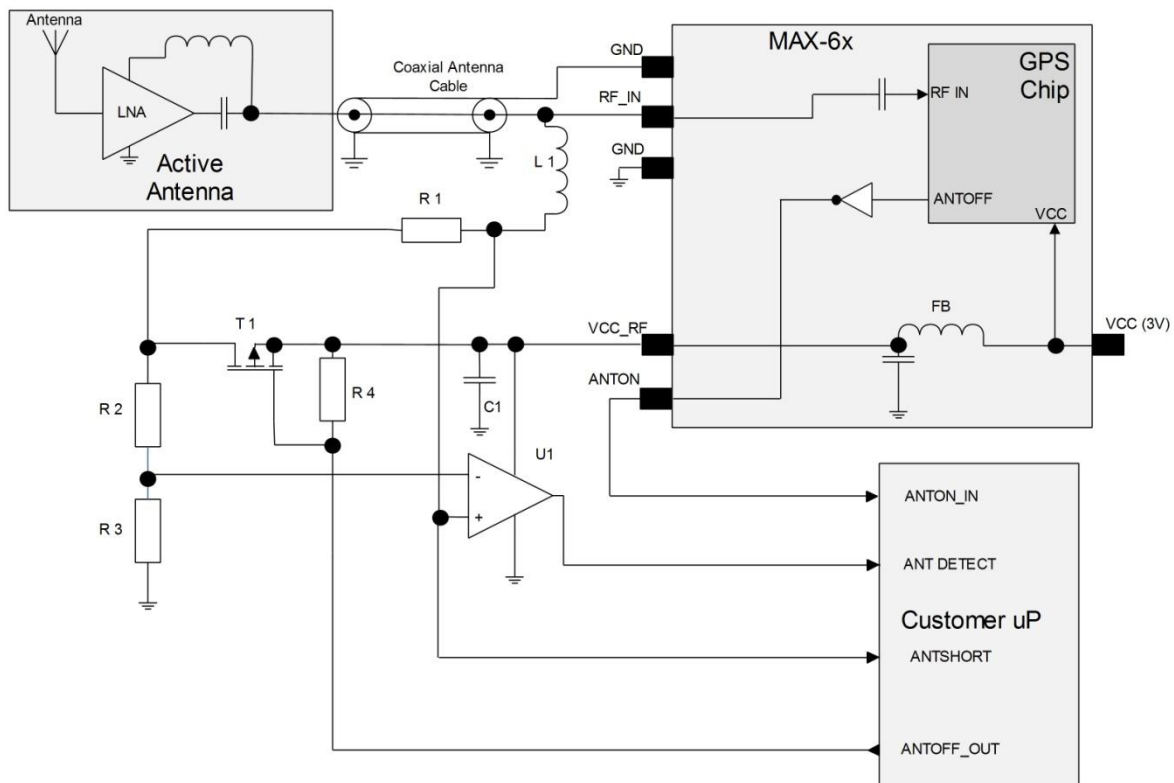


Figure 52: External active antenna supervisor using ANON (MAX-6)

References	Value	Tolerance	Description	Remarks / Manufacturer
R1	10 Ω	$\pm 5\%$	Resistor, min 0.25 W	
R2	560 Ω	$\pm 5\%$	Resistor	
R3	100 k Ω	$\pm 5\%$	Resistor	
R4	100 k Ω	$\pm 5\%$	Resistor	
U1	LT6000		Rail to Rail Op Amp	Linear Technology
T1	Si1016X-T1-E3		Transistor	Vishay
L1	LQG15HS27NJ02		Inductor	muRata
C1	X5R 100N 10V	10%	Decoupling Capacitor	muRata

Table 23: Active antenna supervisor, bill of material

$$I = \frac{\left(\frac{R2}{R2 + R3} \right)}{R1} \cdot V_{CC_RF}$$

Equation 3: Calculation of threshold current for open circuit detection



State diagram of active antenna supervisor see Figure 45. When using an external LNA in PSM on / off mode, pin 17 can be programmed as ANTOFF (see 1.7.7). Use ANTOFF_IN and ANTOFF_OUT signals to command antenna power supply when going into Power Save Mode (Backup mode).

2.6.8 External active antenna control (NEO-6)

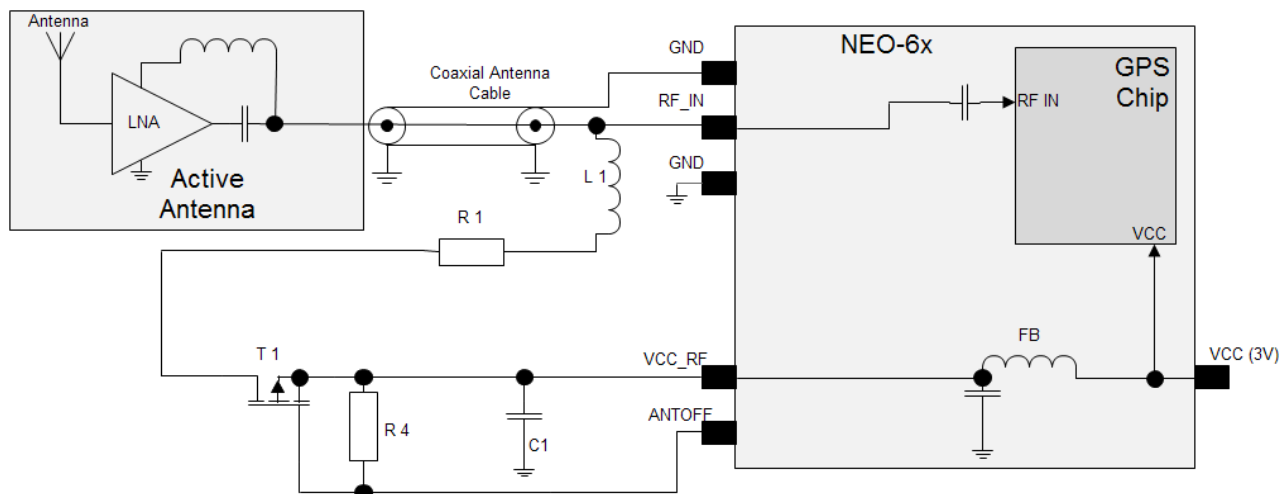


Figure 53: External active antenna control (NEO-6)



When using an external LNA in PSM on / off mode, pin 17 can be programmed as ANTOFF (see 1.7.7).



Use caution when implementing ANTOFF configuration since forward compatibility is not guaranteed

References	Value	Tolerance	Description	Remarks / Manufacturer
R1	10 Ω	$\pm 5\%$	Resistor, min 0.25 W	
R4	100 k Ω	$\pm 5\%$	Resistor	
T1	Si1016X-T1-E3		Transistor	Vishay
L1	LQG15HS27NJ02		Inductor	muRata
C1	X5R 100N 10V	10%	Decoupling Capacitor	muRata

Table 24: Active antenna control, bill of material

2.6.9 External active antenna control (MAX-6)

ANTON Signal can be used to turn on and off an external LNA. This reduces power consumption in Power Save Mode (Backup mode).

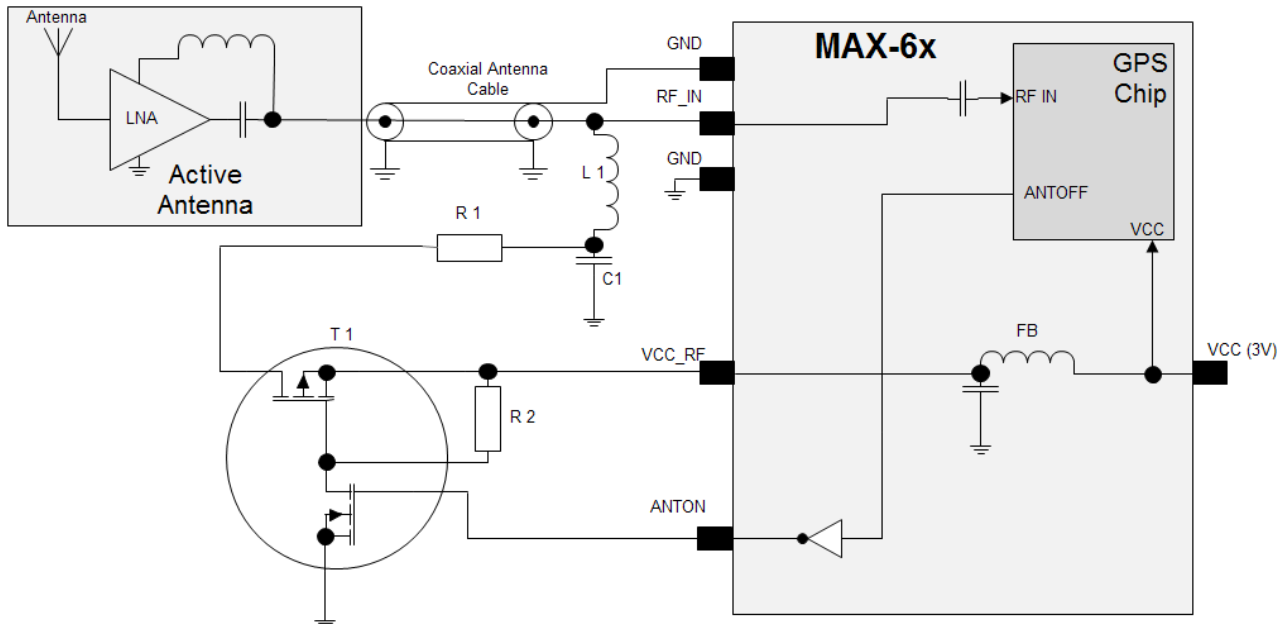


Figure 54: External active antenna control (MAX-6)

References	Value	Tolerance	Description	Remarks / Manufacturer
R1	10 Ω	$\pm 5\%$	Resistor, min 0.25 W	
R2	100 k Ω	$\pm 5\%$	Resistor	
T1	Si1040X		Power MOSFET	Vishay
L1	LQG15HS27NJ02		Inductor	muRata
C1	X5R 1N 10V	10%	Decoupling Capacitor	muRata

Table 25: Active antenna control, bill of material

2.6.10 GPS antenna placement for LEA-6R

For an optimum ADR navigation performance, the following setup recommendations should be considered.

GPS antenna placement, gyro placement and single tick origin

Due to geometric and dynamic aspects of driving vehicles, it is important to correctly place the GPS antenna and the external sensors - from a geometric point of view - in order to get consistent measurement information from the different sensors.

For standard road vehicles: The GPS antenna should be placed above the middle of the rear (unsteered) axis. The gyro can be placed anywhere on the vehicle. Single ticks should origin from the rear (unsteered) wheels. For articulated busses, the sensors should be placed on the front car as if this was a standard road vehicle. In case the GWT solution is used for rail vehicles: The GPS antenna should be placed in the middle of a wagon, while the gyro can be placed anywhere on the same wagon and the single ticks can origin from any wheels of the same wagon.



Large geometrical deviations from the optimal placement - especially of the GPS antenna (e.g. when placing it above the front axis of a long bus) - can result in significant performance degradation!

3 Product handling

3.1 Packaging, shipping, storage and moisture preconditioning

For information pertaining to reels and tapes, Moisture Sensitivity levels (MSD), shipment and storage information, as well as drying for preconditioning see the data sheet of the specific u-blox 6 GPS module.

3.2 Soldering

3.2.1 Soldering paste

Use of "No Clean" soldering paste is strongly recommended, as it does not require cleaning after the soldering process has taken place. The paste listed in the example below meets these criteria.

Soldering Paste: OM338 SAC405 / Nr.143714 (Cookson Electronics)
 Alloy specification: Sn 95.5/ Ag 4/ Cu 0.5 (95.5% Tin/ 4% Silver/ 0.5% Copper)
 Melting Temperature: 217°C
 Stencil Thickness: 150 µm for base boards

The final choice of the soldering paste depends on the approved manufacturing procedures.

The paste-mask geometry for applying soldering paste should meet the recommendations in section 2.5.1.



The quality of the solder joints on the connectors ('half vias') should meet the appropriate IPC specification.

3.2.2 Reflow soldering

A convection type-soldering oven is strongly recommended over the infrared type radiation oven. Convection heated ovens allow precise control of the temperature and all parts will be heated up evenly, regardless of material properties, thickness of components and surface color.

Consider the "IPC-7530 Guidelines for temperature profiling for mass soldering (reflow and wave) processes, published 2001".

Preheat phase

Initial heating of component leads and balls. Residual humidity will be dried out. Please note that this preheat phase will not replace prior baking procedures.

- Temperature rise rate: max.3°C/s If the temperature rise is too rapid in the preheat phase it may cause excessive slumping.
- Time: 60 – 120 s If the preheat is insufficient, rather large solder balls tend to be generated. Conversely, if performed excessively, fine balls and large balls will be generated in clusters.
- End Temperature: 150 - 200°C If the temperature is too low, non-melting tends to be caused in areas containing large heat capacity.

Heating/ Reflow phase

The temperature rises above the liquidus temperature of 217°C. Avoid a sudden rise in temperature as the slump of the paste could become worse.

- Limit time above 217°C liquidus temperature: 40 – 60 s
- Peak reflow temperature: 245°C

Cooling phase

A controlled cooling avoids negative metallurgical effects (solder becomes more brittle) of the solder and possible mechanical tensions in the products. Controlled cooling helps to achieve bright solder fillets with a good shape and low contact angle.

- Temperature fall rate: max 4°C / s



To avoid falling off, the u-blox 6 GPS module should be placed on the topside of the motherboard during soldering.

The final soldering temperature chosen at the factory depends on additional external factors like choice of soldering paste, size, thickness and properties of the base board, etc. Exceeding the maximum soldering temperature in the recommended soldering profile may permanently damage the module.

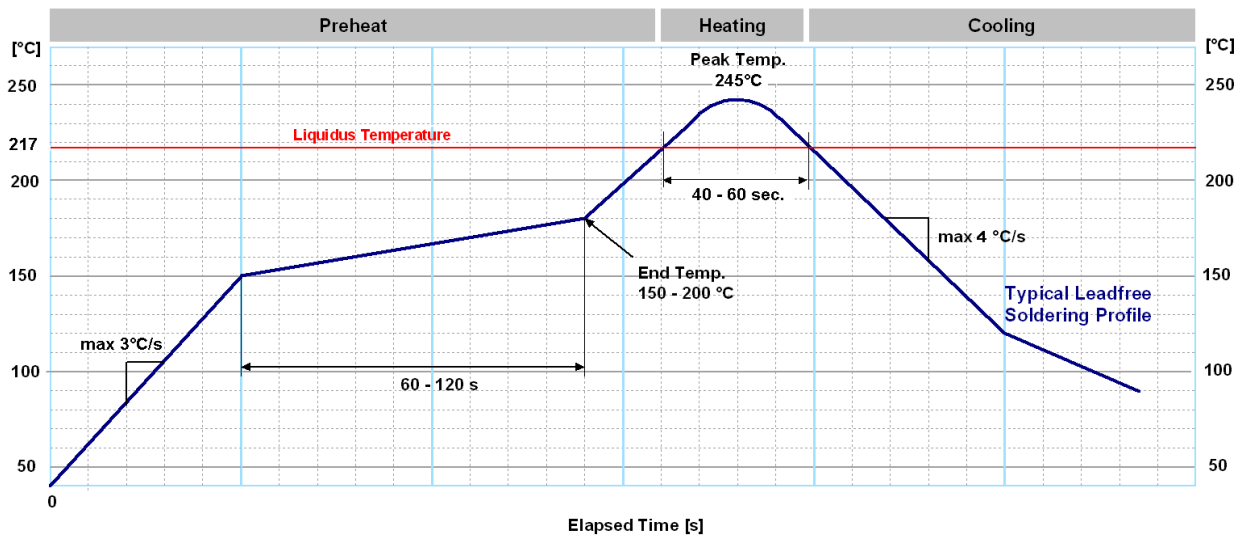


Figure 55: Recommended soldering profile



u-blox 6 modules **must not** be soldered with a damp heat process.

3.2.3 Optical inspection

After soldering the u-blox 6 module, consider an optical inspection step to check whether:

- The module is properly aligned and centered over the pads
- All pads are properly soldered
- No excess solder has created contacts to neighboring pads, or possibly to pad stacks and vias nearby.

3.2.4 Cleaning

In general, cleaning the populated modules is strongly discouraged. Residues underneath the modules cannot be easily removed with a washing process.

- Cleaning with water will lead to capillary effects where water is absorbed in the gap between the baseboard and the module. The combination of residues of soldering flux and encapsulated water leads to short circuits or resistor-like interconnections between neighboring pads.
- Cleaning with alcohol or other organic solvents can result in soldering flux residues flooding into the two housings, areas that are not accessible for post-wash inspections. The solvent will also damage the sticker and the ink-jet printed text.
- Ultrasonic cleaning will permanently damage the module, in particular the quartz oscillators.

The best approach is to use a "no clean" soldering paste and eliminate the cleaning step after the soldering.

3.2.5 Repeated reflow soldering

Only single reflow soldering processes are recommended for boards populated with u-blox 6 modules. u-blox 6 modules should not be submitted to two reflow cycles on a board populated with components on both sides in order to avoid upside down orientation during the second reflow cycle. In this case the module should always be placed on that side of the board which is submitted into the last reflow cycle. The reason for this (besides others) is the risk of the module falling off due to the significantly higher weight in relation to other components.

Two reflow cycles can be considered by excluding the above described upside down scenario and taking into account the rework conditions described in Section 3.2.8.



Repeated reflow soldering processes and soldering the module upside down are not recommended.

3.2.6 Wave soldering

Base boards with combined through-hole technology (THT) components and surface-mount technology (SMT) devices require wave soldering to solder the THT components. Only a single wave soldering process is encouraged for boards populated with u-blox 6 modules.

3.2.7 Hand soldering

Hand soldering is allowed. Use a soldering iron temperature setting equivalent to 350°C and carry out the hand soldering according to the IPC recommendations / reference documents IPC7711. Place the module precisely on the pads. Start with a cross-diagonal fixture soldering (e.g. pins 1 and 15), and then continue from left to right.

3.2.8 Rework

The u-blox 6 module can be unsoldered from the baseboard using a hot air gun. When using a hot air gun for unsoldering the module, max 1 reflow cycle is allowed. In general we do not recommend using a hot air gun because this is an uncontrolled process and might damage the module.



Attention: use of a hot air gun can lead to overheating and severely damage the module. Always avoid overheating the module.

After the module is removed, clean the pads before placing and hand-soldering a new module.



Never attempt a rework on the module itself, e.g. replacing individual components. Such actions immediately terminate the warranty.

In addition to the two reflow cycles manual rework on particular pins by using a soldering iron is allowed. For hand soldering the recommendations in IPC 7711 should be followed. Manual rework steps on the module can be done several times.

3.2.9 Conformal coating

Certain applications employ a conformal coating of the PCB using HumiSeal® or other related coating products. These materials affect the HF properties of the GPS module and it is important to prevent them from flowing into the module. The RF shields do not provide 100% protection for the module from coating liquids with low viscosity; therefore care is required in applying the coating.



Conformal Coating of the module will void the warranty.

3.2.10 Casting

If casting is required, use viscose or another type of silicon pottant. The OEM is strongly advised to qualify such processes in combination with the u-blox 6 module before implementing this in the production.



Casting will void the warranty.

3.2.11 Grounding metal covers

Attempts to improve grounding by soldering ground cables, wick or other forms of metal strips directly onto the EMI covers is done at the customer's own risk. The numerous ground pins should be sufficient to provide optimum immunity to interferences and noise.



u-blox makes no warranty for damages to the u-blox 6 module caused by soldering metal cables or any other forms of metal strips directly onto the EMI covers.

3.2.12 Use of ultrasonic processes

Some components on the u-blox 6 module are sensitive to Ultrasonic Waves. Use of any Ultrasonic Processes (cleaning, welding etc.) may cause damage to the GPS Receiver.



u-blox offers no warranty against damages to the u-blox 6 module caused by any Ultrasonic Processes.

3.3 EOS/ESD/EMI Precautions

When integrating GPS receivers into wireless systems, careful consideration must be given to electromagnetic and voltage susceptibility issues. Wireless systems include components which can produce Electrical Overstress (EOS) and Electro-Magnetic Interference (EMI). CMOS devices are more sensitive to such influences because their failure mechanism is defined by the applied voltage, whereas bipolar semiconductors are more susceptible to thermal overstress. The following design guidelines are provided to help in designing robust yet cost effective solutions.



To avoid overstress damage during production or in the field it is essential to observe strict EOS/ESD/EMI handling and protection measures.



To prevent overstress damage at the RF_IN of your receiver, never exceed the maximum input power (see Data Sheet).

3.3.1 Abbreviations

For a list of abbreviations used see Table 28 in Appendix A.

3.3.2 Electrostatic discharge (ESD)

Electrostatic discharge (ESD) is the sudden and momentary electric current that flows between two objects at different electrical potentials caused by direct contact or induced by an electrostatic field. The term is usually used in the electronics and other industries to describe momentary unwanted currents that may cause damage to electronic equipment.

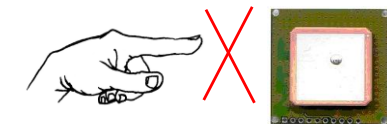
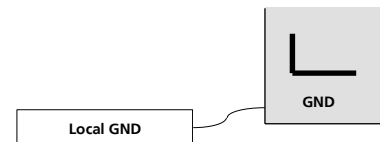


3.3.3 ESD handling precautions

ESD prevention is based on establishing an Electrostatic Protective Area (EPA). The EPA can be a small working station or a large manufacturing area. The main principle of an EPA is that there are no highly charging materials in the vicinity of ESD sensitive electronics, all conductive materials are grounded, workers are grounded, and charge build-up on ESD sensitive electronics is prevented. International standards are used to define typical EPA and can be obtained for example from International Electrotechnical Commission (IEC) or American National Standards Institute (ANSI).

GPS receivers are sensitive to ESD and require special precautions when handling. Particular care must be exercised when handling patch antennas, due to the risk of electrostatic charges. In addition to standard ESD safety practices, the following measures should be taken into account whenever handling the receiver.

- Unless there is a galvanic coupling between the local GND (i.e. the work table) and the PCB GND, the first point of contact when handling the PCB shall always be between the local GND and PCB GND.
- Before mounting an antenna patch, connect ground of the device.
- When handling the RF pin, do not come into contact with any charged capacitors and be careful when contacting materials that can develop charges (e.g. patch antenna ~10 pF, coax cable ~50-80 pF/m, soldering iron, ...)
- To prevent electrostatic discharge through the RF input, do not touch any exposed antenna area. If there is any risk that such exposed antenna area is touched in non ESD protected work area, implement proper ESD protection measures in the design.
- When soldering RF connectors and patch antennas to the receiver's RF pin, make sure to use an ESD safe soldering iron (tip).



Failure to observe these precautions can result in severe damage to the GPS receiver!

3.3.4 ESD protection measures



GPS receivers are sensitive to Electrostatic Discharge (ESD). Special precautions are required when handling.



For more robust designs, employ additional ESD protection measures. Using an LNA with appropriate ESD rating can provide enhanced GPS performance with passive antennas and increases ESD protection.

Most defects caused by ESD can be prevented by following strict ESD protection rules for production and handling. When implementing passive antenna patches or external antenna connection points, then additional ESD measures as shown in Figure 56 can also avoid failures in the field.

Small passive antennas (<2 dBic and performance critical)	Passive antennas (>2 dBic or performance sufficient)	Active Antennas
A LNA with appropriate ESD rating	B	C

Figure 56: ESD Precautions



Protection measure A is preferred because it offers the best GPS performance and best level of ESD protection.

3.3.5 Electrical Overstress (EOS)

Electrical Overstress (EOS) usually describes situations when the maximum input power exceeds the maximum specified ratings. EOS failure can happen if RF emitters are close to a GPS receiver or its antenna. EOS causes damage to the chip structures.

If the RF_IN is damaged by EOS, it's hard to determine whether the chip structures have been damaged by ESD or EOS.

3.3.6 EOS protection measures



For designs with GPS receivers and wireless (e.g. GSM/GPRS) transceivers in close proximity, ensure sufficient isolation between the wireless and GPS antennas. If wireless power output causes the specified maximum power input at the GPS RF_IN to be exceeded, employ EOS protection measures to prevent overstress damage.

For robustness, EOS protection measures as shown in Figure 57 are recommended for designs combining wireless communication transceivers (e.g. GSM, GPRS) and GPS in the same design or in close proximity.

See C26 telematics reference design [9].

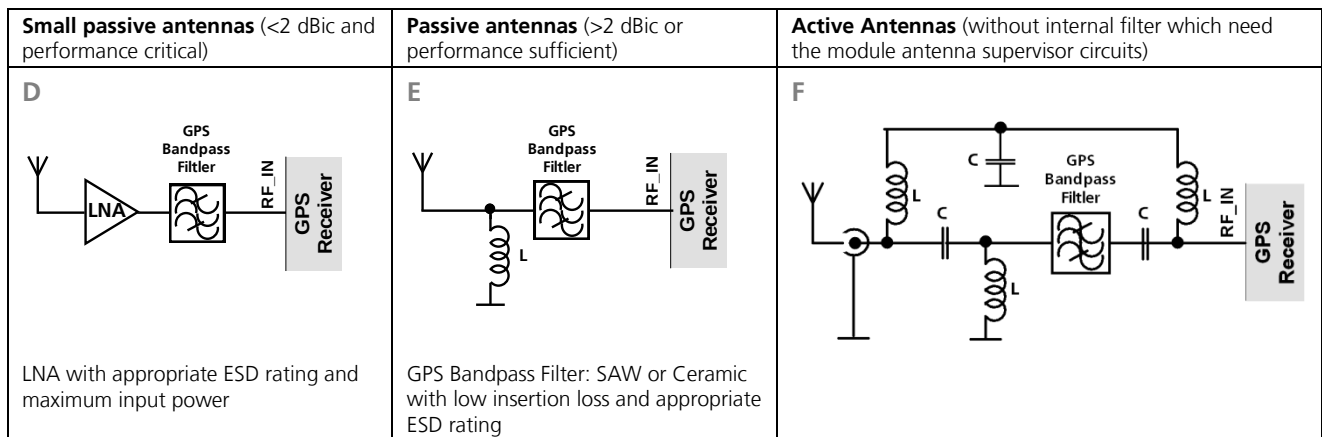


Figure 57: EOS and ESD Precautions

3.3.7 Electromagnetic interference (EMI)

Electromagnetic interference (EMI) is the addition or coupling of energy originating from any RF emitting device. This can cause a spontaneous reset of the GPS receiver or result in unstable performance. Any unshielded line or segment (>3mm) connected to the GPS receiver can effectively act as antenna and lead to EMI disturbances or damage.

The following elements are critical regarding EMI:

- Unshielded connectors (e.g. pin rows etc.)
- Weakly shielded lines on PCB (e.g. on top or bottom layer and especially at the border of a PCB)
- Weak GND concept (e.g. small and/or long ground line connections)

EMI protection measures are recommended when RF emitting devices are near the GPS receiver. To minimize the effect of EMI a robust grounding concept is essential. To achieve electromagnetic robustness follow the standard EMI suppression techniques.

<http://www.murata.com/products/emc/knowhow/index.html>

<http://www.murata.com/products/emc/knowhow/pdf/4to5e.pdf>

Improved EMI protection can be achieved by inserting a resistor (e.g. $R > 20 \Omega$) or better yet a ferrite bead (BLM15HD102SN1) or an inductor (LQG15HS47NJ02) into any unshielded PCB lines connected to the GPS receiver. Place the resistor as close as possible to the GPS receiver pin.

Example of EMI protection measures on the RX/TX line using a ferrite bead:

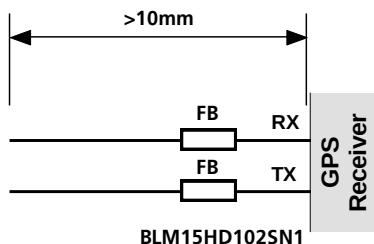


Figure 58: EMI Precautions

VCC can be protected using a feed thru capacitor. For electromagnetic compatibility (EMC) of the RF_IN pin refer to section 3.3.6

3.3.8 Applications with wireless modules LEON / LISA

GSM uses power levels up to 2 W (+33 dBm). Consult the Data Sheet for the absolute maximum power input at the GPS receiver.

3.3.8.1 Isolation between GPS and GSM antenna

In a handheld type design an isolation of approximately 20dB can be reached with careful placement of the antennas. If such isolation can't be achieved, e.g. in the case of an integrated GSM/GPS antenna, an additional input filter is needed on the GPS side to block the high energy emitted by the GSM transmitter. Examples of these kinds of filters would be the SAW Filters from Epcos (B9444 or B7839) or Murata.

3.3.8.2 Increasing jamming immunity

Jamming signals come from in-band and out-band frequency sources.

3.3.8.3 In-band jamming

With in-band jamming the signal frequency is very close to the GPS frequency of 1575 MHz (see Figure 59). Such jamming signals are typically caused by harmonics from displays, micro-controller, bus systems, etc.

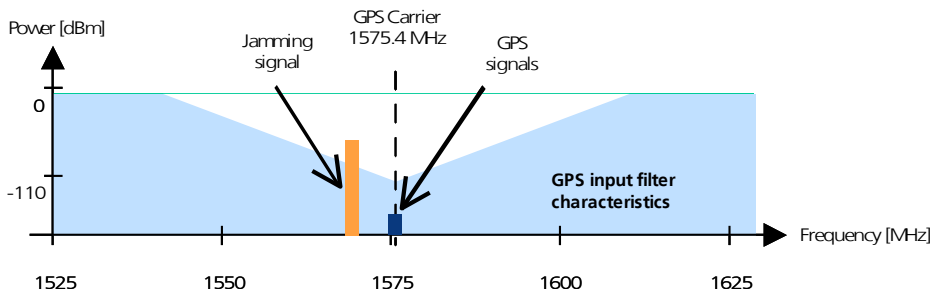


Figure 59: In-band jamming signals

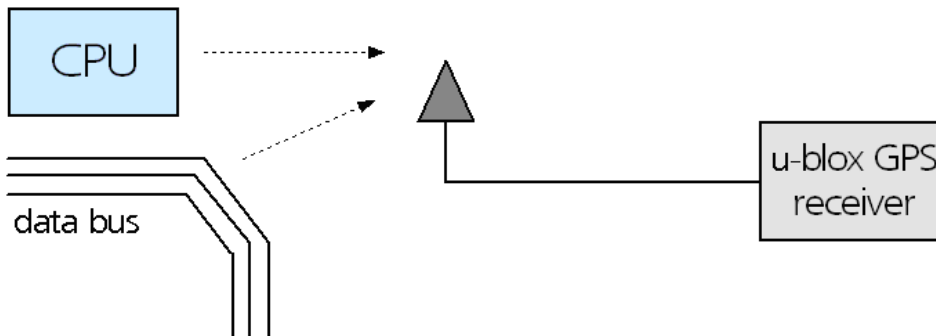


Figure 60: In-band jamming sources

Measures against in-band jamming include:

- Maintaining a good grounding concept in the design
- Shielding
- Layout optimization
- Filtering
- Placement of the GPS antenna
- Adding a CDMA, GSM, WCDMA bandpass filter before handset antenna

3.3.8.4 Out-band jamming

Out-band jamming is caused by signal frequencies that are different from the GPS carrier (see Figure 61). The main sources are wireless communication systems such as GSM, CDMA, WCDMA, WiFi, BT, etc.

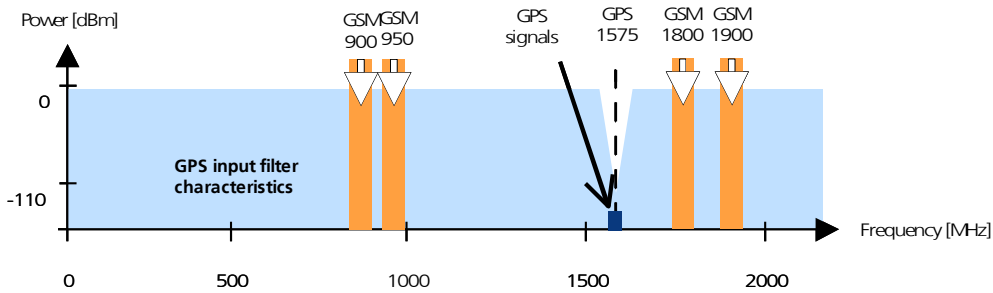


Figure 61: Out-band jamming signals

Measures against out-band jamming include maintaining a good grounding concept in the design and adding a SAW or bandpass ceramic filter (as recommend in Section 3.3.6) into the antenna input line to the GPS receiver (see Figure 62).



Figure 62: Measures against in-band jamming

3.3.8.5 GPS and GSM solution with integrated SMT antennas and chip SIM

An example is available on our C16 telematics reference design [9], that combines LEON-G200 GSM/GPRS modem module with NEO-6Q GPS receiver module.

3.3.9 Recommended parts

	Manufacturer	Part ID	Remarks	Parameters to consider
Diode ON Semiconductor		ESD9R3.3ST5G	(3.3.4 C) Standoff Voltage>3.3 V	• Low Capacitance < 0.5 pF
		ESD9L3.3ST5G	(3.3.4 C) Standoff Voltage>3.3 V	• Standoff Voltage > Voltage for active antenna
		ESD9L5.0ST5G	(3.3.4 C) Standoff Voltage>5 V	• Low Inductance
SAW	Epcos	B9444: B39162-B9444-M410	(3.3.6) 15dBm Max Power Input	
		B9416: B39162-B9416-K610	(3.3.6) Low insertion loss	
	Murata	SAFEA1G57KD0F00	(3.3.6) 1.35x1.05x0.5 mm	
		SAFZE1G57KA0F90	(3.3.6) 2.5x2.0x1.0 mm	
		SAFEB1G57KB0F00	(3.3.6) 1.35x1.05x0.6 mm	
		SAFEA1G57KE0F00	(3.3.6) 1.35x1.05x0.45 mm	• Good wireless band suppression
		SAFFB1G58KA0F0A	GPS and GLONASS	• High attenuation
		SAFEA1G58KA0F00	GPS and GLONASS	• High attenuation
	CTS	CER0032A	(3.3.6) 4.2x4.0x2.0 mm > 8kV ESD HBM	
LNA	Avago	ALM-1106	(3.3.4 A) LNA	pHEMT (GaAs)
		ALM-1412	(3.3.6 D) LNA + FBAR Filter	
		ALM-1712	(3.3.6 D) Filter + LNA + FBAR Filter	
		ALM-2412	(3.3.4 A) LNA + FBAR Filter	
	MAXIM	MAX2659ELT+	(3.3.4 A) LNA	SiGe
	JRC	NJG1143UA2	LNA	
	Infineon	BGM1032N16	Filter + LNA	
		BGM981N11	Filter + LNA + Filter	
		BGM1052N16	LNA + Filter	
	Triquint	TQM640002	Filter + LNA + Filter	
Inductor	Murata	LQG15HS27NJ02	(3.3.6 F) L, 27 nH	Impedance @ freq GPS > 500 Ω
Capacitor	Murata	GRM1555C1E470JZ01	(3.3.6 F) C, 47 pF	
Ferrite Bead	Murata	BLM15HD102SN1	(3.3.7) FB	High IZI @ fGSM
Feed thru Capacitor for Signal	Murata	NFL18SP157X1A3	Monolithic Type	Load Capacitance appropriate to Baud rate
		NFA18SL307V1A45	Array Type	CL < xxx pF
Feed thru Capacitor for VCC	Murata	NFM18PC	0603 2A	Rs < 0.5 Ω
		NFM21P....	0805 4A	

Table 26: Recommended parts for ESD/EOS protection

3.3.9.1 Recommended GPS & GLONASS active antenna (A1)

Manufacturer	Order No.	Comments
Inpaq (www.inpaq.com.tw)	GPSGLONASS03D-S3-00-A	3 V 6 mA
Taoglas (www.taoglas.com)	AA.160.301111	36*36*4mm, 3-5V 30mA

Table 27: Recommend GPS & GLONASS active antenna (A1)

4 Product testing

4.1 u-blox in-series production test

u-blox focuses on high quality for its products. To achieve a high standard it's our philosophy to supply fully tested units. Therefore at the end of the production process, every unit is tested. Defective units are analyzed in detail to improve the production quality.

This is achieved with automatic test equipment, which delivers a detailed test report for each unit. The following measurements are done:

- Digital self-test (Software Download, verification of FLASH firmware, etc.)
- Measurement of voltages and currents
- Measurement of RF characteristics (e.g. C/No)

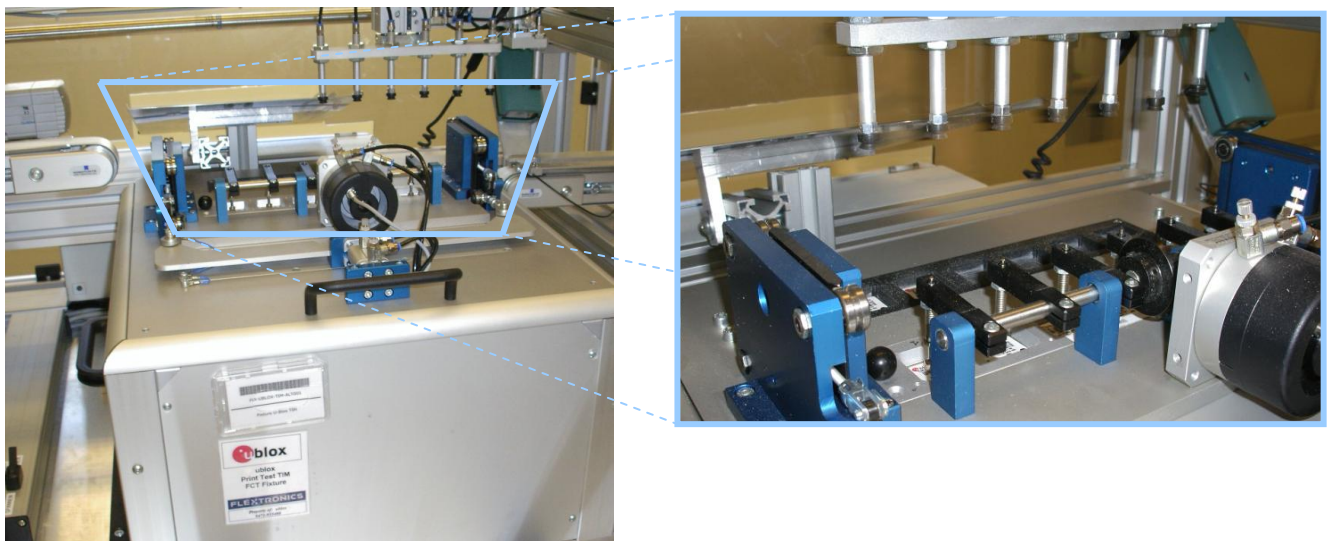


Figure 63: Automatic Test Equipment for Module Tests

4.2 Test parameters for OEM manufacturer

Because of the testing done by u-blox (with 100% coverage), it is obvious that an OEM manufacturer doesn't need to repeat firmware tests or measurements of the GPS parameters/characteristics (e.g. TTFF) in their production test.

An OEM manufacturer should focus on:

- Overall sensitivity of the device (including antenna, if applicable)
- Communication to a host controller

4.3 System sensitivity test

The best way to test the sensitivity of a GPS device is with the use of a 1-channel GPS simulator. It assures reliable and constant signals at every measurement.



Figure 64: 1-channel GPS simulator

u-blox recommends the following Single-Channel GPS Simulator:

- Spirent GSS6100
Spirent Communications Positioning Technology
(previously GSS Global Simulation Systems)

www.positioningtechnology.co.uk

4.3.1 Guidelines for sensitivity tests

1. Connect a 1-channel GPS simulator to the OEM product
2. Choose the power level in a way that the "Golden Device" would report a C/No ratio of 38-40 dBHz
3. Power up the DUT (Device Under Test) and allow enough time for the acquisition
4. Read the C/No value from the NMEA GSV or the UBX-NAV-SVINFORM message (e.g. with u-center)
5. Compare the results to a "Golden Device" or a u-blox 6 Evaluation Kit.

4.3.2 'Go/No go' tests for integrated devices

The best test is to bring the device to an outdoor position **with excellent sky view** (HDOP < 3.0). Let the receiver acquire satellites and compare the signal strength with a "Golden Device".



As the electro-magnetic field of a redistribution antenna is not homogenous, indoor tests are in most cases not reliable. These kind of tests may be useful as a 'go/no go' test but not for sensitivity measurements.

4.3.3 Testing LEA-6R designs



When testing the design ensure that no GPS signals are being received or delete the calibration after the tests. Failure to do so can result in operation errors.

4.3.3.1 Direction signal

This input shall be set once to high level and once to low level. In both states the software parameters are read back with the UBX-NAV-EKFSTATUS. The direction flag shall read forward for a high level at the FWD input and backward for a low level at the FORWARD input.

4.3.3.2 Speedpulse signal

A rectangular waveform with 2 kHz frequency shall be fed into the SPEED input. The result can be read back with the UBX-NAV-EKFSTATUS message: Speed Ticks: 1800...2400

4.3.3.3 Gyroscope (rate) input

Do not move the device and check UBX-ESF-MEAS: 2 > Gyro Z: > -2

Quickly turn the device to the right (clockwise), check UBX-ESF-MEAS: Gyro Z: > 50

Quickly turn the device to the left (counterclockwise), check UBX-ESF-MEAS: Gyro Z: < -50



The rate input can only be tested if an A/D converter is connected to LEA-6R.

4.3.3.4 Temperature sensor

The temperature measured by the temperature sensor connected to the LEA-6R shall be read with the UBX-ESF-MEAS message. The measurement tolerance is in the order of about $\pm 5^\circ$.

4.3.3.5 Erase calibration

To erase the calibration send a CFG-EKF command with the appropriate clearing flags set.

4.3.4 Testing NEO-6V designs

The NEO-6V ADR algorithm supports a variety of sensors (such as wheel ticks and gyroscope) and receives the sensor data via UBX messages from the application processor. Digital sensor data is available on the vehicle bus. No extra sensors are required for Dead Reckoning functionality. ADR is completely self-calibrating.

For more details on GWT protocol see u-blox 6 Receiver Description Including Protocol Specification [10]

For more details on DWT protocol contact u-blox. (Contact)

Appendix

A Abbreviations

Abbreviation	Definition
ANSI	American National Standards Institute
CDMA	Code Division Multiple Access
EMC	Electromagnetic compatibility
EMI	Electromagnetic interference
EOS	Electrical Overstress
EPA	Electrostatic Protective Area
ESD	Electrostatic discharge
GND	Ground
GPS	Global Positioning System
GSM	Global System for Mobile Communications
IEC	International Electrotechnical Commission
PCB	Printed circuit board

Table 28: Explanation of abbreviations used

B Migration to u-blox 6 receivers

Migrating ANTARIS®4 and u-blox 5 designs to a u-blox 6 receiver module is a fairly straightforward procedure. Nevertheless there are some points to be considered during the migration.



Not all of the functionalities available with ANTARIS®4 are supported by u-blox 6. These include:

- RTCM is supported in FW7.0x but not in ROM6.02 and FW6.02 versions.
- UTM (Universal Transverse Mercator Projection)

B.1 Checklist for migration

Have you chosen the optimal module?

- ☐ For best GPS performance (i.e. better sensitivity level and acquisition time) select a LEA-6H, LEA-6S, NEO-6Q or NEO-6G for the advantage of TCXO performance.
- ☐ If TCXO performance is not required, choose a LEA-6A or NEO-6M.
- ☐ For active antenna applications, choose a LEA-6H, LEA-6S or LEA-6A, since an antenna supply circuit is already built in or see chapter 1.4 and chapter 2.6.
- ☐ For the ability to upgrade the firmware, choose a LEA-6H.
- ☐ For precision timing choose a LEA-6T or NEO-6T
- ☐ For dead reckoning choose a LEA-6R

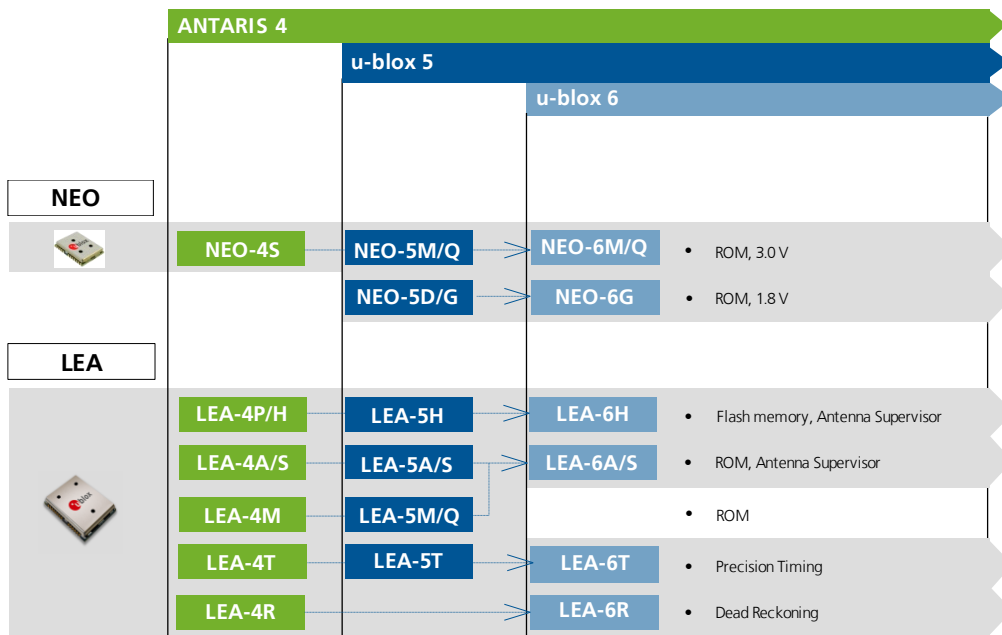


Figure 65: u-blox 6 module migration made easy

Check u-blox 6 Hardware Requirements:

- ☐ Check the battery power to supply the battery backup pin, since u-blox 6 draws higher current in comparison to ANTARIS 4 receivers.
- ☐ Compare the u-blox 6 module peak current consumption (~70 mA) with the specification of the power supply.
- ☐ u-blox 6 modules can be operated in three different modes: Max. Performance mode, Eco mode or Power Save mode.
- ☐ NEO-6Q, NEO-6G and NEO-6M feature a Configuration Pin that allows switching between the power modes: Max Performance mode and Eco mode.
- ☐ For more information on u-blox6 Power supply specifications and power modes, see the LEA-6 Data Sheet [1] and NEO-6 Data Sheet [2].
- ☐ If you use an active antenna supervisor circuitry to detect open conditions, you need to verify resistor reference recommendations in our integration manuals.
- ☐ See chapter 3.3 EOS/ESD/EMI Precautions.
- ☐ If you use the USB interface, the external series resistor values in USB_DM and USB_DP line should be adjusted, see 1.6.2.

Check u-blox 6 Software Requirements:

- ☐ Not all of the functionalities available with ANTARIS 4 are supported by u-blox 6 Firmware version 6.02. These include:
 - o FixNow Mode: Low power modes are supported via mapping to the Power Save mode of FW 7.0x or ROM 6.02. For migration of FXN functionalities consult the u-blox 6 Firmware Version 7.0x Release Note [5], respectively the u-blox 6 Receiver Description including Protocol Specification [3]
 - o No UTM (Universal Transverse Mercator Projection).
 - o No RTCM protocol for DGPS support (ROM6.02, FW6.02).
 - o Raw Data support with LEA-6T only.
- ☐ Check B.2 Software migration

B.2 Software migration

B.2.1 Software migration from ANTARIS 4 or u-blox 5 to a u-blox 6 GPS receiver

Software migration from ANTARIS 4 or u-blox 5 to a u-blox 6 GPS receiver is a straightforward procedure. Nevertheless there are some differences to be considered with u-blox 5 firmware version 5.00. Like its ANTARIS 4 and u-blox 5 predecessors, u-blox 6 technology supports UBX and NMEA protocol messages. Backward compatibility has been maintained as far as possible. New messages have been introduced for new functions. Only minor differences have to be expected in the UBX-NAV and UBX-AID classes of the UBX protocol and for the standard NMEA messages such as **GGA**, **GLL**, **GSA**, **GSV**, **RMC**, **VTG** and **ZDA**.

ANTARIS 4	u-blox6	Remarks
UBX-CFG-NAV2	UBX-CFG-NAV5	UBX-CFG-NAV2 has been replaced by UBX-CFG-NAV5. The new message has additional features. The default dynamic platform is "Portable". This platform is rather generic and allows the receiver to be operated in a wide dynamic range covering pedestrians, cars as well as commercial aircrafts. Automotive applications such as first-mount navigation systems may better utilize the "Automotive" platform, which is better geared to the dynamics of land vehicles but is only of limited use in airborne and high-dynamics environments. UBX-CFG-NAV5 does not support following features: - Almanac Navigation - Navigation Input filters UBX-CFG-NAV5 has a message length of 36 Bytes (40 Bytes for UBX-CFG-NAV2) UBX-CFG-NAV5 FixMode is set by default to "Auto 3D/2D" as for ANTARIS4. Check the u-blox 6 Receiver Description including Protocol Specification [3]. if this mode needs to be changed.
UBX-CFG-MSG	UBX-CFG-MSG	No support for multiple configurations in one UBX-CFG-MSG command
UBX-CFG-RXM	N/A	Contrary to ANTARIS 4, u-blox6 does not need selecting GPS acquisition sensitivity mode (Fast, Normal, High Sens and Auto mode) since the acquisition engine is powerful enough to search all satellite in one go. FixNow mode is not available anymore. Contact your local u-blox support team should you need further information.
PUBX,01	N/A	Other UBX or NMEA messages can be used to replace this message
UBX-NAV-POSUTM	N/A	
UBX-CFG-TP	UBX-CFG-TP	u-blox 6 maintains this message for backwards compatibility only. For new designs use UBX-CFG-TP5.
	UBX-CFG-TP5	This is a new u-blox 6 message, for information see u-blox 6 Receiver Description including Protocol Specification [3].
UBX-CFG-ANT	UBX-CFG-ANT	Antenna Open Circuit Detection: The default setting for LEA-4S and LEA-4A was "enabled". With all LEA-6 modules the default setting is "disabled". Automatic Short Circuit Recovery: With ANTARIS 4 this was "disabled" by default. With u-blox 6 the default setting is "enabled".
UBX-CFG-RATE	UBX-CFG-RATE	Set to 1 with u-blox 6
UBX-CFG-TMODE	UBX-CFG-TMODE	With u-blox 6 FW 6.02 and above it is no longer necessary to configure the number of satellites in UBX-CFG-NAV to 1 to enable the timing mode. This is performed automatically.
UBX-MON-HW	UBX-MON-HW	Message length has changed as the number of pins is different with u-blox6.
0s Leap second by default	FW 6.02 and FW7.0x: 15 s Leap second by default	
UBX-CFG-RATE	UBX-CFG-RATE	Disable SBAS services to achieve 4Hz navigation
UBX-NAV-EKFSTATUS	UBX-NAV-EKFSTATUS	This message is only provided for backwards compatibility and should not be utilized for future designs. Instead, the messages ESF-STATUS and ESF-MEAS should be used. For u-blox 6 firmware the gyroscope value (gyroMean) is only output if the gyroscope is used in the navigation solution. This message is only available on LEA-4R and LEA-6R GPS Receivers.

Table 29: Main differences between ANTARIS 4 and u-blox 6 software for migration

The default NMEA message set for u-blox 6 is **GGA, GLL, GSA, GSV, RMC** and **VTG**. Contrary to ANTARIS 4, **ZDA** is disabled by default.

Firmware update is supported by all of these interfaces. The firmware update mechanism of u-blox 6 is more sophisticated than with ANTARIS 4. It is now based on UBX protocol messages. Customers, who implemented firmware download in their application processor, will need to replace the software. A template is available from your u-blox support team.

In case migrating from LEA-4T or LEA-5T to LEA-6T-0, the command to save the configuration (UBX-CGF-CFG) changes. This is because in LEA-6T-0 a serial Flash at the SPI is used instead of the parallel Flash (LEA-4T and LEA-5T). So for the LEA-4T, LEA-5T and LEA-6T-1 the target to save the configuration has to be set to "devFlash", but for the LEA-6T-0 it has to be set to "devSpiFlash".

Please refer to the u-blox 6 Receiver Description including Protocol Specification [3] for more information. This document is available on the [u-blox website](#).

B.2.2 Software migration from 6.02 to 7.03

Timing Survey-in Mode:

Customers currently using survey-in need to review their accuracy limit setting with CFG-TMODE or CFG-TMODE2.

For migration from 6.02 to 7.03 consult the u-blox 6 Firmware Version 7.0x Release Note [5], respectively the u-blox 6 Receiver Description including Protocol Specification [3]

B.3 Hardware Migration

B.3.1 Hardware Migration: ANTARIS 4 → u-blox 6

u-blox 6 modules have been designed with backward compatibility in mind but some minor differences were unavoidable. These minor differences will however not be relevant for the majority of the ANTARIS 4 designs.

Good performance requires a clean and stable power supply with minimal ripple. Care needs to be exercised in selecting a strategy to achieve this. Avoid placing any resistance on the Vcc line. For better performance, use an LDO to provide a clean supply at Vcc and consider the following:

- Special attention needs to be paid to the power supply requirements. (currents & backup current see datasheet for further details)
- Wide power lines or even power planes are preferred.
- Place LDO near the module.
- Avoid resistive components in the power line (e.g. narrow power lines, coils, resistors, etc.).



Placing a filter or other source of resistance at Vcc can create significantly longer acquisition times.

B.3.2 Hardware Migration: u-blox 5 → u-blox 6

Check the pins RxD1 and EXTINT0 regarding the input voltage threshold.

Serial termination resistors: Recommendation has changed from 27 Ω to 22 Ω . See chapter 1.6.2.1.

For more information see the LEA-6 Data Sheet [1] and LEA-5 Data Sheet [7].

B.4 Migration of LEA modules

B.4.1 Migration from LEA-4 to LEA-6

See also the migration Table in the u-blox5 Hardware Integration Manual.

For u-blox6 the Input Voltage thresholds on the pins RXD1 and EXTINT0 have changed.

The Safeboot functionality is inverted compared to Antaris receivers.

VCC_OUT is now VCC and not 1.8 V as on Antaris Modules.

Also check your power supply requirements with the datasheet (for VCC and VBCKP).

Pin	LEA-4H/LEA-4P/LEA-4T		LEA-6H/LEA-6T		Remarks for Migration
	Pin Name	Typical Assignment	Pin Name	Typical Assignment	
1	Reserved	VDDIO level I/O; not connected	SDA2	NC	
2	Reserved	VDDIO level I/O; not connected	SCL2	NC	
3	TXD1	VDDIO level I/O	TxD1	Output	
4	RXD1	VDDIO level I/O	RxD1	Input	Leave open if not used.
5	VDDIO	1.65 – 3.60 V	NC	Connect to VCC	Can be left open, but connection to VCC is recommended for compatibility reason. With LEA-6H the I/O voltage is always VCC.
6	VCC	2.70 – 3.30 V	VCC	2.70 – 3.60 V	Extended power supply range, higher peak supply current.
7	GND	GND	GND	GND	No difference
8	VDD18OUT	NC	VCC_OUT	NC	Internally connected to VCC, if you have circuitry connected to this pin, check if it withstands the VCC voltage.
9	Reserved	NC	Reserved	NC	
10	RESET_N	1.8 V	RESET_N	NC	Input only, do not drive high. Internal pull up to VCC.
11	V_BAT	1.50 – 3.6 V	V_BCKP	1.4 – 3.6 V	Wider voltage range but needs more current. Check your backup supply, regarding the higher consumption.
12	BOOT_INT	NC	Reserved	NC	Do not drive low.
13	GND	GND	GND	GND	No difference
14	GND	GND	GND	GND	No difference
15	GND	GND	GND	GND	No difference
16	RF_IN	RF_IN	RF_IN	RF_IN	No difference
17	GND	GND	GND	GND	No difference
18	VCC_RF	VCC - 0.1 V	VCC_RF	VCC - 0.1 V	No difference
19	V_ANT	3.0 V – 5.0 V	V_ANT	2.7 V – 5.5 V	wider range
20	AADET_N	NC	AADET_N	NC	check resistor R5 in Active antenna supervisor Figure 47
21	EXTINT1	NC	NC	NC	
22	Reserved	NC	NC	NC	
23	Reserved	NC	NC	NC	
24	VDDUSB	Connected to GND or VDD_USB	VDDUSB	Connected to GND or VDD_USB	Do not leave open. (VDD_USB is 3.3V regulated power supply from VBUS.)
25	USB_DM	NC	USB_DM	NC	New serial termination resistors recommended: 22 Ω
26	USB_DP	NC	USB_DP	NC	
27	EXTINT0	NC	EXTINT0	NC	
28	TIMEPULSE	VDDIO level I/O	TIMEPULSE	Output	

Table 30: Pin-out comparison LEA-4H/LEA-4P/LEA-4T vs. LEA-6H/LEA-6T

Pin	LEA-4A/LEA-4S		LEA-6A/LEA-6S		Remarks for Migration
	Pin Name	Typical Assignment	Pin Name	Typical Assignment	
1	TxD2	3.0 V out	SDA2	NC	
2	RxD2	1.8 - 5.0 V	SCL2	NC	
3	TxD1	3.0 V out	TxD1	Output	
4	RxD1	1.8 - 5.0 V in	RxD1	Input	Leave open if not used. Max. 5 V
5	VDDIO	VCC	NC	Connect to VCC	Leave open for only LEA-6x design. Connect to VCC for backward compatibility to LEA-5x.
6	VCC	2.70 – 3.30 V	VCC	2.70 – 3.60 V	Extended power supply range, higher peak supply current.
7	GND	GND	GND	GND	No difference
8	VDD18OUT	1.8 V out	VCC_OUT	NC	Internally connected to VCC, if you have circuitry connected to this pin, check if it withstands the VCC voltage.
9	GPSPMODE6	NC (GND or VDD18OUT)	CFG_COM1	NC	
10	RESET_N	ACTIVE LOW	RESET_N	NC	Input only, do not drive high. Internal pull up to VCC.
11	V_BAT	1.50 – 3.6 V	V_BCKP	1.4 – 3.6 V	Wider voltage range but needs more current. Check your backup supply, regarding the higher consumption.
12	BOOT_INT	NC	Reserved	NC	Do not drive low.
13	GND	GND	GND	GND	No difference
14	GND	GND	GND	GND	No difference
15	GND	GND	GND	GND	No difference
16	RF_IN	RF_IN	RF_IN	RF_IN	No difference
17	GND	GND	GND	GND	No difference
18	VCC_RF	VCC - 0.1 V	VCC_RF	VCC - 0.1V	No difference
19	V_ANT	3.0 V - 5.0 V	V_ANT	2.7V -5.5V	wider range
20	AADET_N	NC (1.8 to 5.0 V)	AADET_N	NC	check resistor R5 in Active antenna supervisor Figure 47
21	GPSPMODE5	NC (GND or VDD18OUT)	NC	NC	
22	GPSPMODE2 GPSPMODE2 3	NC (GND or VDD18OUT)	NC	NC	
23	GPSPMODE7	NC (1.8 to 5.0 V)	NC	NC	
24	VDDUSB	3.0 –3.6V/ GND	VDDUSB	Connected to GND or VDD_USB	Do not leave open. (VDD_USB is 3.3 V regulated power supply from VBUS.)
25	USB_DM	VDDUSB I/O	USB_DM	NC	New serial termination resistors recommended: 22 Ω
26	USB_DP	VDDUSB I/O	USB_DP	NC	
27	EXTINT0	NC (1.8 to 5.0 V)	EXTINT0	NC	Max. 5 V
28	TIMEPULSE	VDDIO level output	TIMEPULSE	Output	

Table 31: Pin-out comparison LEA-4A/LEA-4S vs. LEA-6A/LEA-6S

B.4.2 Migration of LEA-4R designs to LEA-6R

LEA-6R module has been designed with backward compatibility in mind, but some incompatibilities were unavoidable. These minor differences will, however, not be relevant for the majority of ANTARIS 4 designs.

Please check in your design the following points carefully to assure a safe migration:

- ☐ For u-blox 6 the Input Voltage thresholds on the pins RxD1 and Extint0 have changed.
- ☐ VCC_OUT is now VCC and not 1.8 V as on Antaris Modules.
- ☐ Pin5 is now NC instead of VDDIO. All I/O Voltages are referenced to VCC.
- ☐ Check your power supply requirements with the datasheet (for VCC and VBCKP), u-blox 6 has a higher peak current and backup current than ANTARIS 4 modules

- ❑ The SPI is now running at 870 kHz (against 460 kHz on ANTARIS 4), check that your design supports the signal path
 - If you do a redesign, ensure the signal paths of the SPI to support a bandwidth of 4 MHz.

B.4.3 Migration from LEA-5 to LEA-6

For u-blox6 only the Input Voltage thresholds on the pins RXD1 and EXTINT0 have changed.

Be aware, that with u-blox 6 there is no LEA anymore, which supports SPI interface. For SPI consider NEO-6 form factor.

B.5 Migration of NEO modules

B.5.1 Migration from NEO-4S to NEO-6

For u-blox6 the Input Voltage thresholds on the pins RXD1 and EXTINT0 have changed.

The Safeboot functionality is inverted compared to Antaris receivers.

Also check your power supply requirements with the datasheet (for VCC and VBCKP).

Also check the setting of the configuration pins.

The pin-outs of NEO-4S and NEO-6M/NEO-6Q differ slightly. Table 32 compares the modules and highlights the differences to be considered.

Pin	NEO-4S		NEO-6Q/NEO-6M		Remarks for Migration
	Pin Name	Typ. Assignment	Pin Name	Typ. Assignment	
1	BOOT_INT	NC	Reserved	NC	Do not drive low.
2	SELECT	VDDIO level I/O; not connected	SS_N	NC	
3	TIMEPULSE	VDDIO level I/O	TIMEPULSE	Output	
4	EXTINT0	NC	EXTINT0	NC	
5	USB_DM	NC	USB_DM	NC	New serial termination resistors recommended: 22 Ohms
6	USB_DP	NC	USB_DP	NC	
7	VDDUSB	Connected to GND or VDD_USB	VDDUSB	Connected to GND or VDD_USB	Do not leave open. (VDD_USB is 3.3V regulated power supply from VBUS.)
8	Reserved	NC	Reserved	NC	Pins 8 and 9 must be connected.
9	VCC_RF	VCC-0.1 V	VCC_RF	VCC-0.1 V	No difference
10	GND	GND	GND	GND	No difference
11	RF_IN	RF_IN	RF_IN	RF_IN	No difference
12	GND	GND	GND	GND	No difference
13	GND	GND	GND	GND	No difference
14	MOSI	NC	MOSI/CFG_COM0	NC	The function of the CFG pin has changed. See Section 2.2.3.1 for more details.
15	MISO	NC	MISO/CFG_COM1	NC	
16	SCK/ CFG_USB	NC	CFG_GPS0 /SCK	NC	Leave open if not used. The function of the CFG pin has changed. See Section 2.2.3.1 for more details.
17	NCS	NC	Reserved	NC	No difference
18	Reserved	NC	SDA2	NC	
19	Reserved	NC	SCL2	NC	
20	TXD1	VDDIO level I/O	TxD1	Output	
21	RXD1	VDDIO level I/O	RxD1	Input	Leave open if not used.
22	V_BAT	1.5-3.6 V	V_BCKP	1.4-3.6 V	Wider voltage range but needs more current. Check your backup supply, regarding the higher consumption.
23	VCC	2.7-3.3 V	VCC	2.7-3.6 V	Higher peak supply current
24	GND	GND	GND	GND	No difference

Table 32: Pin-out comparison NEO-4S vs. NEO-6

B.5.2 Migration from NEO-5 to NEO-6

For u-blox 6 only the Input Voltage thresholds on the pins RXD1 and EXTINT0 have changed.

Also check the setting of the configuration pins.

Serial termination resistors: Recommendation has changed from 27 Ω to 22 Ω . See chapter 1.6.2.1.

C Interface Backgrounder

C.1 DDC Interface

Two wires, serial data (SDA) and serial clock (SCL), carry information between the devices connected to the bus. These lines are connected to all devices on the DDC. SCL is used to synchronize data transfers and SDA is the data line. Both SCL and SDA lines are "open drain" drivers. This means that DDC devices can only drive them low or leave them open. The pull-up resistor (R_p) pulls the line up to V_{DD} if no DDC device is pulling it down to GND. If the pull-up resistors are missing, the SCL and SDA lines are undefined and the DDC bus will not work. For most DDC systems the low and high input voltage level thresholds of SDA and SCL depend on V_{DD} . See the LEA-6 Data Sheet [1] or NEO-6 Data Sheet [2] for the applicable voltage levels.

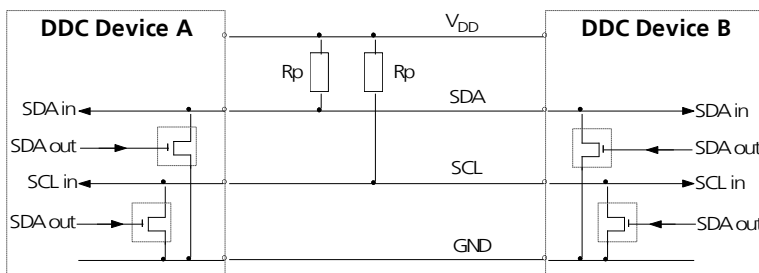


Figure 66: A simple DDC connection

The signal shape and the maximum rate in which data can be transferred over SDA and SCL is limited by the values of R_p and the wire and I/O capacitance (C_p). Long wires and a large number of devices on the bus increase C_p , therefore DDC connections should always be as short as possible. The resistance of the pull-up resistors and the capacitance of the wires should be carefully chosen.

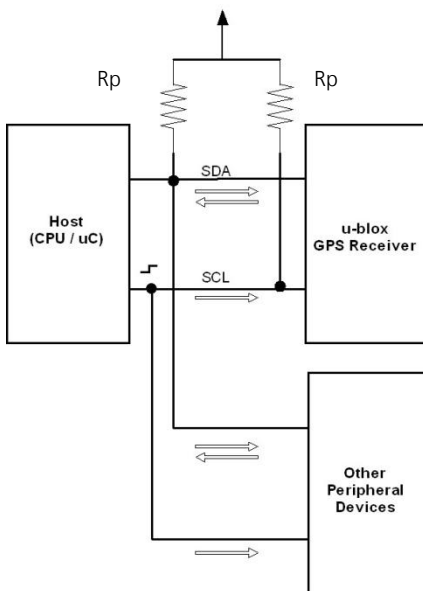


Figure 67: DDC block diagram

C.1.1 Addresses, roles and modes

Each device connected to a DDC is identified by a unique 7-bit address (e.g. whether it's a microcontroller, EEPROM or D/A Converter, etc.) and can operate as either a transmitter or receiver, depending on the function

of the device. The default DDC address for u-blox GPS receivers is set to 0x42. Setting the mode field in the CFG-PRT message for DDC accordingly can change this address.



The first byte sent is comprised of the address field and R/W bit. Hence the byte seen on the bus 0x42 is shifted by 1 to the left plus R/W bit thus being 0x84 or 0x85 if analyzed by scope or protocol analyzer.

In addition to transmitters and receivers, devices can also be considered as masters or slaves when performing data transfers. A master is the device which initiates a data transfer on the bus and generates the clock signals to permit that transfer. At that time, any device addressed is considered a slave. The DDC-bus is a multi-master bus, i.e. multiple devices are capable of controlling the bus. Such architecture is not permanent and depends on the direction of data transfer at any given point in time. A master device not only allocates the time slots when slaves can respond but also enables and synchronizes designated slaves to physically access the bus by driving the clock. Although multiple nodes can assume the role of a master, only one at any time is permitted to do so. Thus, when one node acts as master, all other nodes act as slaves. Table 33 shows the possible roles and modes for devices connected to a DDC bus.

	Transmit	Receive
Master: sends the clock and addresses slaves	Sends data to slave	Receives data from slave
Slave: receives the clock and address	Sends data to master	Receives data from master

Table 33: Possible roles and modes of devices connected to DDC bus

u-blox 6 GPS receivers normally run in the slave mode. There is an exception when an external EEPROM is attached. In that case, the receiver attempts to establish presence of such a non-volatile memory component by writing and reading from a specific location. If EEPROM is present (assumed to be located at a fixed address 0xA0), the receiver assumes the role of a master on the bus and never changes role to slave until the following start-up (subject to EEPROM presence). This process takes place only once at the start-up, i.e. the receiver's role cannot be changed during the normal operation afterward. This model is an exception and should not be implemented if there are other participants on the bus contending for the bus control (μ C / CPU, etc.).

As a slave on the bus, the u-blox 6 GPS receiver cannot initiate the data transfers. The master node has the exclusive right and responsibility to generate the data clock, therefore the slave nodes need not be configured to use the same baud rate. For the purpose of simplification, if not specified differently, SLAVE denotes the u-blox 6 GPS receiver while MASTER denotes the external device (CPU, μ C) controlling the DDC bus by driving the SCL line.



u-blox GPS receivers support Standard-Mode I²C-bus specification with 7-bit addressing and a data transfer rate up to 100 kBit/s and a SCL clock frequency up to 100 kHz.

C.1.2 DDC troubleshooting

Consider the following questions when implementing I²C in designs:

- Is there a stable supply voltage Vdd? Often, external I²C devices (like I²C masters or monitors) must be provided with Vdd.
- Are appropriate termination resistances attached between SDA, SCL and Vdd? The voltage level on SDA and SCL must be Vdd as long as the bus is idle and drop near GND if shorted to GND. [Note: Very few I²C masters exist which drive SCL high and low, i.e. the SCL line is not open-drain. In this case, a termination resistor is not needed and SCL cannot be pulled low. These masters will not work together with other masters (as they have no multi-master support) and may not be used with devices which stretch SCL during transfers.]
- Are SDA and SCL mixed up? This may accidentally happen e.g. when connecting I²C buses with cables or connectors.
- Do all I²C devices support the I²C supply voltage used on the bus?
- Do all I²C devices support the maximum SCL clock rate used on the bus?
- If more than one I²C master is connected to the bus: do all masters provide multi-master support?

- Are the high and low level voltages on SDA and SCL correct during I²C transfers? The I²C standard defines the low level threshold with 0.3 Vcc, the high level threshold with 0.7 Vcc. Modifying the termination resistance R_p, the serial resistors R_s or lowering the SCL clock rate could help here.
- Are there spikes or noise on SDA, SCL or even Vdd? They may result from interferences from other components or because the capacitances C_p and/or C_c are too high. The effects can often be reduced by using shorter interconnections.



For more information about DDC implementation refer to the u-blox 6 Receiver Description including Protocol Specification [3].

C.2 SPI Interface

C.2.1 SPI basics

Devices communicate in master/slave mode where the master device provides the clock signal (SCK) and determines the state of the chip select (SCS/SS_N) lines, i.e. it activates the slave it wants to communicate with. The slave device receives the clock and chip select from the master. Multiple slave devices are allowed with individual slave select (chip select) lines. This means that there is one master, while the number of slaves is only limited by the number of chip selects. In addition to reliability and relatively high speed (with respect to the conventional UART), the SPI interface is easy to use and requires no special handling or complex communication stack implementation in the software.

The standard configuration for a slave device (see Figure 68) uses two control and two data lines. These are identified as follows:

- SCS — Slave Chip Select (control: output from master, usually active low)
- SCK — Serial Clock (control: output from master)
- MOSI — Master Output, Slave Input (data: output from master)
- MISO — Master Input, Slave Output (data: output from slave)



Alternative naming conventions are also widely used. Confirm the pin/signal naming with specific components used.

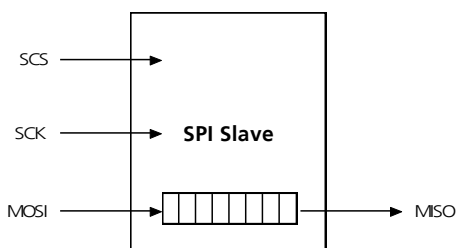


Figure 68: SPI slave

SPI always follows the basic principle of a shift register. During an SPI transfer, command codes and data values are simultaneously transmitted (shifted out serially) and received (shifted in serially). The data is entered into a shift register and then internally available for parallel processing. The length of the shift registers is not fixed, but can vary from device to device. Normally the shift registers are 8Bit or integral multiples thereof. However, they can also have an odd number of bits. For example two cascaded 9Bit EEPROMs can store 18Bit data.

When an SPI transfer occurs, an 8-bit character is shifted out one data pin while a different 8-bit character is simultaneously shifted in a second data pin. Another way to view this transfer is that an 8-bit shift register in the master and another 8-bit shift register in the slave are connected as a circular 16-bit shift register. When a transfer occurs, this distributed shift register is shifted eight bit positions; thus, the characters in the master and slave are effectively exchanged.

The serial clock (SCK) line synchronizes shifting and sampling of the information on the two serial data lines (MOSI and MISO). The chip select (SCS/SS_N) line allows individual selection of a slave SPI device. If an SPI slave

device is not selected (i.e. its chip select is not activated), its data output enters a high-impedance state (hi-Z) and does not interfere with SPI bus activities.

The data output MISO functions as the data return signal from the slave to the master.

Figure 69 shows a typical block diagram for an SPI master with several slaves. Here, the SCK and MOSI data lines are shared by all of the slaves. Also the MISO data lines are linked together and led back to the master. Only the chip selects are separately brought to each SPI device.

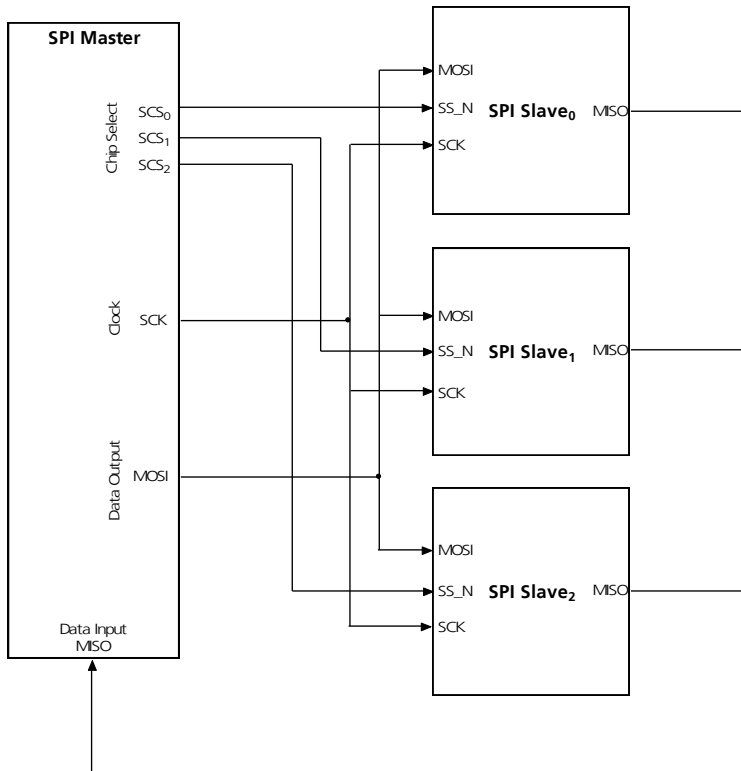


Figure 69: Master with independent slaves

SPI allows multiple microcontrollers to be linked together. These can be configured according to single or multiple master protocols. In the first variant the microcontroller(s) designated as slave(s) behave like a normal peripheral device. The second variant allows for several masters and allows each microprocessor the possibility to take the role of master and to address another microprocessor. In this case one microcontroller must permanently provide the clock signal.

There are two SPI system errors. The first occurs if several SPI devices want to become master at the same time. The other is a collision error that occurs for example when SPI devices work with different polarities.



Systems involving multiple microcontrollers are beyond the scope of this document.



Cascading slave peripherals is not supported.

Four I/O pin signals are associated with SPI transfers: the SCK, the MISO data line, the MOSI data line, and the active low SCS/SS_N pin. In the unselected state the MISO lines are hi-Z and therefore inactive. The master decides with which peripheral device it wants to communicate. The clock line SCK provides synchronization for data communication and is brought to the device whether or not it is selected.

The majority of SPI devices provide all four of these lines. Sometimes MOSI and MISO are multiplexed, or else one is missing. A peripheral device, which must not or cannot be configured, requires no input line but only a data output. As soon as it gets selected it starts sending data. In some ADCs therefore the MOSI line is missing. Some devices have no data output (e.g. LCD controllers which can be configured, but cannot send data or status messages).

The following rules should answer the most common questions concerning these signals:

- **SCK:** The SCK pin is an output when the SPI is configured as a master and an input when the SPI is configured as a slave. When the SPI is configured as a master, the SCK signal is derived from the internal bus clock. When the master initiates a transfer, eight clock cycles are automatically generated on the SCK pin. When the SPI is configured as a slave, the SCK pin is an input, and the clock signal from the master synchronizes the data transfer between the master and slave devices. Slave devices ignore the SCK signal unless the slave select pin is active low. In both the master and slave SPI devices, data is shifted on one edge of the SCK signal and is sampled on the opposite edge where data is stable. Edge polarity is determined by the SPI transfer protocol.
- **MISO/MOSI:** The MISO and MOSI data pins are used for transmitting and receiving serial data. When the SPI is configured as a master, MISO is the master data input line, and MOSI is the master data output line. When the SPI is configured as a slave, these pins reverse roles.
- **SCS/SS_N:** In master mode, the SCS output(s) select external slaves (e.g. SCS1_N, SCS2_N). In slave mode, SS_N is the slave select input. The chip select pin behaves differently on master and slave devices. On a slave device, this pin is used to enable the SPI slave for a transfer. If the SS_N pin of a slave is inactive (high), the device ignores SCK clocks and keeps the MISO output pin in the high-impedance state. On a master device, the SCS pin can serve as a general-purpose output not affecting the SPI.

D DR calibration

D.1 Constraints

The calibration of the DR sensors is a transparent and continuously ongoing process during periods of good GPS reception:

- Gyroscope Bias Voltage level of the gyroscope while driving a straight route or not moving
- Gyroscope Scale Factor Adjusted while in left and right turns; gyro sensitivity
- Speed Pulse Scale Factor Used to calibrate odometer pulse frequency to GPS speed over ground
- Temperature Compensation The gyroscope is a temperature-dependent device that requires temperature compensation

When a new GPS receiver is installed in a vehicle, the accuracy is only **moderately good** until sufficient **calibration** data has been collected, e.g. during a first drive. With time, continuous calibration results in continuous improvement of dead reckoning accuracy. Small discontinuities, like deviating wheel diameters after exchanging tires (summer vs. snow tires) or aging of the sensors, will be balanced out by ongoing automatic calibration.

Calibration parameters must be **reset**, if

- a DR module is transferred to a **different vehicle** and/or a **different gyroscope** is connected
- the sensor integrity check has reported any failure from the sensors and set itself into **GPS only mode**

Calibration can be reset with UBX message UBX – CFG (Config) – EKF (Extended Kalman Filter).

D.2 Initial calibration drive

For **optimum navigation performance** the system needs some learning time and distance for calibrating the various sensors inputs. The following driving directions are recommended to achieve an efficient calibration so dead reckoning yields high accuracy after the shortest possible period of time.

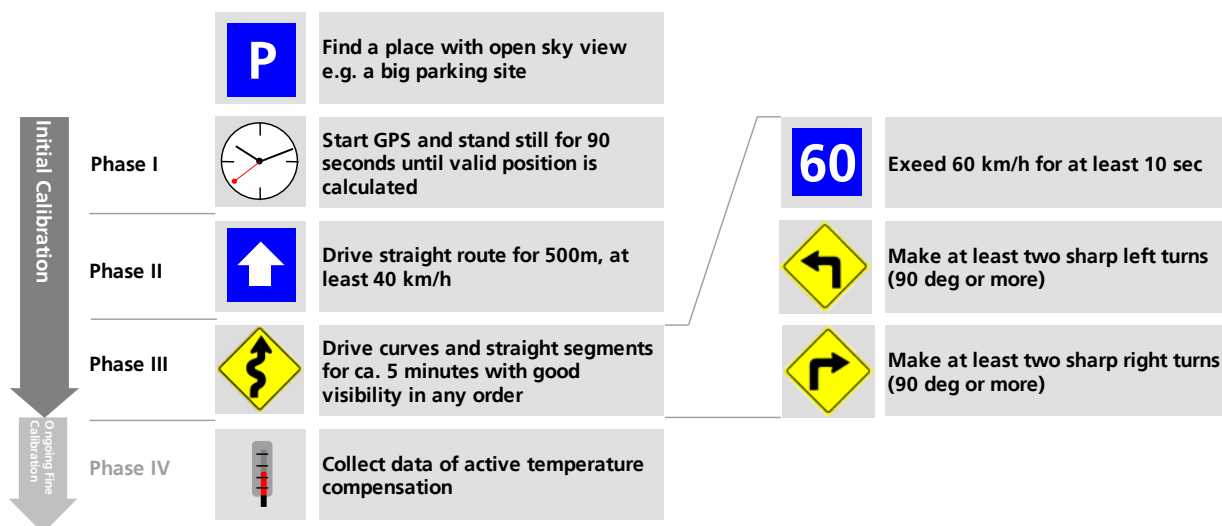


Figure 70: Initial DR Calibration Drive

The mentioned distances and durations are typical values, a better indication are the sensor calibration status given in UBX – ESF (External Sensor Fusion) – STATUS (Status). The status values indicate which phase of the initial calibration the receiver is in (calibrating, coarse calibration, fine calibration). In **Phase IV** good DR

performance can already be expected, as all sensors are calibrated. Still further fine calibration will be ongoing with good GPS reception and, if a temperature sensor is available, the temperature compensation table will be filled. Gyro offset values are measured as soon as the car stops for more than 3 seconds and are stored in the temperature compensation table to further increase the performance.



The above instructions result in a calibration status within the shortest period of time. Should traffic, road and regulatory conditions not allow such a calibration drive, the time until optimum calibration will increase. However navigation results are already satisfactory after a relatively short driving distance and time.



The above instructions shall not be made a rule towards any end user. They shall only be applied in a testing environment where sufficient care is taken that these driving instructions can be carried out without creating any risk of accidents or violation of regulations.

Consequences of a bad/wrong calibration procedure

u-blox SFDR Technology needs well-calibrated sensors to have optimal performance. A poorly calibrated system will report wrong positions and headings during GPS loss. Also the performance is degraded during good GPS performance, as the position output with good GPS performance will be combined with the poor data from the sensors.

As long as the miscalibration is minor (e.g. change of tires from summer to winter tires), the system will recover itself. If the miscalibration leads to a 'sensor integrity check error' (the receiver reports GPS only solutions), a reset of the calibration data and new initial calibration is required.

Related documents

- [1] LEA-6 Data Sheet, Docu. No GPS.G6-HW-09004
- [2] NEO-6 Data Sheet, Docu. No GPS.G6-HW-09005
- [3] u-blox 6 Receiver Description including Protocol Specification, Docu. No GPS.G6-SW-10018
- [4] GPS Antenna Application Note, Docu. No GPS-X-08014
- [5] u-blox 6 Firmware Version 7.0x Release Note, Docu. No GPS.G6-SW-10024
- [6] u-blox 6 Firmware Version 6.02 Release Note, Docu. No GPS.G6-SW-10003
- [7] LEA-5 Data Sheet, Docu. No GPS.G5-M55-07026
- [8] MAX-6 Data Sheet, Docu. No GPS.G6-HW-10106
- [9] C26 telematics reference design
- [10] u-blox 6 Receiver Description Including Protocol Specification (NDA version), Docu. No GPS.G6-SW-10019. This document requires an NDA. For more information or to obtain this document contact your nearest u-blox sales representative.

Unless otherwise stated, all these documents are available on our homepage (<http://www.u-blox.com>).



Additional information is available in the FAQ section of our website (<http://www.u-blox.com/en/faq.html>).



For regular updates to u-blox documentation and to receive product change notifications please contact our local support.

Revision history

Revision	Date	Name	Status / Comments
-	24/03/2010	tgri	Initial release
A	20/07/2010	mdur	Preliminary; updated 1.6.3 DDC, 1.7.1 RESET_N, 1.7.4 Config Pins, 2.1.1 Layout, 3.2 Soldering
B	12/22/2010	jfur	new: Integration LEA-6R (1.2), FW7, Data ready indicator (1.7.6), Second time pulse for LEA-6T (1.7.5), Figure 41 and Figure 46, External active antenna supervisor NEO-6 (2.6.6), D DR calibration updated: Rework (3.2.8), Recommended parts (3.3.9), RTCM (B), Migration of LEA-4R designs to LEA-6R (B.4.2), Checklist (2.1), 2.2.1 LEA-6 passive antenna design, 2.3.1 Passive antenna design (NEO-6)
C	2/3/2011	jfur	MAX-6x integration, External active antenna supervisor, ANTOFF, External active antenna control (NEO-6), GPS antenna placement for LEA-6R
D	5/18/2011	jfur	Repeated Reflow Soldering updated (3.2.5), Soldering u-blox 6 modules in a leaded process removed, updated for FW 7.03/DR2.0, LEA-6T-1 integration
E	8/8/2011	jfur	LEA-6x / NEO-6x updated for FW 7.03 Chapter added: 1.7.8 LEA-6T-0 antenna supervision signals Chapter 1.6.1 UART maximum baud rate. Chapter 3.3 EOS / ESD / EMI updated
F	9/14/2011	jfur	NEO-6V integration 1.6.4 SPI FLASH For new designs Chapter added: 2.2.2 LEA-6H GLONASS ready module integration B.2.2 Software migration from 6.02 to 7.03 3.3.9 Recommended parts, SAW
G	11/25/2011	jfur	new Chapter 2.1.3 Dead Reckoning Chapter 2.2.2.1 updated NEO-6P and NEO-6T integration 1.7.1 RESET_N updated Section 2.1.3 Updated: Automotive Dead Reckoning (ADR) solutions new Chapter 2.6.7 External active antenna supervisor using ANTON (MAX-6) Chapter 1.3.2.3 Power Save mode updated
H	12/7/2011	jfur	Chapter 2.1.3 Automotive Dead Reckoning updated Added Chapter 2.6.7 External active antenna supervisor using ANTON Chapter 1.3.2.3 Power Save mode updated

Contact

For complete contact information visit us at www.u-blox.com